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Geometric Estimation of Autoregressive Processes and Spectral Scale Selection for Kernel Density Estimation

Abschlussarbeit an der Universität Ulm

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Abstract

Write abstract (150-250 words)

1 Introduction

2 Geometric Estimation of Autoregressive Processes with bounded innovations

The original goal of this chapter was to apply the spectral scale selection developed in [6] to recover the order p of autoregressive time series. Even though the scale selection principle is primarily developed for applications in TDA, the hope was to use the well known sliding window embedding as a bridge between TDA and time series analysis. In the process of studying this problem, an alternative convex-geometric direction emerged, while the TDA approach turned out to be unfruitful. This chapter presents a new AR order selection method using ideas from convex geometry.

In the first section, we give an overview of the theory of time series needed for our purposes. In particular, we briefly review classical order selection and AR coefficient estimation. Next, we introduce the embedding of time series as point clouds in $\mathbb{R}^k$. We show consistency for a certain kind of convex-geometric functional, the minimum width, that can recover the correct AR order p. The projected sub-gradient descent is used to compute the empirical minimum width. A discussion of different kinds of innovation distributions is given, where the focus lies primarily on bounded distributions, of which the Beta distribution is the main example. Finally, we present our numerical results and interpret them.

The main goal of this chapter is to study the connection between time series analysis and convex geometry through sliding window analysis.

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be the underlying probability space.

2.1 Preliminaries on time series

For information on autoregressive moving-average (ARMA) processes, see Brockwell et al [9, Chapter 3].

Definition 1. *Let $\{Z_t\}_{t\in\mathbb{Z}}$ be a family of independent and identically distributed random variables with finite second moment. A process $\{X_t\}$ is an ARMA(p, q) process if*

$$X_t = \sum_{i=1}^{p} \phi_i X_{t-i} + Z_t + \sum_{j=1}^{q} \theta_j Z_{t-j}, \qquad t \in \mathbb{Z}, \tag{2.1}$$

where the polynomials $\phi(z) := 1 - \phi_1 z - \cdots - \phi_p z^p$ and $\theta(z) := 1 + \theta_1 z + \cdots + \theta_q z^q$ are required to have no common factors. Additionally, $\phi_p \neq 0$ and $\theta_q \neq 0$.

In case of $\theta \equiv 1$ we recover the autoregressive (AR) process of order $p \in \mathbb{N}$ and X_t has the form

$$X_t = \sum_{i=1}^{p} \phi_i X_{t-i} + Z_t, \tag{2.2}$$

where $\phi_i \in \mathbb{R}$, $i = 1, \ldots, p$ are the coefficients with $\phi_p \neq 0$.

It is well known that there exists a stationary process $\{X_t\}_{t\in\mathbb{Z}}$ that can be written in the form (2.1) if $\phi(z) \neq 0$ for all $z \in \mathbb{C}, |z| = 1$. Another important property that we will need for the geometric interpretation of the embedded AR process is causality: An ARMA process $\{X_t\}_{t\in\mathbb{Z}}$ is called causal if there exist constants ψ_j, $j \geq 0$ with $\sum_{j\geq 0} |\psi_j| < +\infty$, such that $X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$ for all $t \in \mathbb{Z}$. It turns out that this property is stronger than stationarity, as an ARMA process is causal if and only if $\phi(z) \neq 0$ for all $z \in \mathbb{C}, |z| \leq 1$. In what follows, we will exclusively work with causal ARMA processes.

An important second order object for stationary sequences X is its auto covariance function $\gamma_X(h) = \mathrm{Cov}[X_{t+h}, X_t]$. The auto covariance matrix Γ_k of $(X_t, \ldots, X_{t+k-1})$

is

$$\Gamma_k = \begin{pmatrix} \gamma(0) & \gamma(1) & \dots & \gamma(k-1) \\ \gamma(1) & \gamma(0) & \dots & \gamma(k-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(k-1) & \gamma(k-2) & \dots & \gamma(0) \end{pmatrix}$$

The empirical analog of Γ_k is obtained by plugging in the sample auto covariance function $\hat{\gamma}$. The sample covariance matrix $\hat{\Gamma}_k$ is positive semi-definite.

2.2 Estimation methods

We present the two classical coefficient estimation methods Yule-Walker and Maximum Likelihood. As a third estimation method we present Horváth's L^∞ estimator which is specifically designed for AR processes with innovations having uniform distribution.

Yule-Walker

The Yule-Walker Equations are given by

$$\Gamma_p \phi = \gamma_p, \quad \sigma^2 = \gamma(0) - \phi^T \gamma_p, \quad \gamma_p = \begin{pmatrix} \gamma(1) \\ \vdots \\ \gamma(p) \end{pmatrix}$$

We obtain estimates $\hat{\phi}$ and $\hat{\sigma}^2$ by plugging in the empirical analogues $\hat{\Gamma}_p$ and $\hat{\gamma}_p$ into the Yule-Walker equations and solving for ϕ and σ^2.

In particular, the Yule-Walker estimate $\hat{\phi}$ is obtained by solving the linear system of equations $\hat{\Gamma}_p \hat{\phi} = \hat{\gamma}_p$. This can be solved efficiently by using the Durbin-Levinson algorithm which exploits the Toeplitz structure of $\hat{\Gamma}_k$.

Maximum-Likelihood

Write the joint density of the realization of an AR(p) process

$$f(x_1,\dots,x_T) = f_{(X_1,\dots,X_p)}(x_1,\dots,x_p)\prod_{t=p+1}^{T}\underbrace{f_{X_t|X_1,\dots,X_{t-1}}(x_t|x_1,\dots,x_{t-1})}_{=f_{X_t|X_{t-p},\dots,X_{t-1}}(x_t|x_{t-p},\dots,x_{t-1})}$$

Conditioning on the first p observations gives

$$\begin{aligned} f_{X_{p+1},\dots,X_T|X_1,\dots,X_p}(x_{p+1},\dots,x_T|x_1,\dots,x_p) &= \frac{f(x_1,\dots,x_T)}{f_{(X_1,\dots,X_p)}(x_1,\dots,x_p)} \\ &= \prod_{t=p+1}^{T} f_{X_t|X_{t-p},\dots,X_{t-1}}(x_t|x_{t-p},\dots,x_{t-1}) \end{aligned} \tag{2.3}$$

We use conditional ML because otherwise we would have to specify the joint distribution of $(X_1,\dots,X_p)$.

Now we know that $X_t|X_{t-p},\dots,X_{t-1}$ is distributed as $Z_t+\mu_t$, where $\mu_t = \sum_{j=1}^{p}\phi_j X_{t-j}$. At this point, we can assume different distribution classes for Z_t and continue to evaluate with the assumed distribution.

Assuming that the innovations are normally distributed, we obtain the quasi-conditional maximum likelihood function

$$f(x_{p+1},\dots,x_T|x_1,\dots,x_p) = \prod_{t=p+1}^{T} f_{\mu_t,\sigma^2}(x_t), \quad f_{\mu_t,\sigma^2} \sim N(\mu_t,\sigma^2).$$

The conditional log-likelihood function is then

$$l(\phi,\sigma^2) = -\frac{T-p}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{t=p+1}^{T}(x_t-\mu_t)^2$$

Note that conditional quasi maximum likelihood is equivalent to least squares. Under fixed σ^2 maximizing the expression above is equivalent to minimizing

$$\sum_{t=p+1}^{T}\left(x_t - \sum_{j=1}^{p}\phi_j x_{t-j}\right)^2$$

Horváth

Of course we are in the scenario where we assume that the innovations are bounded. Horváth [31] assumed further that the innovations are $\mathcal{U}[-h,h]$ and found that the conditional likelihood function can then be written as a Linear program.

The Estimation of the coefficients of an AR process with uniform innovations was already considered by Horváth [31] in 1989. His starting point is Equation (2.3), but instead of using the Gaussian density he plugs in the uniform density. The resulting conditional likelihood function becomes

$$l_{x_{p+1},\dots,x_T|x_1,\dots,x_p}(\phi,h) = \frac{1}{(2h)^{T-p}}\mathbf{1}\left\{\left|y_t - \sum_{j=1}^{p}\phi_j x_{t-j}\right| \leq h \text{ for all } t = p+1,\dots,T\right\}. \tag{2.4}$$

This can be written as a linear programming problem, as demonstrated below.

We find a vector $b = (b_0, b_1, \dots, b_n)$ that minimizes $e_1^T b = b_0$ subject to

$$\begin{pmatrix} A \\ -A \end{pmatrix} \begin{pmatrix} b_0 \\ b_1 \\ \vdots \\ b_n \end{pmatrix} \leq \mathbf{1}, \quad A = \begin{pmatrix} x_{p+1} & \dots & x_1 \\ \vdots & & \vdots \\ x_T & \dots & x_{T-p} \end{pmatrix} \in \mathbb{R}^{(T-p)\times(p+1)}.$$

In modern times such a problem can be solved very efficiently by, for example, the open source software HiGHS, which is also available as Python package.

From Equation (2.4) it follows that

$$\hat{\phi} = \arg\min_{\phi}\max_{t}\left|x_t - \sum_{j=1}^{p}\phi_j x_{t-j}\right|$$
$$\hat{h} = \max_{t}\left|x_t - \sum_{j=1}^{p}\hat{\phi}_j x_{t-j}\right|$$

Thus, conditional maximum likelihood under uniform density leads to L^∞-regression analogous to how conditional ML under Gaussian density leads to L^2-regression.

2.3 Order selection methods

There are several established methods for choosing an order $\hat{p}$ given a realization $x = (x_1, \dots, x_T)$ that is assumed to come from an AR(p) process. All of them require to obtain estimates $\hat{\phi}_1^{(k)}, \dots, \hat{\phi}_k^{(k)}$ for each candidate order $k = 1, \dots, K$. This can be done efficiently using the Durbin-Levinson algorithm.

Information criteria selectors

Denote by $l(\hat{\theta})$ the log-likelihood function evaluated at its maximizer $\hat{\theta}$. The class of information criteria selectors has the form

$$\hat{p}_{\mathrm{IC}} = \arg\min_{1 \le k \le K} \mathrm{IC}_T(k), \quad \mathrm{IC}_T(k) = -\frac{2}{T-K} l(\hat{\theta}) + \frac{k c_T}{T-K}.$$

For example, $\hat{\theta} = (\hat{\phi}, \hat{\sigma}^2)$ and $l(\hat{\phi}, \hat{\sigma}_k^2) = -\frac{T-K}{2}(\log(2\pi\hat{\sigma}_k^2) + 1)$ and therefore

$$\mathrm{IC}_T(k) = \log(\hat{\sigma}_k^2) + \frac{k c_T}{T-K}$$

where we have dropped all terms independent of k. Since $\hat{\sigma}_1^2 \ge \hat{\sigma}_2^2 \ge \cdots \ge \hat{\sigma}_K^2$ (the lack of fit decreases as the candidate order increases), it must be counteracted by a penalty term for the candidate order k.

Here, $c_T > 0$ is some penalty term sequence. Different choices of c_T lead to different AR order selectors that are known in the literature. Choosing $c_T = 2$ leads to the Akaike information criterion (AIC), $c_T = \log T$ to the Bayesian information criterion (BIC), and $c_T = 2 \log\log T$ to the Hannan-Quinn information criterion (HQIC). All of these criteria can be motivated in their own sense and serve a different purpose. The AIC is asymptotically efficient, but not strongly consistent. BIC and HQIC are strongly consistent but not asymptotically efficient.

Partial autocorrelation cutoff

The partial autocorrelation function α is given by $\alpha(0) = 1$ and $\alpha(k) = \phi_{kk}$, where $\phi_k = (\phi_{1k}, \dots, \phi_{kk})$ is the coefficient vector obtained as a solution of the Yule-Walker Equations. The sample partial autocorrelation function $\hat{\alpha}$ is defined analogously through empirical analogs. For an AR(p) process we have $\alpha(p) = \phi_p$ and $\alpha(k) = 0$ for $k > p$. For causal AR(p) processes with iid innovations it is known

that $\sqrt{T}\hat{\alpha}(k) \xrightarrow{d} N(0,1)$ as $T \to +\infty$ for each fixed $k > p$. In particular, assuming T is large, we have $\mathbb{P}(\sqrt{T}\hat{\alpha}(k) \in [-b, b]) \approx 2\Phi(b) - 1$. Specifically, setting $b = \Phi^{-1}(0.975) \approx 1.96$, we obtain $\mathbb{P}(\hat{\alpha}(k) \in [-\frac{1.96}{\sqrt{T}}, \frac{1.96}{\sqrt{T}}]) \approx 0.95$. Therefore, we might choose as the AR order the last k that $\hat{\alpha}(k)$ is outside the bounds $\pm\frac{1.96}{\sqrt{T}}$.

2.4 Embedding of time series

The sliding window embedding is the conceptual bridge between time series analysis and other fields in mathematics, for example, Topological Data Analysis (TDA) or Convex Geometry. The time series is embedded into $\mathbb{R}^k$ for some embedding dimension $k \in \mathbb{N}$. In TDA the obtained sliding window point cloud is put into the persistent homology pipeline to look for interesting topological features. See the papers [43]. We will instead study the geometric properties of the sliding window point cloud in Section 2.5 below. Before that, we introduce the sliding window embedding. Let $X = (X_1, \ldots, X_T)$ be one realization of some univariate time series. Then we obtain a point cloud from X by defining:

$$Y_t^{(k)} = \begin{pmatrix} X_t \\ X_{t-1} \\ \vdots \\ X_{t-k+1} \end{pmatrix} \in \mathbb{R}^k, \quad \mathcal{P}_k = \left\{ Y_t^{(k)} : t = K, \ldots, T \right\} \subseteq \mathbb{R}^k$$

Here, $k \in \mathbb{N}$ is the dimensionality of the point cloud $\mathcal{P}_k$ and K a predetermined upper cap, that is, $k \in \{1, \ldots, K\}$. Note that we are discarding observations $X_1, \ldots, X_{K-k}$ in the construction of the point cloud $\mathcal{P}_k$. This is mainly for computational and notational convenience. We choose the "backward" embedding $Y_t^{(k)}$ over the "forward" embedding $\tilde{Y}_t^{(k)} = (X_t, X_{t+1}, \ldots, X_{t+k-1})^T$ for notational convenience throughout this chapter. Geometrically, the resulting point clouds are equivalent up to an orthogonal transformation and a time index shift, that is, $\tilde{Y}_t^{(k)} = R_k Y_{t+k-1}^{(k)}$, where $R_k = \operatorname{adiag}(1, \ldots, 1)$.

2.5 Geometric Approach

In this section, we study how the geometry of the point cloud $\mathcal{P}_k \subseteq \mathbb{R}^k$ changes with increasing embedding dimension k. For bounded innovation distributions, like uniform or beta, the point cloud $\mathcal{P}_k$ is characterized by a well known convex–geometric object. The question becomes if we can apply some convex–geometric functional f to these well known objects and detect a transition at $k = p + 1$, where p is the AR order of the embedded time series.

In order to be able to talk about the "geometry of a point cloud" in a rigorous way, we first need to introduce some concepts from probability theory. Let Y be a k-dimensional random vector with density f_Y. The *support* of Y is given by

$$\operatorname{supp}(Y) = \{x \in \mathbb{R}^k : \mathbb{P}(Y \in B_r(x)) > 0 \text{ for all } r > 0\}.$$

In the case where f_Y is continuous, it holds $\operatorname{supp}(Y) = \overline{\{y \in \mathbb{R}^k : f_Y(y) > 0\}}$.

Let $X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$ be the MA(∞)-representation of an AR(p) process, where $\{Z_t\}_{t\in\mathbb{Z}}$ are the independent and identically distributed innovations, and where the sequence $\{\psi_j\}_{j\in\mathbb{N}_0}$ is given recursively by $\psi_{-(p-1)} = \cdots = \psi_{-1} = 0$, $\psi_0 = 1$ and $\psi_j = \phi_1\psi_{j-1} + \cdots + \phi_p\psi_{j-p}$ for all $j \geq 1$. Now each point $Y_t^{(k)} \in \mathcal{P}_k$ can be written as

$$Y_t^{(k)} = \begin{pmatrix} X_t \\ X_{t-1} \\ \vdots \\ X_{t-k+1} \end{pmatrix} = \sum_{j=0}^{\infty} g_j^{(k)} Z_{t-j}, \tag{2.5}$$

where $g_j^{(k)} = (\psi_j, \psi_{j-1}, \ldots, \psi_{j-k+1})^T$. Due to stationarity of $\{X\}_{t\in\mathbb{Z}}$, the distribution of $Y_t^{(k)}$ does not depend on t and we write $Y_t^{(k)} \sim \mu_k$ which means $\mu_k(B) = \mathbb{P}(Y_t^{(k)} \in B)$ for all $B \in \mathcal{B}(\mathbb{R}^k)$.

Bounded innovation distributions lead to $Y_t^{(k)}$ having compact support. Thus, in our work, we focus on innovations Z_t that have support $[a, b]$, $a < b$. Possible ideas for extending our approach to innovation distributions with unbounded support are given in the outlook chapter 5. The most prominent example is the uniform distribution $\mathcal{U}[a, b]$. More generally, we can consider the Beta distribution. A random

variable X follows the Beta distribution Beta(α, β) if its density is given by

$$f_{\alpha,\beta}(x) = \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}\mathbf{1}\{x \in (0,1)\}, \quad \alpha, \beta > 0,$$

where $B(\alpha,\beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$ and Γ is the Gamma function. For $\alpha = \beta = 1$ we have Beta(α,β) = Beta$(1,1) = \mathcal{U}[0,1]$.

We can define a Beta distributed random variable on $[a,b]$ by setting $Z = a + (b-a)U$, where $U \sim$ Beta(α,β).

In our setting, the innovations Z_t will be of the form $(b-a)U_t+a$ with $U_t \sim$ Beta(α,β). Since we want Z_t to be centered and $\mathbb{E}[U_t] = \alpha/(\alpha+\beta)$, we must choose a and b so that $b\alpha + a\beta = 0$. Since this does not determine a and b uniquely we take Var$[Z_t]$ = Var$[U]$, where $U \sim \mathcal{U}[-1,1]$ is the default case, thus Var$[Z_t] = 1/3$. This second condition forces $a = -L\alpha/(\alpha+\beta)$ and $b = L\beta/(\alpha+\beta)$ with $L = (\alpha+\beta)\sqrt{(\alpha+\beta+1)/(3\alpha\beta)}$. We thus have centered Beta distributed innovations with variance always equal to the default case. Setting $\alpha = \beta = 1$ recovers $Z_t \sim \mathcal{U}[-1,1]$.

Initially, just uniformly distributed innovations were considered as a means simplify the study of the shape of the resulting sliding window point cloud. Later, a generalization to Beta distributed innovations was considered.

2.5.1 Convex–Geometric Definitions and Connection to sliding-window support

A convex body $K \subseteq \mathbb{R}^k$ is called centrally symmetric if there exists a $C \in \mathbb{R}^k$ (the center of symmetry) such that $K - C = -(K - C)$.

We will start exploring the geometry of the sequence of point clouds $\{\mathcal{P}_k\}_{k=1}^K$ by considering the case where the innovations are uniformly distributed, that is, $Z_t \sim \mathcal{U}[-1,1]$. To this end, we first introduce some common definitions from convex geometry. A good survey on convex geometry is the book [27].

Definition 2 (Minkowski sum)**.** *Let A and B be two subsets of a vector space V. Their Minkowski sum is given by the set $A + B = \{a + b : a \in A, b \in B\}$.*

Definition 3 (Hausdorff distance)**.** *Let (M, d) be a metric space and X, Y two nonempty subsets of M. The Hausdorff distance d_H between X and Y is defined by*

$$d_H(X,Y) = \max\left\{\sup_{x\in X} d(x,Y), \sup_{y\in Y} d(X,y)\right\},$$

where $d(x,Y) = \inf\{d(x,y) : y \in Y\}$.

Definition 4 (Zonotope)**.** *A zonotope is the Minkowski sum of finitely many line segments. It is said to be generated by $w_1, \ldots, w_n \in \mathbb{R}^d$ if it has the form*

$$\{a_1 w_1 + \cdots + a_n w_n : -1 \le a_j \le 1 \text{ for all } j\}$$

Definition 5 (Zonoid)**.** *Let $w = (w_n)_{n\in\mathbb{N}_0}$ be a sequence of vectors in $\mathbb{R}^d$. A set $\mathcal{Z}$ is called a zonoid generated by w if $d_H(\mathcal{Z}, \mathcal{Z}_n) \to 0$ for $n \to \infty$, where $\mathcal{Z}_n$ is a zonotope generated by $w_0, w_1, \ldots, w_n$.*

Using Equation (2.5) we see that the support of $Y_t^{(k)}$ is given by a zonoid generated by $\{g_j^{(k)}\}_{j\in\mathbb{N}_0}$. We denote this zonoid by $\mathcal{Z}_0^{(k)}$. By Definition 5 the zonoid $\mathcal{Z}_0^{(k)}$ can be approximated by the zonotopes $Z_n^{(k)}$ generated by $g_0^{(k)}, \ldots, g_n^{(k)}$.

Let $[a, b]$, $a < b$ be the support of the innovations $\{Z_t\}$. By writing $[a, b] = c + d[-1, 1]$, where $c = (a + b)/2$ and $d = (b - a)/2$, we can pull everything back to the special case where the support is just the interval $[-1, 1]$. We have

$$\mathcal{Z}^{(k)} := \operatorname{supp}(Y_t^{(k)}) = \sum_{j=0}^{\infty} g_j^{(k)} \operatorname{supp}(Z_{t-j}) = \underbrace{c\sum_{j=0}^{\infty} g_j^{(k)}}_{=:C_k} + \underbrace{d\sum_{j=0}^{\infty} g_j^{(k)}[-1,1]}_{=\mathcal{Z}_0^{(k)}},$$

that is, we have translated and scaled the original zonoid $\mathcal{Z}_0^{(k)}$.

In fact, we can derive the following convergence speed result. It tells us that the approximating zonotope converges to the zonoid faster if the underlying AR process is “more causal”.

Proposition 1. *Let $\mathcal{Z}_0^{(k)}$ and $\mathcal{Z}_n^{(k)}$ be like above coming from an causal autoregressive process where the impulse sequence $\{\psi_j\}$ fulfills $|\psi_j| \le C\rho^j$ for some*

constants $C > 0$ *and* $\rho \in (0,1)$*. Then*

$$d_H(\mathcal{Z}_n^{(k)}, \mathcal{Z}_0) \leq C \frac{\sqrt{k}}{1-\rho} \frac{1}{\rho^{k-2}} \rho^n.$$

In particular, $d_H(\mathcal{Z}_n^{(k)}, \mathcal{Z}_0^{(k)}) = \mathcal{O}(\rho^n)$*. That is,* $d_H(\mathcal{Z}_n^{(k)}, \mathcal{Z}_0^{(k)}) \to 0$ *is exponentially fast.*

Proof. See Appendix Section 6.1.1. □

A convex body is a compact convex set with a nonempty interior. As a convex geometric functional on a convex body K we use its minimum width. For a rigorous notion of width, we introduce the support function.

Definition 6 (Support function)**.** *Let* $K \subseteq \mathbb{R}^d$ *be a convex nonempty set. The support function of* K *is* $h_K : \mathbb{R}^d \to \mathbb{R}$*,* $h_K(u) = \sup\{\langle x, u\rangle : x \in K\}$*.*

We can express the Hausdorff distance between convex bodies using their support functions.

Lemma 1. *Let* $K, L \subseteq \mathbb{R}^k$ *be two convex bodies. Then*

$$d_H(K,L) = \sup_{u \in \mathbb{S}^{k-1}} |h_K(u) - h_L(u)| = \|h_K - h_L\|_{\infty, \mathbb{S}^{k-1}}.$$

Proof. See [23, Section 4]. □

The width of a convex body $K \subseteq \mathbb{R}^d$ in direction u is given by $w_K(u) = h_K(u) + h_K(-u)$ and if K is centrally symmetric with the origin as center of symmetry, then $w_K(u) = 2h_K(u)$. The minimum width is $m(K) = \min\{w_K(u) : u \in \mathbb{S}^{d-1}\}$. We write $w_k(u) = w_{\mathcal{Z}^{(k)}}(u)$ and $m_k = m(\mathcal{Z}^{(k)})$ for the width in direction u and the minimum width of the zonoid $\mathcal{Z}^{(k)}$, respectively.

The support function of a zonoid $\mathcal{Z}_0^{(k)}$ generated by $\{g_j^{(k)}\}_{j \in \mathbb{N}_0}$ is given by

$$h_{\mathcal{Z}_0^{(k)}}(u) = \sum_{j=0}^{\infty} |\langle u, g_j^{(k)}\rangle|, \quad u \in \mathbb{S}^{k-1}.$$

The support function of $\mathcal{Z}^{(k)}$ in general is given by

$$h_{\mathrm{supp}(Y_t^{(k)})}(u) = \langle C_k, u\rangle + \frac{b-a}{2}\sum_{j=0}^{\infty}|\langle u, g_j^{(k)}\rangle|.$$

Next, we formulate a result that states that the relation $k \mapsto m_k$ has an elbow point at $k = p+1$.

Proposition 2. *Let $\mathcal{P}_k$ be the sliding window embedding of a causal AR(p) process with coefficients $\phi_1, \ldots, \phi_p$, $\phi_p \neq 0$ and $\mathcal{Z}^{(k)}$ the corresponding support zonoid. In the case $k \geq p+1$ it holds*

$$m_k = \min_{u\in\mathbb{S}^{k-1}} w_k(u) = \frac{b-a}{\sqrt{1+\phi_1^2+\cdots+\phi_p^2}},$$

where the minimum is attained at the finite set of points

$$\left\{\pm\frac{q_r^{(k)}}{\|q_\phi\|_2} : r = 0, \ldots, k-(p+1)\right\} \tag{2.6}$$

with $q_\phi = (1, -\phi_1, \ldots, -\phi_p)$ and

$$q_r^{(k)} = (\underbrace{0, \ldots, 0}_{r}, 1, -\phi_1, \ldots, -\phi_p, 0, \ldots, 0) \in \mathbb{R}^k.$$

In the case $k < p+1$ it holds

$$m_{k-1} \geq m_k > \frac{b-a}{\sqrt{1+\phi_1^2+\cdots+\phi_p^2}}, \quad \forall k = 2, \ldots, p.$$

Proof. See Appendix Section 6.1.2. □

Remark 1 (about Proposition 2)**.** *For $k = p+1$ the minimizing direction is $u = (u_0, u_1, \ldots, u_p) = q_\phi/\|q_\phi\|$. We can recover the AR coefficients by $\phi_j = -(q_\phi)_j = -\|q_\phi\|u_j = -u_j/u_0$, $j = 1, \ldots, p$. If $\hat{u}$ is a practically obtained minimizer (for example, via gradient descent, see below), we get coefficient estimates $\hat{\phi}_j = -\hat{u}_j/\hat{u}_0$.*

2.5.2 The empirical minimum width and its consistency

An empirical version of the width $w_k(u)$ is obtained by inserting $\mathrm{conv}(\mathcal{P}_k)$ in place of $\mathcal{Z}^{(k)}$. Consequently, we obtain

$$\begin{aligned}\hat{w}_{k,T}(u) &= h_{\mathrm{conv}(\mathcal{P}_k)}(u) + h_{\mathrm{conv}(\mathcal{P}_k)}(-u) \\ &= \max_{x\in\mathcal{P}_k}\langle x, u\rangle - \min_{x\in\mathcal{P}_k}\langle x, u\rangle \\ &= \max_t \langle Y_t^{(k)}, u\rangle - \min_t \langle Y_t^{(k)}, u\rangle\end{aligned}$$

and $\hat{m}_{k,T} = \min\{\hat{w}_k(u) : u \in \mathbb{S}^{k-1}\}$.

The next results establish the consistency of the empirical minimum width, that is, $\hat{m}_{k,T} \to m_k$ for $T \to \infty$. In the next Lemma we show that the convex hull of a point cloud converges to its support $\mathcal{Z}$ almost surely in Hausdorff distance given that the point cloud is dense in $\mathcal{Z}$ with probability 1.

Lemma 2. *Let $\{Y_t\}_{t\geq 1}$ be a $\mathbb{R}^k$ valued process and $\mathcal{P}_T = \{Y_1, \ldots, Y_T\}$. Let $\mathcal{Z} \subseteq \mathbb{R}^k$ be nonempty, compact and convex. Assume*

$$\mathbb{P}\left(\overline{\{Y_t : t \geq 1\}} = \mathcal{Z}\right) = 1$$

Then

$$d_H(\mathrm{conv}(\mathcal{P}_T), \mathcal{Z}) \overset{T\to+\infty}{\to} 0, \quad \textit{almost surely}$$

Proof. See Appendix Section 6.1.3. □

There exists work on the convergence speed of the sequence $d_H(\mathrm{conv}(\mathcal{P}_T), \mathcal{Z})$ for $T \to +\infty$, but only under certain assumptions. Brunel [10] proves $\mathbb{E}[d_H(\mathcal{P}_T, \mathcal{Z})] = \mathbf{O}((\frac{\log n}{n})^{1/\alpha})$ under the assumption that "enough probability mass is put at the caps of $\mathcal{Z}$". He also assumes that $Y_1, \ldots, Y_T$ are independent and identically distributed. Louhichi [40] instead considers stationary points $Y_1, \ldots, Y_T$ but assumes a stronger "a-b-condition" in addition with other specific conditions. She then proceeds to show the same kind of convergence speed as Brunel.

We state the "enough probability mass" condition from Brunel and show that the sliding window point cloud fails this condition. The ϵ-cap of a convex body $\mathcal{Z} \subseteq \mathbb{R}^k$

in direction $u \in \mathbb{S}^{k-1}$ is the set $C_{\mathcal{Z}}(u, \epsilon) = \{x \in K : \langle x, u \rangle \geq h_{\mathcal{Z}}(u) - \epsilon\}$. Brunel's condition is that there exist $\alpha, L > 0$ and $0 < \epsilon_0 \leq 1$ such that $\mu_k(C_{\mathcal{Z}}(u, \epsilon)) \geq L\epsilon^\alpha$ for all $u \in \mathbb{S}^{k-1}$ and $\epsilon \in [0, \epsilon_0]$.

Louhichi's "a-b condition" is given as follows. There exist $a, b, \epsilon_0 > 0$ such that $\inf_{y \in \mathcal{Z}^{(k)}} \mathbb{P}(Y_t^{(k)} \in B(y, \epsilon)) \geq a\epsilon^b$ for all $0 < \epsilon \leq \epsilon_0$. This is a stronger condition as Brunel's condition, because $Y_t^{(k)} \in B(y, \epsilon)$ implies $Y_t^{(k)} \in C_{\mathcal{Z}}(u, \epsilon)$.

The next Proposition shows that in our setting, Brunel's condition fails and therefore also Louchichi's condition.

Proposition 3. *Let $\mathcal{Z}^{(k)}$ be the zonoid from above. Then we have*

$$\lim_{\epsilon \to 0} \frac{\mu_k(C_{\mathcal{Z}^{(k)}}(e_1, \epsilon))}{\epsilon^q} = 0$$

for all $q > 0$. In particular, Brunel's condition is not fulfilled.

Proof. See Appendix Section 6.1.5. □

The next Lemma now shows that the support of a stationary and ergodic process is in fact dense in $\mathcal{Z}$ with probability 1.

Lemma 3. *Let $\{Y_t\}$ be stationary and ergodic. Furthermore, let $\mathcal{Z} = \operatorname{supp}(Y_t)$ be compact and convex. Then $\{Y_t : t \geq 1\}$ is dense in $\mathcal{Z}$ almost surely.*

Proof. See Appendix Section 6.1.4. □

Remark 2. *A causal AR process has the representation $X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$. Such processes are stationary and ergodic given that the innovations $\{Z_t\}$ are stationary and ergodic. Furthermore, the embedded process $t \mapsto Y_t^{(k)}$ inherits the stationarity and ergodicity.*

Corollary 1. *We have $\hat{m}_{k,T} \to m_k$ for $T \to \infty$ almost surely.*

Proof. For once we have

$$|\hat{m}_{k,T} - m_k| = |\min_u \hat{w}_{k,T}(u) - \min_u w_k(u)| \leq \sup_u |\hat{w}_{k,T}(u) - w_k(u)|$$

By Lemma 1 we have

$$\sup_{u\in\mathbb{S}^{k-1}} |\hat{w}_{k,T}(u) - w_k(u)| \leq 2d_H(\mathrm{conv}(\mathcal{P}_k), \mathcal{Z}^{(k)})$$

Finally, applying Lemmas 2 and 3 and Remark 2 proves the statement. □

The proof of Corollary 1 in particular establishes uniform almost sure convergence of $\hat{w}_{k,T}$ to w_k on $\mathbb{S}^{k-1}$. This allows us to establish an argmin consistency result.

Corollary 2. *Let $\mathcal{U}_k$ be the set of minimizing directions of w_k and $\hat{\mathcal{U}}_{k,T}$ the set of minimizing directions of $\hat{w}_k$. Then we have*

$$\sup_{\hat{u}\in\hat{\mathcal{U}}_{k,T}} \mathrm{dist}(\hat{u}, \mathcal{U}_k) \to 0, \quad T \to +\infty, \quad \textit{almost surely},$$

that is, every empirical minimizer becomes arbitrarily close to the set of the population minimizing directions.

Proof. Let $\hat{u} \in \hat{\mathcal{U}}_{k,T}$. For every $\epsilon > 0$ define the set $A_\epsilon := \{u \in \mathbb{S}^{k-1} : \mathrm{dist}(u, \mathcal{U}_k) \geq \epsilon\}$. Note that A_ϵ is a compact set. Now fix $\epsilon > 0$ and assume that A_ϵ is nonempty, that is, there exists $u_0 \in \mathbb{S}^{k-1}$ such that $\|u_0 - \tilde{u}\| \geq \epsilon$ for all $\tilde{u} \in \mathcal{U}_k$. By continuity of w_k there exists $u_\epsilon \in A_\epsilon$ with $w_k(u_\epsilon) = \min_{u_0\in A_\epsilon} w_k(u_0)$. Define $\delta_\epsilon := w_k(u_\epsilon) - m_k \geq 0$. Since $\delta_\epsilon = 0$ would be a contradiction, we have $\delta_\epsilon > 0$.

We already know that $0 \leq w_k(\hat{u}_{k,T}) - m_k \to 0$ for $T \to \infty$. In particular, $w_k(\hat{u}_{k,T}) - m_k < \delta_\epsilon$ for all sufficiently large T (all $T \geq T_0$ for some T_0).

Now $\hat{u}_{k,T} \in A_\epsilon$ would imply $w_k(\hat{u}_{k,T}) - m_k \geq \delta_\epsilon$, which would be a contradiction. Therefore, $\hat{u}_{k,T} \notin A_\epsilon$ for all $T \geq T_0$. This means $\mathrm{dist}(\hat{u}_{k,T}, \mathcal{U}_k) < \epsilon$. Since $\epsilon > 0$ was arbitrarily chosen, this proves $\mathrm{dist}(\hat{u}_{k,T}, \mathcal{U}_k) \to 0$ for $T \to \infty$. Finally, we get $\sup_{\hat{u}\in\hat{\mathcal{U}}_{k,T}} \mathrm{dist}(\hat{u}, \mathcal{U}_k) \to 0$ for $T \to \infty$. □

We can specify this result in the case where $k = p + 1$. Then we know from Proposition 2 that $\mathcal{U}_{p+1} = \{\pm\|q_\phi\|_2^{-1} q_\phi\}$. Consequently, we would get

$$\sup_{\hat{u}\in\hat{\mathcal{U}}_{p+1,T}} \min_{s\in\{-1,1\}} \|\hat{u} - s\|q_\phi\|_2^{-1} q_\phi\|_2 \to 0, \quad T \to \infty.$$

The first coordinate of $\|q_\phi\|_2^{-1} q_\phi$ is $(1 + \phi_1^2 + \cdots + \phi_p^2)^{-1/2}$ (nonzero), and by the convergence result above, the first coordinate of $\hat{u}_{k,T}$ is also eventually nonzero almost surely. The map $g : (u_0, u_1, \ldots, u_p) \mapsto (-u_1/u_0, \ldots, -u_p/u_0)$ is continuous on $\mathbb{S}^p \setminus \{u = (u_0, \ldots, u_p) : u_0 = 0\}$ and even. By Remark 1 we get

$$\sup_{\hat{u} \in \hat{\mathcal{U}}_{k,T}} \|g(\hat{u}) - \phi\|_2 \to 0, \quad T \to \infty.$$

Thus, we have shown that a continuous transformation of every empirical minimizer is a strongly consistent estimator for the AR coefficients ϕ.

This is reminiscent of the classical argmax theorem which can be found, for example, in [49].

2.5.3 Minimization via Projected sub-gradient descent

The idea now is to minimize $\hat{w}_{k,T}(u)$ over all $u \in \mathbb{S}^{k-1}$ and examine the relation $k \mapsto \hat{m}_{k,T}$. The hope is that the theoretically established elbow point that occurs at $k = p + 1$ will also be detectable at the empirical level.

Before discussing the minimization of $\hat{w}_k = \hat{w}_{k,T}$ we collect some properties of this function.

Proposition 4. *1. $\hat{w}_k$ is Lipschitz continuous on S^{k-1} with Lipschitz constant $\operatorname{diam}(\mathcal{P}_k) = \max_{s,t} \|Y_s^{(k)} - Y_t^{(k)}\|$*

2. $\hat{w}_k$ is convex on $\mathbb{R}^k$

3. The set of minimizers $\mathcal{X}_k = \arg\min\{\hat{w}_k(u) : u \in \mathbb{S}^{k-1}\}$ is nonempty and compact.

4. Denote by $\hat{m}_k$ the minimum of $\hat{w}_k$ on $\mathbb{S}^{k-1}$. If $T \geq K + k$, then $\hat{m}_k > 0$ almost surely.

5. $\hat{w}_k$ is piecewise linear and differentiable almost everywhere

6. The set of non-differentiable points is given by

$$\mathcal{N} = \bigcup_{(s,t) \neq (p,q)} \{u : \langle Y_s^{(k)} - Y_t^{(k)} + Y_q^{(k)} - Y_p^{(k)}, u \rangle = 0\}$$

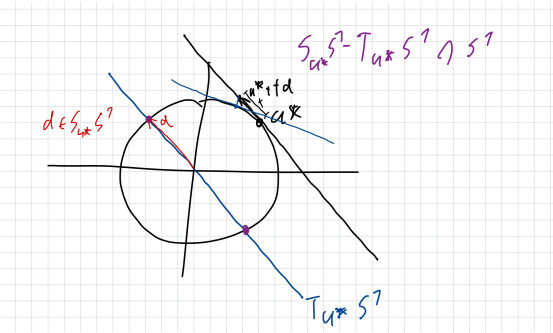


Figure 2.1: One step of the projected sub-gradient descent from Algorithm 1

7. *The gradient of $\hat{w}_k$ in direction $u \notin \mathcal{N}$ is given by*

$$\nabla_u \hat{w}_k = Y^{(k)}_{s_0(u)} - Y^{(k)}_{t_0(u)}, \quad (s_0(u), t_0(u)) = \arg\max_{s,t} \langle Y^{(k)}_s - Y^{(k)}_t, u \rangle$$

Proof. See Appendix Section 6.1.6. □

A classical minimization approach is the projected sub-gradient descent; see Algorithm 1 and Figure 2.1 which visualizes one step of the projected sub-gradient descent from Algorithm 1. [38] provide convergence guaranties. We show that our setting fulfills the assumptions stated in [38, Theorem 4] needed for the linear convergence of Algorithm 1 if the initial point u_0 is close to $\mathcal{X}_k$. But first, we state this result in our notation.

Algorithm 1 Projected gradient descent

Require: $P = \{x_1, \dots, x_n\} \subseteq \mathbb{R}^d$, $u_0 \in \mathbb{S}^{d-1}$, $\{\gamma_t\}_{t \geq 0}$, $\gamma_t = \beta^t \gamma_0$ for some $\gamma_0 > 0$ and $\beta \in (0, 1)$

Ensure: $\hat{u}$

1: Initialize $\hat{w} \leftarrow \hat{w}_k(u_0)$, $\hat{u} \leftarrow u_0$, $\hat{\phi} \leftarrow -\hat{u}[2:d]/\hat{u}_1$
2: **for** $t = 0, 1, \dots$ **do**
3: $\quad (i_t, j_t) \leftarrow \arg\max_{(i,j)} \langle x_i - x_j, u \rangle$
4: $\quad g_t \leftarrow x_{i_t} - x_{j_t}$
5: $\quad \xi_t \leftarrow (I - u_t u_t^T) g_t$
6: $\quad u_{t+1} \leftarrow (u_t - \gamma_t \xi_t)/\|u_t - \eta_t \xi_t\|$
7: $\quad$ if $\hat{w}_k(u_{t+1}) < \hat{w}$, set $\hat{u} \leftarrow u_{t+1}$, $\hat{\phi} \leftarrow -\hat{u}[2:d]/\hat{u}_1$
8: **end for**
9: **return** $\hat{u}$

Theorem 1 ([38, Theorem 4])**.** *Consider the minimization problem* $\min f(x)$*,* $x \in \mathbb{S}^{k-1}$*, where* f *is convex on* $\mathbb{R}^k$ *and Lipschitz-continuous with constant* $L > 0$*. Let* $\{u_k\}_{k\geq 0}$ *be the sequence obtained from Algorithm 1. Let* $\mathcal{X} = \arg\min\{f(x) : x \in \mathbb{S}^{k-1}\}$ *be the set of minimizing directions of* f *and let* f^* *be the minimum of* f *over* $\mathbb{S}^{k-1}$*. Assume that the weak sharp minima condition holds, that is,*

$$\exists \alpha, \rho > 0 \text{ such that } \forall u \in \mathbb{S}^{k-1}, \operatorname{dist}(u, \mathcal{X}) < \rho : f(u) - f^* \geq \alpha \operatorname{dist}(u, \mathcal{X}). \quad (2.7)$$

Define the sequence of step sizes by $\gamma_t = \beta^t \gamma_0$*,* $t = 0, 1, \ldots$*, where* β *and* γ_0 *satisfy certain conditions. If* $u_0 \in \mathbb{S}^{k-1}$ *fulfills* $\operatorname{dist}(u_0, \mathcal{X}) < \min\{\alpha/L, \rho\}$*, then* $\operatorname{dist}(u_t, \mathcal{X}) \leq \beta^t e_0$ *for all* $t \geq 0$*.*

We already established the convexity and Lipschitz-continuity of $f = \hat{w}_k$ in Proposition 4. It turns out that the weak sharp minima condition from Equation (2.7) is also fulfilled in our setting. We record this fact in the following Lemma.

Lemma 4. *There exist* $\alpha > 0$ *and* $\rho > 0$ *such that* $\hat{w}_k(u) - \hat{m}_k \geq \alpha \operatorname{dist}(u, \mathcal{X}_k)$ *for all* $u \in \mathbb{S}^{k-1}$ *with* $\operatorname{dist}(u, \mathcal{X}_k) < \rho$*.*

Proof. See Appendix Section 6.1.7. □

In practice, the parameters γ_0 and β need to be tuned as there is no chance of obtaining explicit values for α, ρ in general. Note that this geometrically diminishing step size schedule is equivalently known as exponentially decaying step size schedule: $\gamma_t = \gamma_0 e^{-dt}$ with $d := -\log \beta > 0$ for $\beta \in (0, 1)$.

2.6 Numerical Results

For our experiments, we will limit ourselves to AR(p) time series with short-term dependence, that is, $1 \leq p \leq 5$, while larger orders can be addressed in future work.

We test our newly developed geometric estimators on simulated and real data. Regarding the experiments with simulated data we will investigate different scenarios: first we will simulate AR(p) processes, "forget" what the order p is and try to recover

it using our geometric approach coupled with an elbow method. We will compare it against the classical established methods BIC and PACF cutoff, that were presented in Section 2.3. The second scenario that we consider is, that we generate AR(p) processes and keep the knowledge of the true AR(p) order. We then compare our estimation method (projected sub-gradient descent, see Remark 1) with the classical coefficient estimation methods presented in Section 2.2.

Choosing the AR coefficients

For our experiments, we want to sample the coefficients $\phi_1, \ldots, \phi_p$ in a theoretically grounded way. Jones [36] gives one such approach. There, an algorithm is given which samples $\phi = (\phi_1, \ldots, \phi_p)$ uniformly from

$$C_p = \{\phi \in \mathbb{R}^p : 1 - \phi_1 z - \cdots - \phi_p z^p \neq 0 \text{ for all } z \in \mathbb{C}, |z| \leq 1\}$$

This allows for fair and principled numerical experiments.

Choosing the starting direction

The choice of the starting direction is arguably the most critical step in the algorithm because Theorem 1 guarantees fast convergence only if the starting direction is already close to the set of minimizers.

A general approach for choosing the starting direction u_0 is to sample it uniformly from $\mathbb{S}^3$, see [38, Section 7]. As a second option we try out the simple deterministic starting direction $e_1 = (1, 0, 0, 0)$.

Termination condition

We implement a maximum number of iterations $\text{iter}_{\max} = 2000$ and an improvement patience $\text{pat} = 200$, that is, if the function value has not improved for pat, the descent is terminated prematurely.

Tuning experiments

The goal is to determine γ_0 and β that will work reasonably well. We parametrize β via $\beta^H = 1/2$ where H is the number of iterations after which the step size halves. We consider the grid $H \in \{2, 5, 10, 20, 50, 100, 200, 1000\}$. Regarding the initial step size, we compare two options: 1) not dependent on the data: $\gamma_0 \in 0.05, 0.1, 0.2, 0.4, 0.8$ and 2) data-dependent: $\gamma_0 = cL^{-1}$, where $L = \operatorname{diam}(\mathcal{P}_k)$ is the Lipschitz-constant of the width $\hat{w}_k$ and $c \in 0.05, 0.1, 0.2, 0.4, 0.8$. Precedence

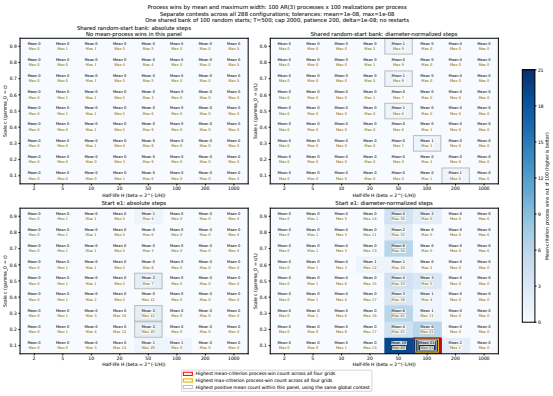


Figure 2.2

for dividing the step size by the Lipschitz-constant can be found in [38, Theorem 4] or [11, Theorem 3.2].

We sample 100 different AR(3) processes uniformly using Jones' Algorithm. For each AR(3) process, L independent realizations are generated and the best configuration is looked for. Afterwards we have $100L$ best configurations and we choose the one configuration which was best most often.

The best configuration (β, γ_0) is the one that achieves the lowest objective value most often. The results of our tuning experiment are given in Figure 2.2.

These tuning experiments show that normalizing the step size by dividing by L leads to better performance. Also the deterministic starting direction e_1 performed better than a fixed random starting direction. The best configuration is $H = 100$ and $c = 0.1$.

Repeat under many u_0?

There exists an interesting probabilistic interpretation for why the starting direction e_1 worked better than uniformly drawn u_0. We know from Proposition 2 that the optimal direction u^* can be expressed using the AR coefficients as $u^* = (1, -\phi)/\sqrt{1 + \|\phi\|^2}$. The angle between u^* and e_1 is $\arctan \|\phi\|$ while the expected angle between u^* and $u_0 \sim U(\mathbb{S}^3)$ is $\mathbb{E}[\angle(u_0, u^*)] = \pi/4 + 1/\pi \approx 1.1037$. If we draw ϕ uniformly from the causality region then $\mathbb{E}[\arctan \|\phi\|] \approx 0.7556$.

All subsequent experiments are performed using different seeds.

2.6.1 Simulated Data

Coefficient estimation experiments

Here we compare the four presented coefficient estimation methods presented in Section 2.2, namely, OLS (L_2), Yule-Walker, Horváth's L_∞ linear program and our minimum-width approach.

Our experiment looks like this: We sample $L = 100$ causal AR(p)-processes for $p \in \{1, 2, 3, 4, 5\}$. We assume that we know the true order p and compute the estimates $\hat{\phi}_1^{(l)}, \ldots, \hat{\phi}_p^{(l)}$, $l = 1, \ldots, L$. As a measure of error we use $\mathrm{MAE} = L^{-1} \sum_{l=1}^{L} \|\hat{\phi}^{(l)} - \phi^{(l)}\|_1$ (mean absolute error) and median absolute error.

As innovation distribution, we choose the Beta distribution Beta(α, β) on the interval $[-1, 1]$ with different choices of $\alpha > 0$ and $\beta > 0$:

$$(\alpha, \beta) \in \{(1, 1), (1/2, 1/2), (2, 2), (1/2, 2), (2, 1/2)\}.$$

Figure 2.3 shows on the left side 5 AR(1) processes with Beta(α, β)-distributed innovations for different α and β. On the right the corresponding sliding window point clouds are shown. We see that the parameters α and β control the position of the point cloud inside the support zonoid, whose borders are marked using a black dashed line.

Table 2.1 shows the results. Figure 2.4 illustrates the results.

We see that the L^∞-regression approach works best across all orders and that for all methods except the Minimum width approach, the error increases for larger orders p. See Figure **??**. Strikingly the minimum width approach fails for orders $p \geq 3$. This leads us to an investigation of why this is the case. From the table we also see that the median absolute errors of the Minimum width are much less of a failure. In fact, they beat OLS and Yule-Walker, but are still below L^∞-regression. This indicates that there are some specific runs where the Minimum width approach failed badly.

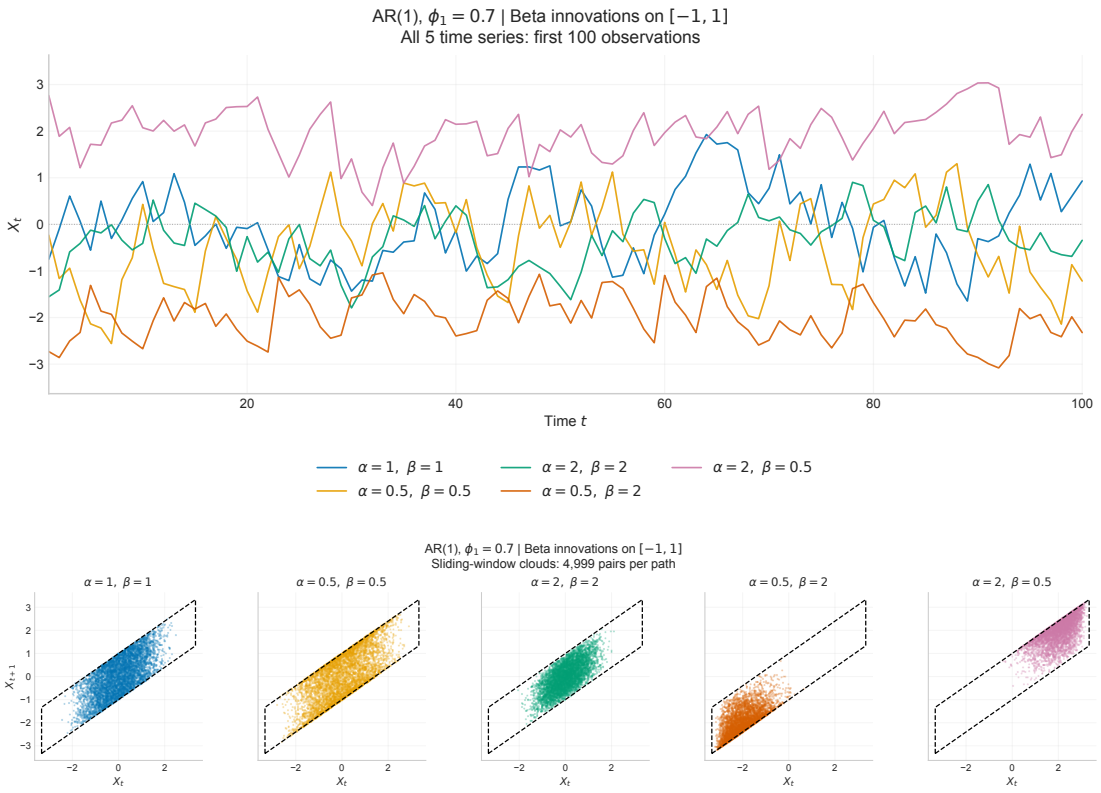


Figure 2.3: AR(1) time series with different Beta innovation distributions on $[-1, 1]$ (top) and their corresponding sliding-window point clouds (bottom). Black dashed lines indicate the support boundary.

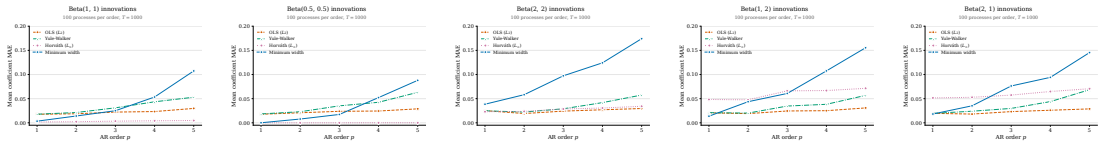


Figure 2.4: Mean coefficient MAE versus AR order p for five Beta innovation distributions. Each panel compares OLS, Yule–Walker, Horváth's L_∞ estimator, and minimum width.

Table 2.1: Mean coefficient MAE over 100 processes per order and innovation distribution, with $T = 1000$. Innovations are normalized to mean zero and variance $1/3$. Values are rounded to four decimal places. Bold values indicate the lowest unrounded mean MAE within each distribution and order.

Estimator	AR(1)	AR(2)	AR(3)	AR(4)	AR(5)
$\mathrm{Beta}(1, 1)$					
OLS (L_2)	0.0177	0.0187	0.0222	0.0233	0.0298
Yule–Walker	0.0178	0.0213	0.0310	0.0435	0.0530
Horváth (L_∞)	**0.0016**	**0.0023**	**0.0034**	**0.0042**	**0.0047**
Minimum width	0.0034	0.0141	0.0250	0.0534	0.1074
$\mathrm{Beta}(0.5, 0.5)$					
OLS (L_2)	0.0177	0.0208	0.0238	0.0244	0.0289
Yule–Walker	0.0184	0.0229	0.0351	0.0425	0.0631
Horváth (L_∞)	**0.0000**	**0.0000**	**0.0000**	**0.0001**	**0.0001**
Minimum width	0.0001	0.0078	0.0173	0.0522	0.0879
$\mathrm{Beta}(2, 2)$					
OLS (L_2)	0.0252	**0.0192**	**0.0242**	**0.0270**	**0.0296**
Yule–Walker	0.0248	0.0223	0.0290	0.0418	0.0575
Horváth (L_∞)	**0.0230**	0.0236	0.0286	0.0306	0.0342
Minimum width	0.0385	0.0582	0.0973	0.1242	0.1742
$\mathrm{Beta}(1, 2)$					
OLS (L_2)	0.0207	**0.0190**	**0.0245**	**0.0251**	**0.0308**
Yule–Walker	0.0210	0.0201	0.0347	0.0383	0.0568
Horváth (L_∞)	0.0483	0.0476	0.0662	0.0668	0.0714
Minimum width	**0.0137**	0.0437	0.0608	0.1075	0.1554
$\mathrm{Beta}(2, 1)$					
OLS (L_2)	0.0196	**0.0181**	**0.0232**	**0.0260**	**0.0288**
Yule–Walker	0.0195	0.0240	0.0298	0.0439	0.0685
Horváth (L_∞)	0.0516	0.0528	0.0574	0.0648	0.0706
Minimum width	**0.0182**	0.0352	0.0763	0.0941	0.1454

Order selection experiments

An alternative possibility for the order selection experiments in Section 2.6.1 below is to take $u_0 = (\hat{u}_k, 0) \in \mathbb{S}^k$ or $u_0 = (0, \hat{u}_k) \in \mathbb{S}^k$ as starting direction for minimum width computation in dimension $k+1$, where $\hat{u}_k$ is the obtained directions from dimension k.

Elbow point selection

rework completely (put below coefficient estimation experiments?)

The last step is to employ an elbow point selector with the property that it would always pick the correct kink in the population case. That is, $\hat{k} = p + 1$ is chosen given that $m_1 \geq \cdots \geq m_p \geq m_{p+1} = \cdots = m_K$.

Second derivative

A natural break point selector is the discrete version of the second derivative since it detects changes in slope.

Descent chain

The descent chain is motivated by the manuscript [6] where different kinds of Markov chains are constructed whose stationary distribution provides inside on which scale is the most distinct. We ask if there is a construction of a Markov chain using the observations $m_1 \geq \ldots m_p \geq m_{p+1} \geq \cdots \geq m_K$ whose stationary distribution peaks at the break point m_{p+1}.

$$\hat{k}_{\mathsf{DC}} = \min \arg \max_{2 \leq k \leq K-1} \frac{\hat{m}_{k-1} - \hat{m}_k}{\hat{m}_k - \hat{m}_K}$$

Table 2.2 summarizes the results from 50 simulations for each true order $p \in \{1, 2, 3, 4, 5\}$. We compare the classical order selectors BIC and PACF cutoff with our minimum width approach, where as elbow selector, the descent chain rule is used.

Table 2.2: AR-order selections for 50 processes per true order, pooled across five Beta innovation shapes. Each entry reports the number of exact selections, followed by (underselections/overselections). $T = 1000$ and $K = 10$.

Selector	AR(1)	AR(2)	AR(3)	AR(4)	AR(5)
Empirical minimum width (descent chain)	32/50 (0/18)	22/50 (4/24)	18/50 (7/25)	22/50 (10/18)	18/50 (14/18)
BIC	**45/50** (5/0)	**42/50** (8/0)	**46/50** (4/0)	**40/50** (10/0)	**42/50** (7/1)
PACF cutoff	33/50 (4/13)	29/50 (5/16)	35/50 (4/11)	38/50 (4/8)	29/50 (5/16)

2.7 Literature positioning and novelty assessment

In this final Section of the Chapter we assess the novelty of our approach and position it among existing literature.

First of all, special estimation methods for auto-regressive processes with bounded innovations have been considered before, the main precedent being Horváth's paper where the maximum likelihood estimation with uniform density is formulated as Linear Program (LP). In the time after, exact and very fast LP solvers have been developed.

Another precedent for the study of auto-regressive processes with bounded innovations is [19]. For a causal AR(1) process with the innovation distribution having bounded support $[x_0, x_1]$ and the cumulative distribution function F of the innovations being regularly varying at x_0 and x_1, they define two statistics $T_{1,n}$ and $T_{2,n}$ with $T_{1,n} \leq \phi \leq T_{2,n}$. It is shown that $a_n^{-1}(\phi - T_{1,n})$ and $a_n^{-1}(T_{2,n} - \phi)$ convergence in distribution towards independent Weibull random variables W_1 and W_2, where a_n is some appropriately chosen scaling sequence with $a_n \to 0$. In particular, this implies that both $T_{1,n}$ and $T_{2,n}$ converge to ϕ in probability at rate a_n. The estimator $\hat{\phi}_n := (T_{2,n} + T_{1,n})/2$ can be defined and the nondegenerate limit distribution $a_n^{-1}(\hat{\phi}_n - \phi) \xrightarrow{d} (W_2 - W_1)/2$ holds.

The conditional maximum likelihood approach with innovations assumed to be from

$U[-h,h]$ leads to L^∞-regression:

$$\min_{\phi\in\mathbb{R}^p}\max_t\left|\underbrace{X_t-\sum_{j=1}^p\phi_jX_{t-j}}_{=\langle Y_t^{(p+1)},q_\phi\rangle}\right|=\min_{\phi\in\mathbb{R}^p}\max_t|X_t-\phi^T(X_{t-1},\dots,X_{t-p})|$$

This is a L^∞-analog of ordinary least squares.

Define a hyperplane $H_{\beta,b}$ by $H_{\beta,b}:=\{x\in\mathbb{R}^k:\langle x,\beta\rangle=b\}$. The distance of a point $z\in\mathbb{R}^{(k)}$ from the hyperplane $H_{\beta,b}$ is $d(z,H_{\beta,b})=\inf_{x\in H_{\beta,b}}\|z-x\|=(|\langle\beta,z\rangle-b|)/\|\beta\|$. This allows us to rewrite the above L^∞-minimization in the following way:

$$\min_{\phi\in\mathbb{R}^p}\sqrt{1+\|\phi\|^2}\max_t d(Y_t,H_{q_\phi,0})$$

Geometrically, this is a hyperplane fitting problem, where the optimal hyperplane minimizes the maximum distance along the response axis to any point. The enclosing parallel slab has minimal vertical half width.

The map $\phi\mapsto q_\phi$ maps each $\phi\in\mathbb{R}^p$ inside the hyperplane $H_{e_1,1}\subseteq\mathbb{R}^{p+1}$.

A radial projection on $S\subseteq\mathbb{R}^k\setminus\{0\}$ is a map $r:S\to\mathbb{S}^{k-1}$, $r(x)=x/\|x\|$. The radial projection of $q_\phi\in H_{e_1,1}$ is given by $r(q_\phi)=(1,-\phi)/(1+\|\phi\|^2)$. The inverse r^{-1} on $\mathbb{S}_+^{k-1}$ is given by $r^{-1}(u)=u/u_0$, where $u=(u_0,\dots,u_{k-1})\in\mathbb{S}_+^{k-1}$. This is the coefficient recovery mentioned in Remark 1.

The minimum width approach can also be formulated in terms of a hyperplane fitting problem:

$$\frac{1}{2}\left(\max_t\langle Y_t,u\rangle-\min_t\langle Y_t,u\rangle\right)=\min_c\max_t d(Y_t,H_{u,c})$$

and therefore

$$\begin{aligned}\min_{u\in\mathbb{S}^{k-1}}\hat{w}_k(u)&=2\min_{u\in\mathbb{S}^{k-1}}\frac{1}{2}\left(\max_t\langle Y_t,u\rangle-\min_t\langle Y_t,u\rangle\right)\\&=2\min_{c\in\mathbb{R},u\in\mathbb{S}^{k-1}}\max_t d(Y_t,H_{u,c})\\&=2\min_H\max_t d(Y_t,H)\end{aligned}$$

The minimum width approach can be thus thought of as fitting a hyperplane inside the sliding-window point cloud such that the maximum distance of this hyperplane to any point Y_t is minimized.

It is a L^∞-analog of Total Least Squares.

Combining the above formulations, we can also write the minimum width problem like this:

$$\min_{\phi \in \mathbb{R}^p} \frac{\max_t \langle Y_t, q_\phi \rangle - \min_t \langle Y_t, q_\phi \rangle}{\sqrt{1 + \|\phi\|^2}}$$

A visualization is given in Figure 2.5.

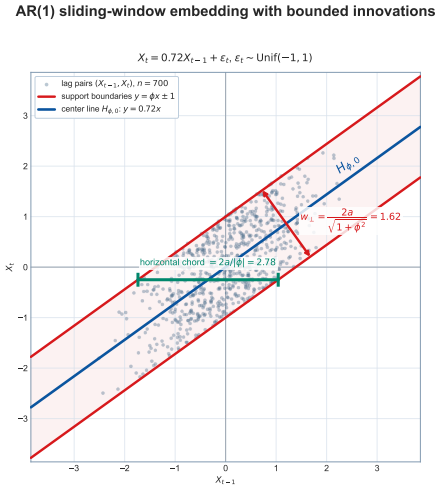


Figure 2.5: A visualization of the different minimization problems

We can write the L^∞-problem in terms of the width function w_k as well. Namely, the problem can be reformulated as

$$\frac{1}{2} \min_{u \in \mathbb{S}^{k-1}, u_0 \neq 0} \frac{w_k(u)}{|u_0|} = \frac{b-a}{2} \min_{q \in \mathbb{R}^k, q_0 = 1} h_{\mathcal{Z}_0^{(k)}}(q) = \frac{b-a}{2} \rho_{\mathcal{Z}_0^{(k)}}(e_0),$$

where $\rho_{\mathcal{Z}_0^{(k)}}(v) := \max\{r \geq 0 : rv \in \mathcal{Z}_0^{(k)}\}$ is the radial (gauge) function of $\mathcal{Z}_0^{(k)}$. The radial function is the reciprocal of the well known Minkowski-functional.

Using this formulation, we see that is also has the “break point property” like the minimum width (see Proposition 2).

Let $\mathcal{N}_k = \{u \in \mathbb{R}^k : u_0 = 1\}$. Let $u = (u_0, \dots, u_{k-1}) \in \mathcal{N}_k$ and let $\tilde{u}$ be a sequence defined by u at the indices $0, \dots, k-1$ and zero everywhere else. Define the sequence $a = \tilde{u} * \psi$. We already know that $\tilde{u} = q_\phi * a$ and $h_k(u) = 1 + \sum_{j \geq 1} |a_j|$. We have $h_k(u) \geq 1$ and $h_k(u) = 1$ iff $a = \delta_0$ iff $\tilde{u} = q_\phi$.

For $k \leq p$ any $u \in \mathcal{N}_k$ has the padded form $\tilde{u} = (\dots, 0, 1, u_1, \dots, u_{k-1}, 0, \dots)$. In particular, $\tilde{u}_p = 0$, but $(q_\phi)_p \neq 0$ and since $h_k(u) = 1$ iff $\tilde{u} = q_\phi$ we have $h_k(u) > 1$.

Since $\mathcal{N}_k$ is nonempty and closed and h_k is continuous and coercive, the minimum is still attained and we have $\min_{u \in \mathcal{N}_k} h_k(u) > 1$ for $k \leq p$. We still have the monotonicity because $\min_{u \in \mathcal{N}_{k+1}} h_{k+1}(u) \leq h_{k+1}((u, 0)) = h_k(u)$ for all $u \in \mathcal{N}_k$ and hence $\min_{u \in \mathcal{N}_{k+1}} h_{k+1}(u) \leq \min_{u \in \mathcal{N}_k} h_k(u)$.

The problem of determining the minimum width $\min_{\|u\|=1} w(u)$, $w(u) = \max_{1 \leq i \leq n} \langle x_i, u \rangle - \min_{1 \leq i \leq n} \langle x_i, u \rangle$ of a point cloud $P = \{x_1, \dots, x_n\} \subseteq \mathbb{R}^d$ is well known in the computational geometry literature. More specifically, one looks for two parallel hyperplanes enclosing all points with smallest distance to each other. For $d = 2$ the problem can be solved exactly with runtime complexity of $\mathbf{O}(n \log n)$ (see [4,]). [32] gives an algorithm for exact computation in three dimensions with runtime complexity of $\mathbf{O}(n^2)$, although algorithms exist with faster sub-quadratic expected runtime complexity, see [2, Corollary 3.6]. The main reference for this kind of problem for dimensions $d \geq 4$ is [13], where an exact algorithm is presented with runtime efficiency $\mathbf{O}(n^{\lceil d/2 \rceil})$.

find specific passage in book!

As a precedent for a descent based approximative (iterative) approach the work [25] can be given. There, a finite difference steepest descent is used to minimize the span semi-norm $\mathrm{sp}(e) := \max_i e_i - \min e_i$ for an error vector $e = (e_i)$ with e_i being the normal deviation from a measurement (x_i, y_i, z_i) to a reference plane $z = mx + ly + c$. The paper argues why measuring this kind of error is important in metrology.

To the best of our knowledge, there is no prior work that proposes to numerically approximate the minimum width of a point cloud using the projected sub-gradient descent. We identify this topic as being of independent interest, that is, independently of using it to estimate the coefficients of an AR method. Although this thesis establishes convergence guaranties, the success of this method rests on how well the gradient descent algorithm can find the minimum, which in turn relies on good strategies for choosing the starting direction and proper tuning. In addition to that,

more advanced versions of the regular gradient descent (Adam) can be tested in the future. Exact solutions suffer from the curse of dimensionality. The hope is that gradient descent approaches can be a way around this problem while still obtaining good numerical approximations.

3 Spectral Scale Selection for Kernel Density Estimation

In this chapter, we apply the scale selection methodology introduced in [6] to a sequence of kernel density estimates $\{\hat{f}_{h_i}\}_{i=1}^N$. The goal is to develop a methodological connection between the spectral scale selection principle introduced in [6] and the classical problem in nonparametric statistics of bandwidth selection for kernel density estimation. We will explain why the spectral construction selects what it selects and how this selection relates to classical bandwidth selection procedures. More concretely, let h_{oracle} be the bandwidth that minimizes $\mathbb{E}[\|f - \hat{f}_h\|_2^2]$ and define $S_{ij}^{(\lambda)} = e^{-\lambda D_{ij}}$ with $D_{ij} = (1/2)\|\hat{f}_{h_i} - \hat{f}_{h_j}\|_2^2$ for some candidate bandwidths $0 < h_1 < \cdots < h_N$. Does the peak of the Perron vector of S relate to h_{oracle}?

3.1 Preliminaries on Kernel Density Estimation

We have data $X_1, \ldots . X_n \in \mathbb{R}$ sampled from an (unknown) distribution with density f. The kernel density function $\hat{f}_h$ estimates this unknown density. It is defined by

$$\hat{f}_h(x) \coloneqq \frac{1}{nh} \sum_{j=1}^{n} K\left(\frac{x - X_j}{h}\right), \quad x \in \mathbb{R}.$$

Using the shorthand notation $K_h(y) := h^{-1}K(h^{-1}y)$ we can also write $\hat{f}_h(x) = n^{-1}\sum_{j=1}^n K_h(x - X_j)$. The kernel density estimate $\hat{f}_h(x)$ is a Monte Carlo estimate of $(f * K_h)(x) = \mathbb{E}_{X\sim f}[K_h(x - X)]$. Writing $P_n = n^{-1}\sum_{j=1}^n \delta_{X_j}$ for the empirical measure, we see that $(P_n * K_h)(x) = \int P_n(y)K_h(x - y)\,dy = n^{-1}\sum_{j=1}^n K_h(x - X_j) = \hat{f}_h(x)$.

Two common choices for the kernel K are:

- the Epanechnikov kernel $K(u) = \mathbf{1}\{-1 \leq u \leq 1\}3(1-u^2)/4$, $u \in \mathbb{R}$
- the Gaussian kernel $K(u) = (2\pi)^{-1/2}\exp(-u^2/2)$, $u \in \mathbb{R}$

From now on, we always work with the Gaussian kernel, unless indicated otherwise.

The question of how to choose the bandwidth h is widely known in the literature. See reviews []. We present some of the most popular methods below, which we also use to compare with our newly developed spectral selector. Following the review [29], bandwidth selectors can be divided into several classes, including plug-in and cross-validation methods. The methods of both classes minimize some approximation of the (mean) integrated squared error.

Definition 7. *Let f be a density and $\hat{f}_h$ its kernel density estimate. The integrated squared error (ISE) between f and $\hat{f}_h$ is given by*

$$\mathrm{ISE}(h) := \int \left(f(x) - \hat{f}_h(x)\right)^2 dx.$$

The mean integrated squared error (MISE) is $\mathrm{MISE}(h) := \mathbb{E}[\mathrm{ISE}(h)]$.

Cross-validation methods

The ISE can be written as

$$\mathrm{ISE}(h) = \int \hat{f}_h(x)^2\,dx - 2\mathbb{E}_{X\sim f}\left[\hat{f}_h(X)\right] + \int f(x)^2\,dx.$$

The first term $\int \hat{f}_h^2$ can be computed from the data. The third term $\int f^2$ does not depend on the bandwidth and can be ignored when minimizing. We are left to find an (unbiased) estimator for $\mathbb{E}_{X\sim f}[\hat{f}_h(X)]$. Such an estimator is given by $n^{-1}\sum_{i=1}^n \hat{f}_{h,-i}(X_i)$, where $\hat{f}_{h,-i}$ is the leave-one-out kernel density estimate given by

$$\hat{f}_{h,-i}(x) = \frac{1}{(n-1)h}\sum_{j\neq i} K\left(\frac{x-X_j}{h}\right), \quad x \in \mathbb{R}.$$

The Ordinary Least Squares Cross-Validation bandwidth is

$$\hat{h}_{\mathrm{CV}} = \arg\min_{h>0} \mathrm{CV}(h), \quad \mathrm{CV}(h) = \int \hat{f}_h(x)^2 \, dx - \frac{2}{n} \sum_{i=1}^{n} \hat{f}_{h,-i}(X_i).$$

It is known that (add source!!) $\mathrm{ISE}(\hat{h}_{\mathrm{CV}}) - \mathrm{ISE}(\hat{h}_0) = \mathbf{O}(n^{-1})$, that is, the integrated squared error evaluated at the cross-validation selected bandwidth converges to the integrated squared error evaluated at its true minimizer $\hat{h}_0$ to zero at rate n^{-1}.

Plug-in methods

The bias-variance decomposition of the MISE is

$$\mathrm{MISE}(h) = \int \mathrm{Var}[\hat{f}_h(x)] \, dx + \int \mathrm{Bias}(\hat{f}_h(x))^2 \, dx$$

with

$$\mathrm{IV}(h) := \int \mathrm{Var}[\hat{f}_h(x)] \, dx = \frac{\|K\|_2^2}{nh} - \frac{\|K_h * f\|_2^2}{n}, \quad \text{and}$$
$$\mathrm{ISB}(h) := \int \mathrm{Bias}(\hat{f}_h(x))^2 \, dx = \|K_h * f - f\|_2^2.$$

This decomposition gives conditions on $h = h_n$ so that the kernel density estimate $\hat{f}_h$ is consistent. In order for the bias to vanish, it must hold $h_n \to 0$, because then $K_h * f \to f$. For the variance to vanish, we must require $nh_n \to \infty$. An example of such a sequence is $h_n = cn^{-\alpha}$ for some $c > 0$ and $\alpha \in (0, 1)$.

It is important to be convinced that MISE actually has a global minimum. To this end, we refer to Chacón et al. [16, Theorem 1] which tells us that a global minimum always exists if K is a nonnegative kernel that peaks at 0.

Asymptotic expansion of the integrated variance

Note that $\|K_h\|_1 = \int K_h(x) \, dx = \int K(x) \, dx = 1$ for every $h > 0$. By Young's

convolution inequality $\|K_{h_n} * f\| \leq \|K_{h_n}\|_1 \|f\|_2 = \|f\|_2$. Therefore,

$$\frac{\frac{\|K_{h_n} * f\|_2^2}{n}}{\frac{\|K\|_2^2}{nh_n}} = \frac{h_n \|K_{h_n} * f\|_2^2}{\|K\|_2^2} \leq h_n \frac{\|f\|_2^2}{\|K\|_2^2} \to 0$$

and thus $-n^{-1}\|K_{h_n} * f\|_2^2 = \mathbf{o}((nh_n)^{-1}\|K\|_2^2)$. We therefore may omit the term $-n^{-1}\|K_{h_n} * f\|_2^2$ as it is asymptotically negligible.

Asymptotic expansion of the integrated squared bias

For the $\mathrm{ISB}(h)$ term we get

$$\mathrm{ISB}(h) = \frac{h_n^4}{4}\mu_2(K)^2 R(f'') + \mathbf{o}(h_n^4)$$

Asymptotic expansion of $\mathrm{MISE}(h)$

The first non-vanishing asymptotic approximation of the MISE is

$$\mathrm{MISE}(h_n) = \underbrace{\frac{\|K\|_2^2}{nh_n} + \frac{h_n^4}{4}\mu_2(K)^2\|f''\|_2^2}_{=\mathrm{AMISE}(h_n)} + o\left(\frac{1}{nh_n} + h_n^4\right).$$

We have found scaling sequences a_n and b_n such that $a_n \,\mathrm{MISE}(b_n t) \to G(t) := at^{-1} + bt^4$ with constants $a = \|K\|_2^2$ and $b = \mu_2(K)^2\|f''\|_2^2/4$, namely

$$n^{4/5}\,\mathrm{MISE}(n^{-1/5}t) \to \frac{\|K\|_2^2}{t} + \frac{\mu_2(K)^2\|f''\|_2^2}{4}t^4, \quad n \to \infty$$

Minimizing the AMISE leads to

$$h_{\mathrm{AMISE}}(n) = \left(\frac{\|K\|_2^2}{n\mu_2(K)^2\|f''\|_2^2}\right)^{1/5} \tag{3.1}$$

We have $h_{\text{oracle}} = h_{\mathrm{AMISE}} + \mathbf{o}(n^{-1/5})$ (see again [16]), which implies that h_{oracle} and h_{AMISE} are asymptotically equivalent.The optimal bandwidth thus behaves like $n^{-1/5}$.

Plug-in methods use Equation (3.1) as a starting point and try to find estimators $\hat{R}(f'')$ for $R(f'') = \|f''\|_2^2$. Silverman's rule of thumb assumes that f is the density

of a normal distribution $N(0, \sigma^2)$. In this case, $\|f''\|_2^2$ can simply be computed and plugged into Equation (3.1):

$$\|f''\|_2^2 = \frac{3}{8\sqrt{\pi}\sigma^5}, \quad h_{\text{AMISE}} = \left(\frac{4}{3}\right)^{1/5} \sigma n^{-1/5} \approx 1.06\sigma n^{-1/5}$$

Among the many other more sophisticated plug in methods, we present the (improved) Sheather and Jones' plug in selector, since it is used fairly commonly in practice; see for example the documentation of the kde packages of the statistics sowftware R and SAS.

They approximate $R(f'') = \|f''\|_2^2$ by $R(\hat{f}_g'') = \|\hat{f}_g''\|_2^2$, where $\hat{f}_g''$ is the second derivative of the kernel density estimate with pilot bandwidth g, whose choice is a major problem. They choose it so that the leading bias term of the mean squared error $\mathbb{E}[(R(f'')-R(\hat{f}_g''))^2]$ cancels. This choice leads to $g^7 = R(K'')\mu_2(K)^{-1}R(f''')^{-1}n^{-1}$. They then make this depend on h_{AMISE} by solving Equation (3.1) for n and plugging it into the formula for g^7, which results in $g(h) = C_K(R(f'')/R(f'''))^{1/7}h^{5/7}$.

Of course, the ratio $R(f'')/R(f''')$ has to be estimated separately, which is done by $R(\hat{f}_a'')/R(\hat{f}_b''')$ with a and b chosen following Silverman's rule of thumb. The resulting equation $\hat{g}(h) = C_K(R(\hat{f}_a'')/R(\hat{f}_b'''))^{1/7}h^{5/7}$ is plugged into $R(f_g'')$, which in turn is plugged into Equation (3.1). The Equation

$$h = \left(\frac{\|K\|_2^2}{n\mu_2(K)^2\|\hat{f}_{\hat{g}(h)}''\|_2^2}\right)^{1/5}$$

is solved numerically for h. There exist consistency results for this plug in selector. It converges to the oracle bandwidth h_{oracle} at a rate $n^{-5/14}$.

For illustration, the next Figure 3.1 shows 50 points sampled from $N(0,1)$ together with the true density f and the kernel density estimates $\hat{f}_h$ with $h \in \{\hat{h}_{CV}, h_{\text{oracle}}, h_{\text{AMISE}}, h_{\text{Sil}}, h_{\text{SJ}}\}$.

Exact MISE representation for Gaussian mixture model

Exact computations are generally not possible since the true density f of the sampled data $X_1, \ldots, X_n$ is not known. In simulation experiments, of course, we simulate the data ourselves and thus know the density f. For certain choices of the true density f, explicit formulas for the MISE exist.

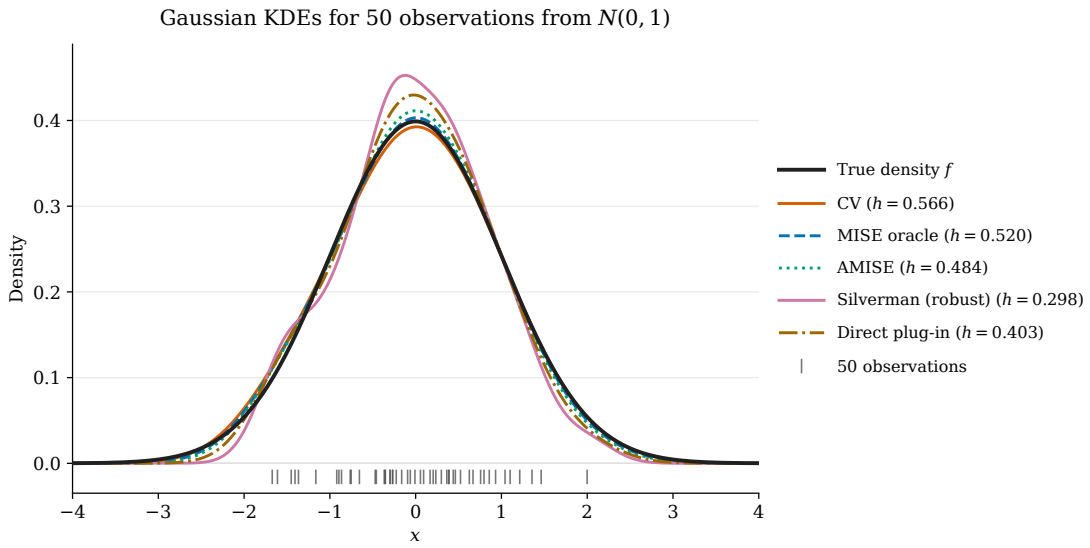


Figure 3.1: Gaussian kernel density estimates based on 50 observations sampled from $N(0, 1)$, using cross-validation, the MISE oracle, AMISE, Silverman's robust rule, and the direct plug-in rule. The black curve shows the true density, and the ticks indicate the observations.

A particularly broad and well studied class is that of Gaussian mixture models. They are characterized by densities f of the form $f = \sum_{l=1}^{L} w_l \phi_{m_l, \sigma_l^2}$, where $w_l > 0$, $\sum_l w_l = 1$, $m_l \in \mathbb{R}$, and $\sigma_l^2 > 0$.

Marron and Wand [41] derive an explicit formula for the MISE between a density f of the Gaussian mixture model and its kernel density estimate with Gaussian kernel of order $r \in \mathbb{N}$. Here we state the formula for the special case $r = 1$, that is, for the standard Gaussian kernel.

Proposition 5. *The mean integrated squared error between a kernel density estimate $\hat{f}_h$ with Gaussian kernel and the density function f of a Gaussian mixture model with weights w_l, variances s_l^2 and means m_l is given by*

$$\mathrm{MISE}(h) = \frac{1}{2\sqrt{\pi}} \frac{1}{nh} + \left(1 - \frac{1}{n}\right) \mathcal{K}(h, 2) - 2\mathcal{K}(h, 1) + \mathcal{K}(h, 0),$$

$$\mathcal{K}(h, q) = \sum_{l,l'=1}^{L} \frac{w_l w_{l'}}{\sqrt{s_l^2 + s_{l'}^2 + qh^2}} \phi\left(\frac{m_l - m_{l'}}{\sqrt{s_l^2 + s_{l'}^2 + qh^2}}\right), \quad q \in \{0, 1, 2\}.$$

Minimizing $\mathrm{MISE}(h)$ yields the "oracle" bandwidth h_{oracle}. We will use the oracle

bandwidth as an ultimate benchmark in all settings where MISE has an explicit representation and thus can be minimized using standard methods.

We already established above that a minimizer of MISE always exists. The question of uniqueness is a more complicated problem. For a general Gaussian mixture model we cannot assume the uniqueness of the minimizer, but we are able to do so in the simple case where $f \sim N(0, \sigma^2)$.

First, notice that $h \mapsto M_n(h) = (2\sqrt{\pi})^{-1}((nh)^{-1} + (1-1/n)/(\sqrt{\sigma^2+h^2}) + \sigma^{-1}) - 2(\sqrt{2\pi}\sqrt{2\sigma^2+h^2})^{-1}$ is continuous on $(0,\infty)$. We set $Q_n(h) = 2\sqrt{\pi}h^2 M_n'(h)$ and see that $Q_n(h) \to 2\sqrt{2}-1 > 0$ for $h \to \infty$ and $Q_n(h) \to -1/n < 0$ for $h \to 0$. Also, we have $Q_n'(h) > 0$ for all $h > 0$, which implies that Q_n only has one zero, and, in turn, the minimizer of M_n is unique.

As a function of σ, the second derivative of M_n at the unique minimizer behaves like σ^{-3}, which means that, for increasing variance, the valley in which the unique minimizer lies becomes flatter.

The integrated squared error between $\hat{f}_h$ and $f \sim N(0,\sigma^2)$ is given by

$$\mathrm{ISE}(h) = \frac{1}{n^2}\sum_{i,j=1}^{n} \phi_{\sqrt{2}h}(X_i - X_j) - \frac{2}{n}\sum_{i=1}^{n} \phi_{\sqrt{h^2+\sigma^2}}(X_i) + \phi_{\sqrt{2}\sigma}(0)$$

The connection between the minimum of ISE and the minimum of MISE is as follows. First, $\min_h \mathrm{ISE}(h) \leq \mathrm{ISE}(h_0)$ for all $h_0 > 0$. Therefore $\mathbb{E}[\min_h \mathrm{ISE}(h)] \leq \mathbb{E}[\mathrm{ISE}(h_0)]$ and since this is true for all $h_0 > 0$, it follows $\mathbb{E}[\min_h \mathrm{ISE}(h)] \leq \min_h[\mathrm{MISE}(h)]$.

3.2 Divergences between distributions

For information on information theory and divergences, see the book [45] or the survey [42].

Let $L^1(\mathbb{R})$ be the set of all absolutely integrable functions $f : \mathbb{R} \to \mathbb{R}$ and $\mathcal{P} \subseteq L^1$ be the set of all absolutely continuous probability distributions. That is,

$$\mathcal{P} = \left\{ f \in L^1 : f \geq 0, \int f(x)\,dx = 1 \right\}$$

Note that $\mathcal{P} \subseteq L^1$ is convex. A divergence is a map $D : \mathcal{P} \times \mathcal{P} \to \mathbb{R}_{\geq 0}$ such that $D(p, q) = 0$ if and only if $p = q$.

The two most important classes of divergences are f-divergences and Bregman divergences.

Computing divergences between densities amounts to evaluating integrals over $\mathbb{R}$. In general, these integrals cannot be computed explicitly, and numerical procedures have to be applied. For a practical guide on numerical integration, see [21, Chapter 9].

3.2.1 f-divergences

We take material on f-divergences mainly from [45, Chapter 7]. There, divergences are defined between two probability distributions P and Q. For our purposes, it is enough to define them directly between probability density functions (Lebesgue integral).

Definition 8. *Let $f : \mathbb{R}_{>0} \to \mathbb{R}$ be a convex function with $f(1) = 0$. The f-divergence is defined as*

$$D_f(p\|q) \coloneqq \int q(x) f\left(\frac{p(x)}{q(x)}\right) dx.$$

The f-divergence has the following invariance property: it holds $D_f(p\|q) = D_g(p\|q)$ whenever $f(x) - g(x) = c(x - 1)$ for some constant $c \in \mathbb{R}$. We will write $f \sim g$ in short.

Since we will use divergences to construct scale similarity matrices, it is important to know under which conditions on f the f-divergence and thus the scale similarity matrix are symmetric. Proposition 6 yields the answer.

Proposition 6. *The f-divergence is symmetric iff $f \sim \tilde{f}$, where $\tilde{f}$ is defined by $\tilde{f}(x) = xf(1/x)$.*

Proof. We make use of [45, Proposition 7.2]. First, assume that $f(x) = xf(x^{-1}) + c(x-1)$. Then $D_f(p\|q) = \int qf(p/q) = \int q((p/q)f(q/p)+c((p/q)-1)) = \int pf(q/p)+$

$c\int(p-q) = D_f(q\|p)$. Now assume that the f-divergence is symmetric on $\mathcal{P}$. We have $D_f(p\|q) = D_f(q\|p) = \int pf(q/p) = \int q(p/q)f(1/(p/q)) = \int q\tilde{f}(p/q) = D_{\tilde{f}}(p\|q)$. Now, let $x, y \in \mathbb{R}$ with $x < 1 < y$. Define $q_1 = (y-1)/(y-x)$ and $q_2 = (1-x)/(y-x)$ and let $g(z) = \delta_{q_1}(z) + \delta_{q_2}(z)$ and $h(z) = \delta_{xq_1}(z) + \delta_{yq_2}$. We get

$$D_f(h\|g) = \int gf(h/g) = q_1 f(x) + q_2 f(y)$$
$$D_{\tilde{f}}(h\|g) = \int g\tilde{f}(h/g) = q_1 \tilde{f}(x) + q_2 \tilde{f}(y)$$

and therefore $(y-1)(f(x) - \tilde{f}(x)) = (1-x)(\tilde{f}(y) - f(y))$ which is equivalent to $(f(x) - \tilde{f}(x))/(x-1) = (f(y) - \tilde{f}(y))/(y-1)$. It follows that $(f(x) - \tilde{f}(x))/(x-1)$ is constant on $\mathbb{R} \setminus \{1\}$ and thus $f(x) - \tilde{f}(x) = c(x-1)$ for all $x \in \mathbb{R}$ (it trivially holds for $x = 1$ too). □

In the following we introduce a few classes of parametrized f-divergences. Each of those parametrized classes has exactly one symmetric divergence among them. The numerical experiments below will make use of just the symmetric divergences since such divergences lead to symmetric scale similarity matrices of which we analyze the Perron vector. A possible extension of our scale selection principle to asymmetric scale similarity matrices is discussed in the Outlook 5. The reason we still introduce the divergences in such generality is to that the ideas in the Outlook can be better appreciated.

α-divergence

A prominent parametrized family of f-divergences is the so called α-divergence (see [18]) where the function $f = f_\alpha$ is given by

$$f_\alpha(t) = \frac{1}{\alpha(\alpha-1)}\left(t^\alpha - \alpha t - (1-\alpha)\right).$$

The actual divergence is then given by

$$D_\alpha(p\|q) = \frac{1}{\alpha(\alpha-1)}\left(\int p(x)^\alpha q(x)^{1-\alpha}\,dx - 1\right). \tag{3.2}$$

Taking the limit $\alpha \to 1$ we recover the Kullback-Leibler divergence, whereas taking

the limit $\alpha \to 0$ recovers the reverse Kullback-Leibler divergence. At $\alpha = 1/2$ we obtain $D_{1/2}(p\|q) = 4D_H(p,q)$, where $D_H(p,q) = (1/2)\int(\sqrt{f}-\sqrt{g})^2 = 1-\int\sqrt{pq}$ is the squared Hellinger distance. In particular, by Proposition 6, the α-divergence is symmetric iff $\alpha = 1/2$ (Hellinger distance). By Hölders inequality $0 \leq C_\alpha(p,q) \leq 1$, where $C_\alpha(p,q) = \int p^\alpha q^{1-\alpha}$ and thus $0 \leq D_\alpha(p\|q) \leq \alpha^{-1}(1-\alpha)^{-1}$ for all $\alpha \in (0,1)$.

For completeness, we mention that in the literature there exists a second parameterization, where $\alpha = (1-\alpha_A)/2$ is plugged into the formula in Equation (3.2) to obtain

$$D_{\alpha_A}(p\|q) = \frac{4}{1-\alpha_A^2}\left(1 - \int p(x)^{\frac{1-\alpha_A}{2}} q(x)^{\frac{1+\alpha_A}{2}}\right).$$

χ^α-divergences (total variation)

Another class of divergences are the χ^α-divergences (see []). They are determined by $f_\alpha(t) = (1/2)|t-1|^\alpha$, where $\alpha \geq 1$. Setting $\alpha = 1$ recovers the well known total variation distance: $d_{\mathsf{TV}}(p,q) = (1/2)\int|p-q|$, which is the only symmetric χ^α-divergence.

Total variation and squared Hellinger are related by the following chain of inequalities:

$$D_H(p,q) \leq d_{\mathsf{TV}}(p,q) \leq 2\sqrt{D_H(p,q)}.$$

Jensen Shannon divergence

Another important divergence that belongs to the class of f-divergences is the Jensen Shannon divergence. The function f is set to be

$$f(t) = \frac{1}{2}\left(t\log t - (t+1)\log\frac{t+1}{2}\right).$$

The divergence is then given by

$$\mathrm{JS}(f,g) = H\left(\frac{f+g}{2}\right) - \frac{1}{2}H(f) - \frac{1}{2}H(g),$$

where $H(f) := -\int f\log f$ is the Shannon entropy.

The JS-divergence is also symmetric. In fact, the JS-divergence is a symmetrization

of the Kullback-Leibler divergence, as it can be written like

$$\mathrm{JS}(p,q) = \frac{1}{2}\left(\mathrm{KL}(p\|\frac{p+q}{2}) + \mathrm{KL}(q\|\frac{p+q}{2})\right), \tag{3.3}$$

that is, the JS-divergence is the arithmetic mean of the KL-divergences of p and q to their equally weighted mixture.

3.2.2 Bregman divergences

The second major class are Bregman divergences. We give a general definition below to show that this concept of distance applies not just to probability densities. The Bregman divergence is a generalization of the L^2 distance and is applied in a variety of instances where the L^2-distance is also used, for example, clustering [5].

Definition 9. *Let $(\mathcal{H}, \langle\cdot,\cdot\rangle)$ be a Hilbert space and $C \subseteq \mathcal{H}$ nonempty, open and convex and $F : C \to \mathbb{R}$ strictly convex, and Fréchet differentiable on C. The Bregman divergence between $x \in C$ and $y \in C$ is defined by*

$$D_F(x\|y) := F(x) - F(y) - \langle x - y, \nabla F(y)\rangle_{\mathcal{H}}. \tag{3.4}$$

The first order Taylor approximation of F around y evaluated at x is $F(y)+DF(y)[x-y]$, where $DF(x) : \mathcal{H} \to \mathbb{R}$ is a bounded linear operator. By the Riesz representation theorem, $DF(y)[h] = \langle \nabla F(y), h\rangle$ for all $h \in \mathcal{H}$ and in particular $DF(y)[x-y] = \langle x - y, \nabla F(y)\rangle_{\mathcal{H}}$. The Bregman divergence $D_F(x\|y)$ as defined in Equation (3.4) is therefore the difference between F evaluated at x and the first order Taylor approximation of F around y evaluated at x.

Similarly to the f-divergence, we have invariance $D_F = D_G$ if and only if $F - G$ is an affine functional, that is, $F(x) - G(x) = \langle a, x\rangle_{\mathcal{H}} + b$ for some $a \in \mathcal{H}$ and $b \in \mathbb{R}$ and all $x \in C$.

If F is twice continuously Fréchet differentiable, there is a symmetry characterization for Bregman divergences similar to the one for f-divergences stated in Proposition 6. Theorem 12 in [48] states that in this case D_F is symmetric if and only

if

$$F(p) = \frac{1}{2}\langle p, Tp\rangle_{\mathcal{H}} + \langle b, p\rangle_{\mathcal{H}} + c, \quad p \in \operatorname{int}(\operatorname{dom}(F)), \tag{3.5}$$

where T is some bounded, self-adjoint and strictly positive operator, $b \in \mathcal{H}$ and $c \in \mathbb{R}$. That is, the Hessian $\nabla^2 F(g)$ of F at $g \in \mathcal{H}$ is T and, in particular, independent of g.

Subsequently, we will consider the inner product space $(L^2, \langle\cdot,\cdot\rangle_2)$ with functions F of the form $F_\phi(p) = \int \phi(p(x))\,dx$, where $\phi : \mathbb{R} \to \mathbb{R}$ is continuously differentiable, strictly convex, ϕ' globally Lipschitz-continuous and $\phi(0) = \phi'(0) = 0$. In this setting, since $\nabla F_\phi(q) = \phi' \circ q$, the Bregman divergence becomes

$$D_\phi(p\|q) := D_{F_\phi}(p\|q) = \int \left(\phi(p(x)) - \phi(q(x)) - (p(x) - q(x))\phi'(q(x))\right)\,dx.$$

We have $D_\phi = D_{\tilde{\phi}}$ if and only if $\phi - \tilde{\phi}$ is an affine function. To see the one direction, we know that this is equivalent to $F_\phi(p) - F_{\tilde{\phi}}(p) = \langle a, p\rangle_2 + b$ for all $p \in L^2$. In particular for $p_t(x) = t\mathbf{1}\{x \in [0,1]\}$. Plugging this particular choice of p_t into the equation yields $\phi(x) - \tilde{\phi}(x) = t\int_0^1 a(x)\,dx + b$, that is, $\phi - \tilde{\phi}$ is affine. The other direction follows from a simple calculation.

We also have that D_ϕ is symmetric if and only if ϕ is a quadratic function given that we additionally assume $\phi \in C^2(\mathbb{R})$. If D_ϕ is symmetric, then $F_\phi(p)$ has the representation in Equation (3.5) for all $p \in L^2$. Plugging in the particular choice $p(x) = x\mathbf{1}\{x \in [0,1]\}$ into Equation (3.5) and recalling the condition $\phi(0) = \phi'(0) = 0$, we arrive at ϕ having the form $\phi(x) = (\kappa/2)x^2$ with $\kappa = \langle \mathbf{1}\{\cdot \in [0,1]\}T\mathbf{1}\{\cdot \in [0,1]\}\rangle$. The converse direction is again a simple calculation.

An example of such a function ϕ is the class of β-divergences (see [18, Equation (46)]) determined by

$$\phi_\beta(t) = \frac{t^{1+\beta}}{\beta(1+\beta)}, \quad \beta > 0.$$

The β-divergence $D_\beta := D_{\phi_\beta}$ is thus given by

$$D_\beta(p\|q) = \frac{1}{\beta(1+\beta)} \left(\int \left(p(x)^{1+\beta} - (1+\beta)p(x)q(x)^\beta + \beta q(x)^{1+\beta} \right) dx \right).$$

By the above symmetry characterization, the β-divergence is symmetric only for $\beta = 1$. In this case, we get

$$D_1(p\|q) = \frac{1}{2}\int (p(x) - q(x))^2 \, dx = \frac{1}{2}\|p-q\|_2^2 = \frac{1}{2}\langle p,p\rangle_{L^2} + \frac{1}{2}\langle q,q\rangle_{L^2} - \langle p,q\rangle_{L^2}.$$

If $p(x) = \hat{f}_h(x) = n^{-1}\sum_j K_h(x - X_j)$ and $q(x) = \hat{f}_s(x) = n^{-1}\sum_j K_s(x - X_j)$ are kernel density estimates with a Gaussian kernel K, we can obtain an explicit formula for $D_1(\hat{f}_h, \hat{f}_s)$. We have

$$D_1(\hat{f}_h, \hat{f}_s) = \frac{1}{2}G_{hh} + \frac{1}{2}G_{ss} - G_{hs}, \tag{3.6}$$

where

$$G_{hs} = \langle \hat{f}_h, \hat{f}_s \rangle_{L^2} = \frac{1}{n^2}\sum_{i,j=1}^{n} \phi_{\sqrt{h^2+s^2}}(X_i - X_j) \tag{3.7}$$

The derivation of the formula uses mainly the Gaussian convolution identity, that is, the fact that for Gaussian densities $f_1 = \phi_{\mu_1,\sigma_1^2}$ and $f_2 = \phi_{\mu_2,\sigma_2^2}$ their convolution is $f_1 * f_2 = \phi_{\mu_1+\mu_2,\sigma_1^2+\sigma_2^2}$.

Numerical Approximation

The L^2 distance between two Gaussian kernel density estimates can be computed extremely fast using the Fourier transform. Although an exact formula is given in Equations (3.6) and (3.7) the computational efficiency is of the order $\mathcal{O}(N^2 n^2)$, where N is the size of the bandwidth grid and n the sample size.

In the following, we will briefly describe the approach following the book [44]. For any function $f \in L^1(\mathbb{R})$ the Fourier transform is defined as $\mathcal{F} : f \mapsto \mathcal{F}[f]$ where $\mathcal{F}[f](\omega) = \int f(x)\exp(-ix\omega)\,dx$. The Fourier transform of the Gaussian density ϕ_σ is famously given by $\mathcal{F}[\phi_\sigma](\omega) = \exp(-\sigma^2\omega^2/2)$. Using the linearity and translation property of the Fourier transform, we obtain the Fourier transform of a kernel density

function with a Gaussian kernel:

$$\mathcal{F}[\hat{f}_h](\omega) = \underbrace{\left(\frac{1}{n}\sum_{j=1}^{n}\exp\left(-iX_j\omega\right)\right)}_{=P_n(\omega)}\exp\left(-h^2\omega^2\right)/2.$$

Next, we use Parseval's identity to get

$$\begin{aligned} G_{r,s} = \int \hat{f}_{h_r}(x)\hat{f}_{h_s}(x)\,dx &= \frac{1}{2\pi}\int \mathcal{F}[\hat{f}_h](\omega)\overline{\mathcal{F}[\hat{f}_s](\omega)}\,d\omega \\ &= \frac{1}{2\pi}\int |P_n(\omega)|^2\exp\left(-(h^2+s^2)\omega^2/2\right)d\omega \\ &= \frac{1}{\pi}\int_0^\infty |P_n(\omega)|^2\exp\left(-(h^2+s^2)\omega^2/2\right)d\omega. \end{aligned}$$

Finally, we approximate G_{rs} by truncating the integral and applying the composite trapezoidal rule:

$$G_{rs} \approx G_{rs}^{(\Omega,K)} := \frac{1}{2\pi}\frac{\Omega}{K}\sum_{k=0}^{K} c_k|P_n(\omega_k)|^2\exp\left(-(h_r^2+h_s^2)\omega_k^2/2\right).$$

We can give an upper bound for the truncation error:

$$\begin{aligned} |G_{rs} - G_{rs}^{(\Omega)}| &\leq \frac{1}{\pi}\int_\Omega^\infty \exp\left(-\frac{(h^2+s^2)\omega^2}{2}\right)d\omega \leq \frac{1}{\pi}\int_\Omega^\infty \exp\left(-h_{\min}^2\omega^2\right)d\omega \\ &= \frac{1}{2\sqrt{\pi}h_{\min}}\operatorname{erfc}(\Omega h_{\min}), \end{aligned}$$

where $\operatorname{erfc}(z) := 2\pi^{-1/2}\int_z^\infty e^{-t^2}\,dt$ is the complementary error function. Fixing $h_{\min} = 0.01$ and $\epsilon = 10^{-8}$ and choosing $\Omega = 450$ yields a truncation error of less than ϵ.

After having determined a well motivated truncation parameter Ω, we determine a suitable number of nodes K experimentally by generating samples $(X_1^{(l)},\ldots,X_n^{(l)})$, $l = 1,\ldots,L$, set $E_K := \max_l \max_{r,s}|G_{r,s} - G_{r,s}^{\Omega,K}|$ and choose K so that $E_K \leq \delta$ for some apriori error δ. See Figure 3.2 for visualization. We choose $K = 1536$.

Having chosen Ω and K we carry out a small experiment comparing the runtime efficiency between exact formula and Fourier approximation. We sample $(X_1^{(l)},\ldots,X_{200}^{(l)}$,

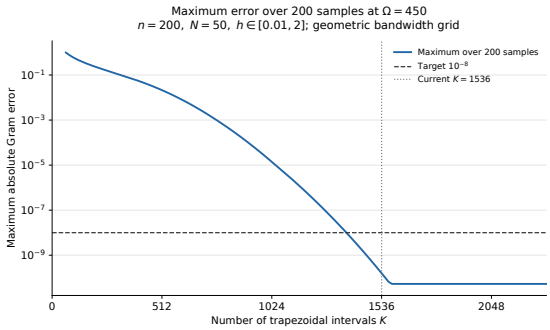


Figure 3.2

$l = 1, \ldots, 100$ and measure the time it took to compute the exact G_{hs} and the approximate $G_{hs}^{(\Omega,K)}$ for all bandwidth pairs h, s. The mean computation times and the maximum error that was made are presented in Table 3.1.

Method	Mean time (s)	Maximum absolute error
Exact formula	0.297500	—
Fourier approximation	0.010111	9.182×10^{-11}

Table 3.1: Runtime comparison over 100 independent standard-normal samples of size $n = 200$, using 50 geometrically spaced bandwidths in $[0.01, 2]$, $\Omega = 450$, and $K = 1536$. Times are arithmetic means for computing the complete Gram matrix. The maximum absolute error is taken over all samples and bandwidth pairs, including diagonal entries, using the exact formula as reference.

Error control is of importance because we must ensure that we control the error on sine of the angle between the Perron vector v of the true similarity matrix S and the Perron vector $\tilde{v}$ of the approximate similarity matrix which is actually computed in practice. In case of the L^2 divergence, the approximation gives a substantial efficiency advantage. Other divergences might not even have closed form expressions (see [30]), and approximation becomes the only choice.

3.3 Assembly of Scale Similarity Matrices

3.3.1 Properties of Divergence matrices and its Perron vector

In this subsection, we give structural results on the divergence matrix D built from the L^2-divergence. In particular we show that D is a so called Robinson matrix. We also show that it fulfills the acute chain condition, which leads to the Perron vector of D peaking at either of the bordering scales. This further motivates the above similarity transformation, because else our scale selection procedure will choose either the first or last scale.

Definition 10 (Robinson matrix)**.** *A matrix D with entries $D_{ij} = D(x_i, x_j)$ is called a (two way) Robinson matrix (cf. [14]) if there exists a permutation π on $\{1, \dots, N\}$ such that for $D^\pi_{ij} = D(x_{\pi(i)}, x_{\pi(j)})$ it holds*

$$D^\pi_{ik} \geq \max\{D^\pi_{ij}, D^\pi_{jk}\} \text{ and } D^\pi_{ki} \geq \max\{D^\pi_{ji}, D^\pi_{kj}\} \tag{3.8}$$

whenever $i < j < k$.

Robinson matrices are generalizations of Min matrices, that is, every Min matrix is a Robinson similarity. Note that for a Robinson similarity, the inequalities in Equation (3.8) are reversed. It is always beneficial to know the kind of matrix one works with to potentially be able to draw from established results about these kinds of matrices. Certain choices of divergences from Section 3.2 lead to Robinson matrices, see Proposition 7 below.

The reason why we do not want to work with the matrix D and instead transform to a similarity matrix $\phi(D)$ is due to the following Lemma. An entry v_i is large if the scale i has a large distance to other scales j. The Perron vector thus gives scales that are far apart from many other scales larger weight. These scales are usually the bordering scales. The next Lemma makes this intuition precise.

Lemma 5. *Let H be a Hilbert space, and let $x_1, \dots, x_N$ be distinct points in $\mathcal{H}$ that form an acute chain, that is,*

$$\langle x_{k+1} - x_k, x_{l+1} - x_l \rangle \geq 0 \quad \text{for all } k, l = 1, \dots, N-1. \tag{3.9}$$

Let $D = (D_{ij}) \in \mathbb{R}^{N\times N}$, $D_{ij} = \|x_i - x_j\|^2$ be the pairwise distance matrix and $v = (v_j) \in \mathbb{R}^N$ its Perron vector. Then v has its largest entries at the border, that is,

$$\arg\max_{1\le j\le N} v_j \subseteq \{1, N\}.$$

Proof. See Appendix Section 6.2.1. □

The "acute chain condition" has precedence in the theory of so called Monge matrices, for reference see, for example [34]. Monge matrices have a special structural property that allows for certain problems to have faster algorithms when assuming this structure. A matrix D is a Monge matrix if $D_{ij}+D_{kl} \le D_{il}+D_{kj}$ whenever $i \le k$ and $j \le l$. In fact, a matrix D with entries $D_{ij} = \|x_i - x_j\|^2$ is Monge if and only if the acute chain condition holds for $x_1, \ldots, x_N$. All symmetric Monge matrices with zero diagonal are Robinson matrices. The acute chain condition has a geometric interpretation: the angle between all the differences of two consecutive points is at most 90 degrees.

We now show that when choosing the Gaussian kernel, the family of kernel density estimates $\{\hat{f}_{h_i}\}_{i=1}^N$ satisfies the acute chain condition and, as a consequence, the matrix $D = (D_{ij})$, $D_{ij} = (1/2)\|\hat{f}_{h_i} - \hat{f}_{h_j}\|_2^2$ is a Robinson matrix.

Proposition 7. *The following two properties hold.*

(a) The family of kernel density estimates $\{\hat{f}_{h_i}\}_{i=1}^N$ satisfies the acute chain condition stated in Equation (3.9)*.*

(b) The divergence matrix $D = (D_{ij})$ with entries given by $D_{ij} = (1/2)\|\hat{f}_{h_i} - \hat{f}_{h_j}\|_2^2$ is a Robinson matrix.

Proof. The Fourier transform $\mathcal{F}[\hat{f}_h]$ of a kernel density estimate $\hat{f}_h$ with a Gaussian kernel is given by $\mathcal{F}[\hat{f}_h](\omega) = P_n(\omega)\exp(-h^2\omega^2/2)$. We set $\delta_k = \hat{f}_{h_{k+1}} - \hat{f}_{h_k}$. By linearity of the Fourier transform, $\mathcal{F}[\delta_k] = P_n(\omega)b_k(\omega)$, where we have set $b_k(\omega) = \exp(-h_{k+1}^2\omega^2/2) - \exp(-h_k^2\omega^2/2)$. Using Parseval's identity, we write

$$\langle \delta_k, \delta_l \rangle = \int \delta_k\delta_l = \frac{1}{2\pi}\int_{\mathbb{R}} |P_n(\omega)|^2 b_k(\omega)b_l(\omega)\, d\omega.$$

Since $h_k < h_{k+1}$, we have $b_k(\omega) < 0$ for all $\omega \neq 0$. Also, notice that $\omega \mapsto P_n(\omega)$ and $P_n(0) = 1$. Thus $P_n(\omega) \neq 0$ for all $\omega \in B_\epsilon(0)$ for some $\epsilon > 0$. It follows that the function $\omega \mapsto |P_n(\omega)|^2 \delta_k(\omega)\delta_l(\omega)$ is strictly positive on $B_\epsilon(0) \setminus \{0\}$. We have found a set with positive Lebesgue measure on which the integrand is positive, which implies that the integral $\langle \delta_k, \delta_l \rangle$ itself is positive, that is, $\langle \delta_k, \delta_l \rangle > 0$.

Now, let $i < j < k$. We have

$$\begin{aligned}
\|\hat{f}_{h_k} - \hat{f}_{h_i}\|_2^2 &= \|\hat{f}_{h_k} - \hat{f}_{h_j} + \hat{f}_{h_j} - \hat{f}_{h_i}\|_2^2 \\
&= \|\hat{f}_{h_k} - \hat{f}_{h_j}\|_2^2 + \|\hat{f}_{h_j} - \hat{f}_{h_i}\|_2^2 + 2\langle \hat{f}_{h_k} - \hat{f}_{h_j}, \hat{f}_{h_j} - \hat{f}_{h_i} \rangle \\
&> \|\hat{f}_{h_k} - \hat{f}_{h_j}\|_2^2 + \|\hat{f}_{h_j} - \hat{f}_{h_i}\|_2^2 \\
&\geq \max\{\|\hat{f}_{h_k} - \hat{f}_{h_j}\|_2^2, \|\hat{f}_{h_j} - \hat{f}_{h_i}\|_2^2\},
\end{aligned}$$

because

$$\begin{aligned}
\langle \hat{f}_{h_k} - \hat{f}_{h_j}, \hat{f}_{h_j} - \hat{f}_{h_i} \rangle &= \langle \sum_{r=j}^{k-1} \underbrace{(\hat{f}_{h_{r+1}} - \hat{f}_{h_r})}_{=\delta_r}, \sum_{s=i}^{j-1} \underbrace{(\hat{f}_{h_{s+1}} - \hat{f}_{h_s})}_{=\delta_s} \rangle \\
&= \sum_{r=j}^{k-1} \sum_{s=i}^{j-1} \langle \delta_r, \delta_s \rangle > 0,
\end{aligned}$$

by the first part of the proof. □

3.3.2 Similarity transformations

A matrix $A \in \mathbb{R}^{n \times n}$ is positive definite if $x^T A x > 0$ for all $x \in \mathbb{R}^n$, $x \neq 0$. A symmetric map $D : X \times X \to \mathbb{R}$ is called conditionally negative definite if

$$\sum_{i,j=1}^{n} a_i a_j D(x_i, x_j) \leq 0$$

for all $x_1, \ldots, x_n \in X$ and $a_1, \ldots, a_n \in \mathbb{R}$ with $\sum_i a_i = 0$. A function $\phi : (0.\infty) \to \mathbb{R}$ is called completely monotone if $\phi \in C^\infty(0, \infty)$ and $(-1)^n \phi^{(n)}(t) \geq 0$ for all $n \in \mathbb{N}_0$ and $t > 0$. The most prominent example of a completely monotone function is $\phi(t) = e^{-t/\tau}$, $\tau > 0$.

The following Theorem connects the concepts presented above and is due to Schoen-

berg; see [7].

Theorem 2 (Schoenberg)**.** *The following statements are equivalent.*

(a) $D : X \times X \to \mathbb{R}$ *is conditionally negative definite with* $D(x, x) = 0$ *for all* $x \in X$

(b) For all $x_1, \ldots, x_n \in X$ *the matrix* $S \in \mathbb{R}^{n \times n}$ *defined by the entries* $S_{ij} = \phi(D(x_i, x_j))$ *is positive semi-definite for all completely monotone functions* $\phi : \mathbb{R}_{\geq 0} \to \mathbb{R}$*.*

(c) There exists some Hilbert space embedding $\Phi : X \to \mathcal{H}$ *such that* $D(x, y) = \|\Phi(x) - \Phi(y)\|_{\mathcal{H}}^2$*.*

The equivalence $(a) \iff (b)$ *can be strengthened to strict positive definiteness by additionally assuming that* $D(x, y) > 0$ *for all* $x \neq y$ *(which is the case for divergences) and requiring* ϕ *to be non-constant and the points* $x_1, \ldots, x_n$ *to be pairwise distinct.*

Proof. Let D be conditionally negative. By [7, Theorem 2.2] this is equivalent to $\tilde{S}_t = (\exp(-tD(x_i, x_j)))_{i,j=1}^n$ being positive semi-definite for all finite choices of points $x_1, \ldots, x_n \in X$. Let ϕ be a completely monotone function. Then by [7, Corollary 6.14], there exists a measure μ on $\mathcal{B}((0, \infty))$ such that $\phi(z) = \int_0^\infty e^{-tz}\, d\mu(t)$. Consequently, we write

$$\begin{aligned}\sum_{i,j=1}^n a_i a_j \phi(D(x_i, x_j)) &= \sum_{i,j=1}^n a_i a_j \int_0^\infty \exp(-tD(x_i, x_j))\, d\mu(t) \\ &= \int_0^\infty \underbrace{\sum_{i,j=1}^n a_i a_j \exp(-tD(x_i, x_j))}_{\geq 0 \text{ by assumption}}\, d\mu(t) \geq 0\end{aligned}$$

for all $a_1, \ldots, a_n \in \mathbb{R}$. This shows that the matrix $S = (\phi(D(x_i, x_j)))_{i,j=1}^n$ is positive semi-definite. If (b) holds, then (a) follows directly by the equivalence stated in [7, Theorem 2.2] since $\exp(-t\cdot)$ is a completely monotone function for all $t > 0$.[7, Proposition 3.2] gives the implication $(a) \implies (c)$, while the reverse direction can

easily be checked by us. It holds

$$\sum_{i,j=1}^{n} c_i c_j \|\Phi(x_i) - \Phi(x_j)\|_{\mathcal{H}}^2 = -2 \sum_{i,j=1}^{n} c_i c_j \langle \Phi(x_i), \Phi(x_j) \rangle = -2 \left\| \sum_{j=1}^{n} c_j \Phi(x_j) \right\|_{\mathcal{H}}^2 \leq 0.$$

For $i \neq j$ we have $0 < D(x_i, x_j) = \|\Phi(x_i) - \Phi(x_j)\|_{\mathcal{H}}^2$. Thus, the points $z_i = \Phi(x_i)$ are pairwise distinct. Invoking [28, Theorem 3.1], we see that the matrix $\tilde{S}_t$ is strictly positive definite, that is, $a^T \tilde{S}_t a > 0$ for all $a \in \mathbb{R}^n$, $a \neq 0$. Since ϕ is assumed to be non-constant, this forces $\mu((0, \infty)) > 0$. Thus, $a^T S a = \int_0^\infty a^T \tilde{S}_t a \, d\mu(t) > 0$, i.e., S is strictly positive definite. □

Fuglede and Topsøe [22] show that the Jensen-Shannon divergence is conditionally negative definite. In fact, using Schoenberg's theorem above, we can also include L^2-divergence, squared Hellinger, and Total Variation in the class of conditionally negative definite divergences. We can use the equivalence $(a) \iff (c)$ from Theorem 2 to show that L^2-divergence, squared Hellinger and Total Variation are conditionally negative definite by giving the isometric Hilbert space embeddings:

- L^2-divergence: $\mathcal{H} = L^2(\mathbb{R})$, $\Phi(p) = p/\sqrt{2}$, $\|\Phi(p) - \Phi(q)\|_{L^2}^2 = (1/2)\|p - q\|_2^2$
- squared Hellinger: $\mathcal{H} = L^2(\mathbb{R})$, $\Phi(p) = \sqrt{p/2}$, $\|\Phi(p) - \Phi(q)\|_{L^2}^2 = (1/2)\|\sqrt{p} - \sqrt{q}\|_2^2$
- Total Variation: $\mathcal{H} = L^2(\mathbb{R}, (0, \infty))$, $\Phi(p)(x, t) = (1/\sqrt{2})\mathbf{1}\{0 < t < p(x)\}$, we have $\|\Phi(p) - \Phi(q)\|_{L^2(\mathbb{R},(0,\infty))}^2 = (1/2) \int |p - q|$
- Jensen Shannon: $\mathcal{H} = L^2(\mathbb{R}, (0, \infty))$, $\Phi(p)(x, t) = (1 - e^{-tp(x)/2})/(\sqrt{2}t)$, $\|\Phi(p) - \Phi(q)\|_{L^2(\mathbb{R},(0,\infty))}^2 = \mathrm{JS}(p, q)$. This can be seen easily:

$$\begin{aligned}
\|\Phi(p) - \Phi(q)\|_{L^2(\mathbb{R},(0,\infty))}^2 &= \int_{-\infty}^{\infty} \int_0^{\infty} \left(\Phi(p)(x,t) - \Phi(q)(x,t)\right)^2 \, dt \, dx \\
&= \int_{-\infty}^{\infty} \int_0^{\infty} \frac{\left(e^{-tq(x)/2} - e^{-tp(x)/2}\right)^2}{2t^2} \, dt \, dx \\
&\overset{(*)}{=} \int_{-\infty}^{\infty} \frac{1}{2} \log \left(\frac{q(x)^{q(x)} p(x)^{p(x)}}{(q(x)/2 + p(x)/2)^{q(x)+p(x)}} \right) dx \\
&= \int_{-\infty}^{\infty} j(p(x), q(x)) \, dx = \mathrm{JS}(p, q),
\end{aligned}$$

where in step $(*)$ we have use a well known integral identity, see, for example [51].

In the context of our scale selection methodology, Theorem 2 gives a general construction principle for a scale similarity matrix $\Sigma^{(N)} = (\Sigma_{ij}^{(N)})$ that is guaranteed to be positive (semi-) definite. Namely, we set $\Sigma_{ij}^{(N)} = \phi(D(\hat{f}_{h_i}, \hat{f}_{h_j}))$ where D is some conditionally negative definite divergence and ϕ a completely monotone function.

Additionally, the transformation $S_{ij} = \exp(-D_{ij}/\tau)$ can be motivated as follows. Let $H(p) = -\sum_j p_j \log p_j$ be the entropy of a probability vector $p \in \mathbb{R}^N$. It can be interpreted as an uncertainty or concentration measure. $H(p)$ is large if p is close to the discrete uniform distribution and maximal for $p = (N^{-1}, \ldots, N^{-1})$ (maximal uncertainty and minimal concentration). It is small if p is concentrated on some entry and minimal for $p = e_j$ for some $j = 1, \ldots, N$, where e_j is the j-th unit vector in $\mathbb{R}^N$ (minimal uncertainty and maximal concentration).

The quantity

$$\sum_{j=1}^{N} p_j D_{ij} - \tau H(p) = \sum_{j=1}^{N} p_j D_{ij} + \tau \sum_{j=1}^{N} p_j \log p_j$$

is minimized by $\hat{p}_i = (\hat{p}_{i1}, \ldots, \hat{p}_{iN})$, where $\hat{p}_{ij} = S_{ij} / \sum_{k=1}^{N} S_{ik}$. See [3] for reference. We can interpret $\hat{p}_i$ as a probability vector that describes how strongly the scale h_i is connected to any other scales.

The minimization of the first term $D_i p = \sum_j p_j D_{ij}$ ($D_i \in \mathbb{R}^{1\times N}$ is the i-th row of D) leads to probability vectors that concentrate on the i-th component since $\min_j D_{ij} = D_{ii} = 0$ (the largest probability weights is put on a bandwidth that is closest to bandwidth h_i with respect to the divergence D). The penalty term $\sum_j p_j \log p_j$ counteracts this concentration by penalizing it, where τ controls the amount of penalization. In that sense, τ can be interpreted as a penalty weight.

This is another instance of the Donsker–Varadhan variational principle.

Taking the limit $\tau \to +\infty$ yields $S = 11^T$, which corresponds to maximal penalization on concentration. Taking the limit $\tau \to 0$ gives $S = I_N$, which corresponds to no penalization on concentration.

Yet another advantage of the exponential transformation is the obvious fact that the entries of the resulting similarity matrix S are all positive and therefore by the Perron-

Frobenius theorem, S has a real valued eigenvalue $\lambda_{\max}$ which is strictly larger than the absolute value of all other eigenvalues. The corresponding eigenvector $v_{\max}$ (called the Perron vector) has all positive entries and is unique up to scaling.

3.3.3 Bandwidth grid

The choice of the correct bandwidth grid is essential. Set

$$\mathcal{H}_N = \{h_1, \dots, h_N\}, \quad 0 < h_{\min} := h_1 < \cdots < h_N =: h_{\max}.$$

The next definition gives a general construction principle for such a bandwidth grid.

Definition 11. *Let $0 < r \leq 1$. A grid $\mathcal{H}_N$ is said to be r-linear if*

$$\frac{1}{r}h_j^r = \frac{1}{r}h_1^r + \underbrace{\frac{h_N^r - h_1^r}{r(N-1)}}_{=:\Delta_{r,N}}(j-1), \quad j = 1, \dots, N. \tag{3.10}$$

Remark 3. *The construction in Equation* (3.10) *is just the Box-Cox transformation of a linear grid $\mathcal{H}_N$, compare with [8, Equation (1)]. Setting $\psi_r(h) = r^{-1}(h^r - 1)$ we see that a grid $\mathcal{H}_N$ is r-linear if $\{\psi_r(h_j)\}_{j=1}^N$ is a linear grid. Taking the limit $r \to 0$ we obtain $\psi_0(h) := \lim_{r\to 0} \psi_r(h) = \log h$. A 0-linear grid can therefore be seen as a geometric grid. Another interpretation is the following: h_j is the weighted power mean between the two endpoints h_1 and h_N with exponent r and weights $w_1 = 1 - (j-1)/(N-1)$ and $w_2 = (j-1)/(N-1)$. For reference, see [12, Chapter 3.1].*

Definition 11 smoothly connects a linear and a geometric grid. Setting $r = 1$ gives a linear grid, that is,

$$h_j = h_{\min} + \Delta_{1,N}(j-1), \quad \Delta_{1,N} = \frac{h_{\max} - h_{\min}}{N-1}.$$

Interpreting $r = 0$ as limit we obtain a geometric grid, that is,

$$\log h_j = \log h_{\min} + \Delta_{0,N}(j-1), \quad \Delta_{0,N} = \frac{\log h_{\max} - \log h_{\min}}{N-1}.$$

The geometric grid can equivalently be written as

$$h_j = h_{\min} q_N^{j-1}, \quad q_N = \left(\frac{h_{\max}}{h_{\min}}\right)^{1/(N-1)}.$$

Taking the limit $N \to +\infty$ increases the fineness of the grid.

Next we formulate a Lemma that gives the first order asymptotic approximation for the spacing $h_{j_N} - h_{i_N}$ of a r-linear grid, where (i_N) and (j_N) are appropriately chosen index sequences. It tells us that for large N the difference between the two scales h_{j_N} and h_{i_N} is approximately $h_{i_N}^{1-r}\delta_{ij}^{r,N}$. In other words, we express the difference between two untransformed bandwidths $h_{j_N} - h_{i_N}$ using the difference between two transformed bandwidths $\psi_r(h_{j_N}) - \psi_r(h_{i_N})$ approximately when N is large and $j_N - i_N = \mathbf{o}(N)$.

Lemma 6. *Let $\mathcal{H}_N = \{0 < h_1 < \cdots < h_N\}$ be a r-linear grid with fixed endpoints $h_1 = h_{\min}$ and $h_N = h_{\max}$, and let $i = (i_N)_{N\in\mathbb{N}}$ and $j = (j_N)_{N\in\mathbb{N}}$ be two index sequences that satisfy $1 \le i_N, j_N \le N$ for all $N \in \mathbb{N}$. Set $\delta_{ij}^{r,N} := (j_N - i_N)\Delta_{r,N}$ (the length of the gap between $\psi_r(h_{i_N})$ and $\psi_r(h_{j_N})$). Assume $|j_N - i_N| = \mathbf{o}(N)$. Then*

$$h_{j_N} - h_{i_N} = h_{i_N}^{1-r}\delta_{ij}^{r,N} + \mathbf{O}((\delta_{ij}^{r,N})^2).$$

Proof. We have $h_{j_N}^r - h_{i_N}^r = r(j_N - i_N)\Delta_{r,N}$ from which follows

$$h_{j_N} = \left(h_{i_N}^r + r\delta_{ij}^{r,N}\right)^{1/r} = h_{i_N}\left(1 + r\underbrace{\frac{\delta_{ij}^{r,N}}{h_{i_N}^r}}_{=:c_N}\right)^{1/r}.$$

Note that $c_N = \mathbf{O}(\delta_{ij}^{r,N})$ because $h_{i_N}^{-r}$ is bounded above by $h_{\min}^{-r}$. In particular, since $|j_N - i_N| = \mathbf{o}(N)$ by assumption, c_N is a null sequence. The first order Taylor approximation of $(1+x)^{1/r}$ around 0 is $(1+x)^{1/r} = 1 + x/r + \mathbf{O}(x^2)$. We get

$$\begin{aligned} h_{j_N} - h_{i_N} &= h_{i_N}\left((1 + rc_N)^{1/r} - 1\right) = h_{i_N}\left(c_N + \mathbf{O}((\delta_{ij}^{r,N})^2)\right) \\ &= h_{i_N}^{1-r}\delta_{ij}^{r,N} + \mathbf{O}((\delta_{ij}^{r,N})^2) \end{aligned}$$

by properties of the $\mathbf{O}$-notation. □

The stronger condition $j_N - i_N = \mathbf{O}(1)$ (the difference between the index sequences stays asymptotically bounded) would lead to the remainder $\mathbf{O}(N^{-2})$.

Motivation for why a geometric grid is the natural choice

The transformation $K \mapsto K_h$ stretches K horizontally while keeping the area under the curve the same. We can measure the width of a kernel K, for example, by $s(K) \coloneqq \sqrt{\mu_2(K)} = (\int x^2 K(x)\, dx)^{1/2}$. The width of the stretched kernel is $s(K_h) = hs(K)$. Increasing h by the factor a increases $s(K_h)$ by the factor a. Choosing a geometric grid means that going from h_j to h_{j+1} increases $s(K_{h_j})$ by the same factor q_N. Thus, each step stretches the kernel by the same factor.

3.4 Numerical Experiments

In our numerical experiments, we restrict ourselves to the L_2 divergence as it is closely related to the MISE criterion for optimal bandwidth selection and it can be computed efficiently using the Fourier transform. We leave the other symmetric divergences (Total variation, squared Hellinger, Jensen-Shannon) and asymmetric parametrized divergences as future work; see Outlook 5.

We use the similarity transformation $S = \exp(-\lambda D)$ and vary the penalty parameter $\lambda = \tau^{-1}$. Specifically, we will look at the limit $\lambda \to \infty$, which corresponds to a decreasing penalization on the concentration of the Perron vector.

In our experiments, we focus first on the case $f \sim N(0, 1)$, that is, normally distributed data. This is the simplest case in which there is only one "optimal" bandwidth (the one minimizing a certain error criterion). We start with this simple case to see if there is even a connection between our spectral pipeline and the notion of "optimal" bandwidth selection. Experimentation with other data distributions is future work; see the Outlook chapter 5. Especially Gaussian mixture models are interesting because in such cases there might not exist a unique optimal bandwidth and the Perron vector could serve as a ranking procedure for multiple candidate bandwidths.

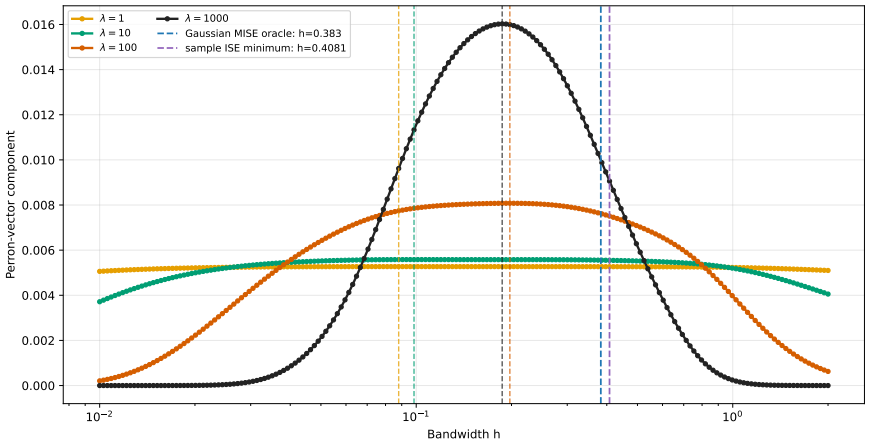


Figure 3.3: $X_i \sim N(0,1)$, $n = 200$, $N = 191$, $h_{\min} = 0.01$, $h_{\max} = 2$, $\lambda \in \{10^k : k = 0, 1, 2, 3\}$, geometric bandwidth grid; the blue line indicates $h_{\text{oracle}} = 0.38304$

3.4.1 Fixed sample experiments

For the first set of experiments, we draw one fixed sample $X = (X_1, \ldots, X_n) \sim f$, $n = 200$ and study the influence of the grid fineness N and the concentration penalty $\lambda = \lambda_N$.

Figure 3.3 shows the relation $i \mapsto v_i$, where v is the Perron vector of $S^{(\lambda)} = (S_{ij}^{(\lambda)})$, $S_{ij}^{(\lambda)} = \exp\left(-\frac{\lambda}{2}\|\hat{f}_{h_i} - \hat{f}_{h_j}\|_2^2\right)$, $\lambda = 1$.

We see that the Perron vector has a flat plateau for $\lambda = 1$ (yellow) and $\lambda = 10$ (green) which means that many bandwidths are considered central. Increasing λ leads to the Perron vector having a more pronounced peak, which is consistent with the interpretation of $\tau = \lambda^{-1}$ being a penalty for how much the Perron vector concentrates.

Of course, a single such image tells us not much and does not give us hints about a possible connection between the oracle bandwidth and the spectrally chosen bandwidth.

Figure 3.4 shows the stability with respect to the grid fineness N. The sample $X = (X_1, \ldots, X_n)$ was kept fixed. We observe a stabilization of the chosen bandwidth for increasing grid fineness.

It can be seen that the grid stability curves for $\lambda = 1$ (yellow) and $\lambda = 10$ (green)

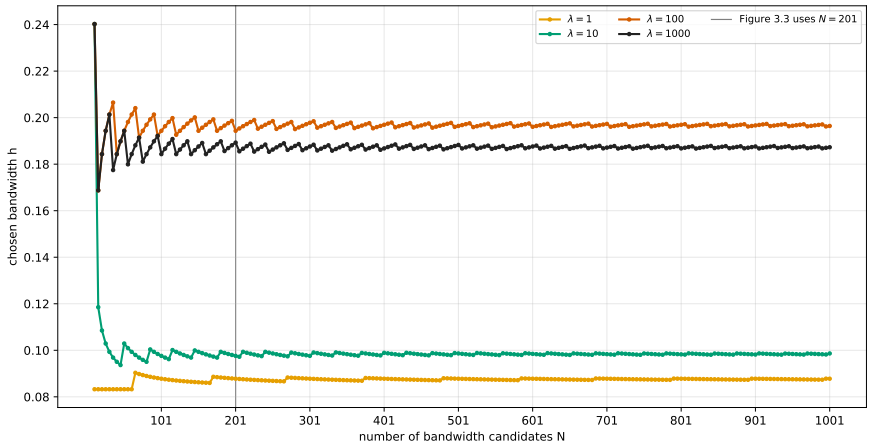


Figure 3.4: For a fixed sample $X = (X_1, \ldots, X_{200})$ the spectral bandwidth selection pipeline was repeated for $N = 11, 16, 21, \ldots, 1001$; $\lambda = 1$

converge towards a smaller bandwidth scale than the curves for $\lambda = 100$ (red) and $\lambda = 1000$ (black). This indicates that some kind of switch is happening between $\lambda = 10$ and $\lambda = 100$. Interestingly, the grid stability plots exhibit a sawtooth pattern that smoothens out.

In Section 3.3.2 we established the interpretation of τ as a penalty weight. Henceforth, we set $\lambda = 1/\tau$ and study $\lambda \to \infty$. This corresponds to a decrease in concentration penalization. Figure 3.5 shows the chosen bandwidth for one fixed sample as λ increases. The path does not approach the oracle bandwidth over the displayed range. We will explain this behavior in Section 3.5 below. Also note that the curve sharply jumps around $\lambda \approx 17.87375$.

The observation regarding the sharp jumps motivates to ask more questions: does it happen under repeated sampling? At which λ does the switch happen each time? Is it an interesting property about this method or just a numerical artifact?

After having examined the fixed λ increasing N behavior in Figure 3.4 and the fixed N increasing λ behavior in Figure 3.5, we would like next to study the joint limit $(N, \lambda_N) \to (\infty, \infty)$. We study the following regimes: $\lambda_N = N^\alpha$, where $\alpha \in \{0, 0.5, 1, 1.5, 2, 3\}$. Figure 3.6 summarizes the results. For $\alpha > 1/2$ the chosen bandwidth seems to converge towards some bandwidth $h_\infty = 0.1764$. In Section 3.5 we explain that this is the minimizer of $\hat{Q}_{200,2}(h) = (h^2/2)\|\partial_h \hat{f}_h\|_2^2$.

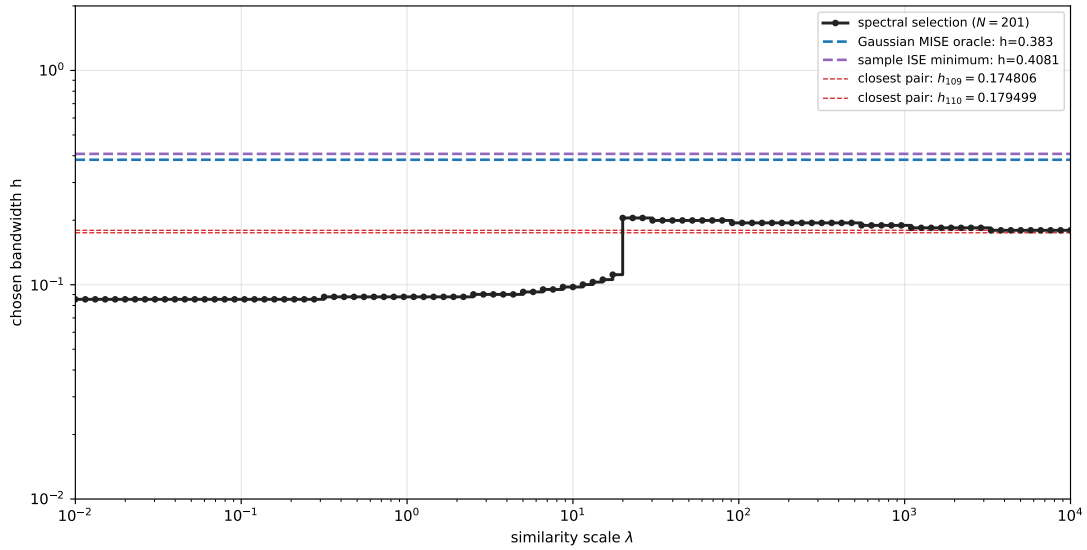


Figure 3.5: Spectrally chosen bandwidth for one fixed $N(0,1)$ sample ($n = 200$) as λ increases; $N = 201$, $h_{\min} = 0.01$, and $h_{\max} = 2$.

explain the sharp jump? (Outlook)

3.4.2 Repeated sampling experiments

Now we draw repeated samples $(X_1^{(i)}, \ldots, X_n^{(i)})$, $i = 1, \ldots, M$ and study the distribution of chosen bandwidths. Figure 3.7 shows the results of studying the distribution of the chosen bandwidths under repeatedly drawn samples $(X_1^{(i)}, \ldots, X_n^{(i)})$, $i = 1, \ldots, M$.

We see a large discrepancy between the distributions of the spectrally chosen bandwidth and the distribution of the minimizer of the energy functional and the minimizer of the ISE.

3.4.3 Increasing sample size experiments

In this section, we want to see what happens when the sample size increases, that is, $n \to \infty$.

It is interesting to ask what happens to this distribution for increased sample size n. The answer is shown in Figure 3.8.

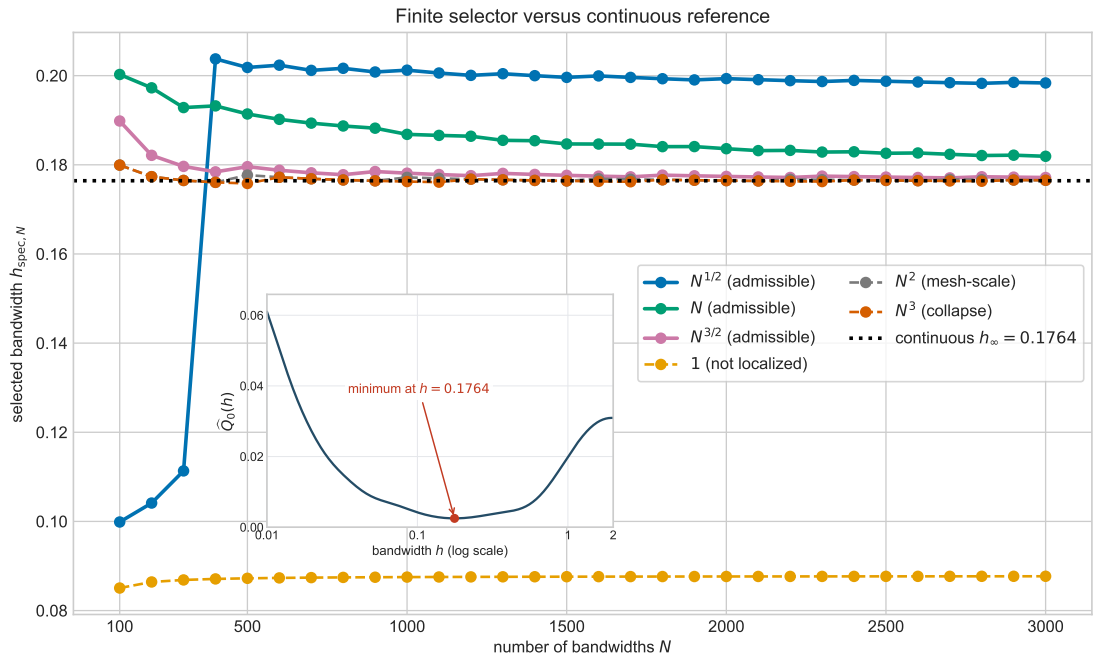
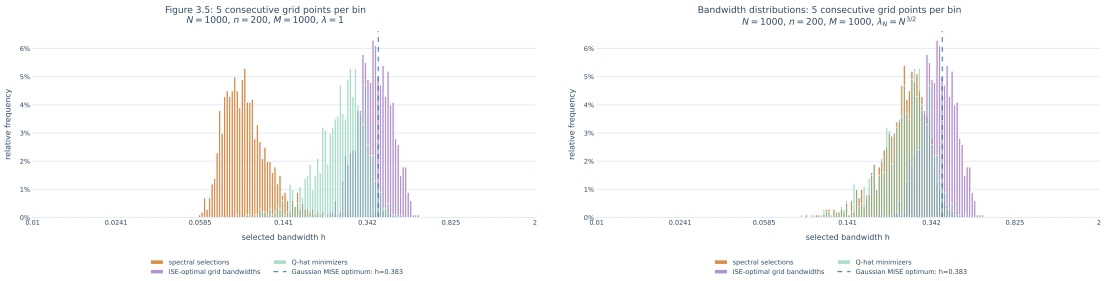


Figure 3.6: For $N \to \infty$ the chosen bandwidth approaches a limit $h_\infty = 0.1764$, for all $\alpha > 1/2$. The convergence is slow for $\alpha = 1$ and faster for $\alpha = 3/2$

explain N^2 and N^3 behavior

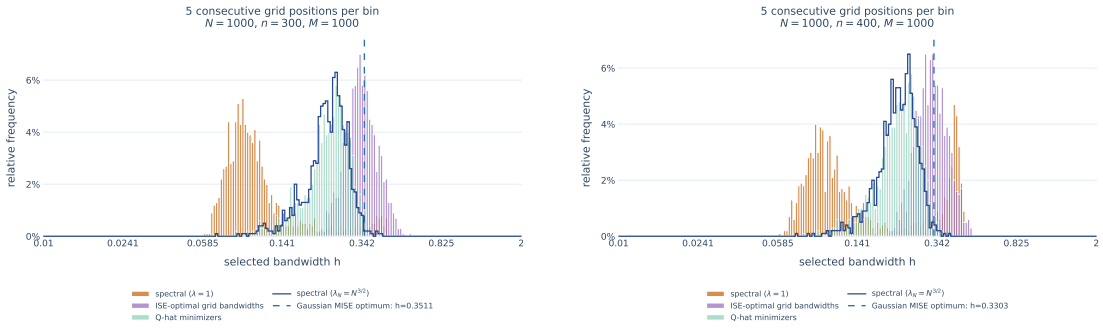
add experiments that show failure for exponents $\alpha > 2$

repeat the experiments for different r and under repeated sampling



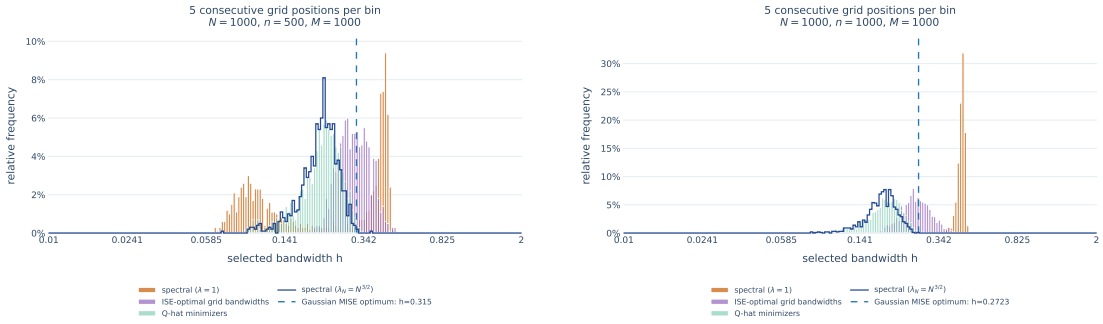
(a) $\lambda = 1$ (b) $\lambda_N = N^{3/2} \approx 31622.78$

Figure 3.7: Distributions of chosen bandwidths for $N = 1000$, $n = 200$, and $M = 1000$ repetitions, combining five consecutive grid positions per bin.



(a) $n = 300$

(b) $n = 400$



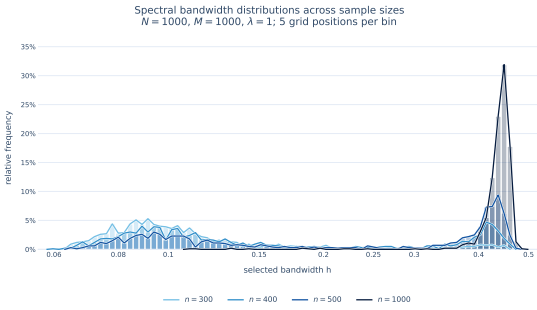
(c) $n = 500$

(d) $n = 1000$

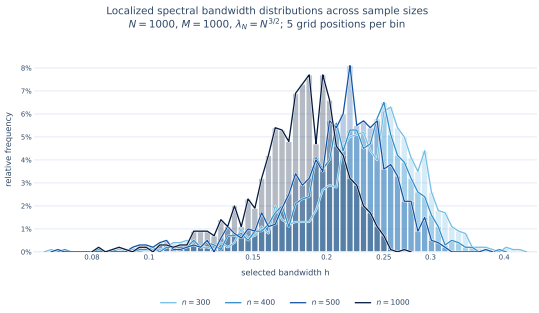
Figure 3.8: Distribution of chosen bandwidths changes for increasing sample size n. A clear transition between two regimes is visible.

Can the two peaks be described?

n_{critical} (finite sample mode switching), gap between two local Perron maxima??



(a) $\lambda = 1$



(b) $\lambda_N = N^{3/2}$

Figure 3.9: Comparison of spectrally selected bandwidth distributions for different sample sizes, with $N = 1000$ and $M = 1000$ repetitions. Note the different vertical scales.

3.5 Explanation of the Results

3.5.1 Convergence to closest adjacent pair

Here we explain the results of the fixed N and $\lambda \to \infty$ experiment.

Let $S^{(\lambda)}$ be the scale similarity matrix with entries $S_{ij}^{(\lambda)} = \exp(-\lambda D_{ij})$. Since the diagonal entries are equal to 1, we can write $S^{(\lambda)} = I_N + A^{(\lambda)}$. The matrices $S^{(\lambda)}$ and $A^{(\lambda)}$ have the same Perron vector. The Perron vector also stays the same if we multiply the matrix by a positive scalar. We therefore define $\tilde{A}^{(\lambda)} = e^{\lambda d_*} A^{(\lambda)}$, where $d_* = \min_{i \neq j} D_{ij}$ is the smallest distance between two distinct scales h_i and h_j. The off diagonal entries of $\tilde{A}^{(\lambda)}$ are given by $\tilde{A}_{ij}^{(\lambda)} = e^{\lambda(d_* - D_{ij})}$, which all lie between 0 and 1. One particular advantage of this construction is that large λ become numerically manageable while the Perron vector does not change, that is, numerical underflow is avoided. A second advantage is due to Davis-Kahan: Let $\tilde{S} = S + E$ for some error matrix where E is a symmetric error matrix with $\|E\| \leq (1/2)\text{gap}(S)$. Let v and $\tilde{v}$ be the Perron vectors of S and $\tilde{S}$ respectively. Then $\sin \angle(v, \tilde{v}) \leq 2\sqrt{2}\|E\|/\text{gap}(S)$. That is, the angular error between v and $\tilde{v}$ is governed by small error matrix norm and large spectral gap of S. We have $\text{gap}(S^{(\lambda)}) = e^{-\lambda d_*}\text{gap}(\tilde{A}^{(\lambda)})$. That is, for large λ the spectral gap of $S^{(\lambda)}$ becomes tiny and the upper bound for the angular error becomes large.

Assuming that $\|E\| \leq \eta\|S\|$ we get $\sin \angle(v, \tilde{v}) \leq 2\sqrt{2}\eta\|S\|/\text{gap}(S)$ and consequently

$$\frac{\|S^{(\lambda)}\|}{\text{gap}(S^{(\lambda)})} = \frac{1 + e^{-\lambda d_*}\mu_1}{e^{-\lambda d_*}(\mu_1 - \mu_2)}, \quad \frac{\|\tilde{A}^{(\lambda)}\|}{\text{gap}(\tilde{A}^{(\lambda)})} = \frac{\mu_1}{\mu_1 - \mu_2},$$

where μ_1 and μ_2 are the largest and second largest eigenvalue of $\tilde{A}^{(\lambda)}$, respectively.

We can interpret $m(S) := \|S\|/\text{gap}(S)$ as a measure (norm-to-gap ratio) for how stable the Perron vector of S can be computed. For large λ, $m(S^{(\lambda)})$ is large, whereas increasing λ influences $m(\tilde{A}^{(\lambda)})$ only through the eigenvalues μ_1 and μ_2 and not through the factor $e^{-\lambda d_*}$ anymore.

In fact, $m(S^{(\lambda)}) \to \infty$ but $m(\tilde{A}^{(\lambda)}) \to 1$ for $\lambda \to \infty$.

In the code, we use numpy linalg eigh, which efficiently computes the eigenvectors

and eigenvalues of symmetric matrices.

Taking the limit $\lambda \to \infty$ we get

$$\lim_{\lambda\to\infty} \tilde{A}_{ij}^{(\lambda)} = \begin{cases} 1 & , D_{ij} = d_* \\ 0 & , D_{ij} > d_*. \end{cases}$$

In the limit, this is an adjacency matrix of a graph on $\{h_1, \ldots, h_N\}$ with edges between h_i and h_j if and only if D_{ij} is minimal. In fact if D is a Robinson matrix (for example if we take the L^2-divergence) this forces $d_* = d_{j,j+1}$ for some $j = 1, \ldots, N-1$.

Assuming that $d_* = d_{j,j+1}$ for some unique $j = 1, \ldots, N-1$, we can show that the Perron vector $v^{(\lambda)}$ of $\tilde{A}^{(\lambda)}$ converges to $(1/2)e_j + (1/2)e_{j+1}$, that is, it concentrates at $\arg\min_{(j,j+1)} D_{j,j+1}$. We write $B = \lim_{\lambda\to\infty} \tilde{A}^{(\lambda)}$ (where the limit is understood entry-wise). The matrix B can be written as $B = e_j^T e_{j+1} + e_{j+1} e_j^T$. Direct calculations show that $e_j + e_{j+1}$ is an eigenvector with eigenvalue 1, $e_j - e_{j+1}$ is an eigenvector with eigenvalue -1 and e_i, $i \neq j, j+1$ are eigenvectors with eigenvalues 0. Therefore, $v = (1/2)e_j + (1/2)e_{j+1}$ is the sum normalized Perron vector of B.

This explains the results shown in Figure 3.5, as well as the yellow line in Figure 3.6. In Figure 3.5 the chosen bandwidth approaches the scales h_{j_*} and h_{j_*+1} with the smallest consecutive jump $D_{j,j+1}$, that is, $j_* \in \arg\min_j D_{j,j+1}$.

3.5.2 Perron vector concentrates at minimizer of normalized curvature-energy functional

To explain the results, we need the concept of a Fréchet-derivative. A good book is Zeidler [50] and Section 4.2 therein.

We introduce the general definition of a Fréchet-derivative. Let X and Y be Banach spaces and $U \subseteq X$ an open nonempty subset. Let $\Psi : U \to Y$ be a map. A linear and bounded operator $d\Psi(u) : X \to Y$ is called Fréchet-derivative of Ψ at the point $u \in U$ if

$$\Psi(u+h) = \Psi(u) + d\Psi(u)[h] + r_u(h), \quad \forall h \in X : u + h \in U,$$

where $r_u(h) : \{h \in X : u+h \in U\} \to Y$ is a remainder that fulfills $\|r_u(h)\|_Y / \|h\|_X \to 0$ for $h \to 0$. Ψ is called continuously differentiable on U if $d\Psi(u)$ exists for all $u \in U$ and if $d\Psi : U \mapsto \mathcal{L}(X, Y)$ is continuous. Here, $\mathcal{L}(X, Y)$ is the set of all linear and bounded operators from X to Y. We write $\Psi \in C^1(U)$.

Assume that the line $\{u + th : t \in [0, 1]\}$ is inside U and define $\gamma : [0, 1] \to Y$, $\gamma(t) = \Psi(u + th)$. We have $\gamma'(t) = d\Psi(u + th)[h]$. Now, if $\Psi \in C^1(U)$ then γ is continuously differentiable and by the fundamental theorem of calculus $\Psi(u + h) - \Psi(u) = \int_0^1 d\Psi(u + th)[h]\, dt$. We can use this to bound the norm of the remainder:

$$\begin{aligned}
\|r_u(h)\|_Y &\leq \int_0^1 \|(d\Psi(u + th) - d\Psi(u))[h]\|_Y \, dt \\
&\leq \|h\|_X \int_0^1 \|d\Psi(u + th) - d\Psi(u)\|_{\mathsf{op}} \, dt \\
&\leq \|h\|_X \sup_{t \in [0,1]} \|d\Psi(u + th) - d\Psi(u)\|_{\mathsf{op}}
\end{aligned}$$

In our setting, $X = \mathbb{R}$, $U = \mathbb{R}_{>0}$, and $Y = L^2(\mathbb{R})$. The map is $\Psi(h) = \hat{f}_h$. The next Lemma gives the Fréchet-derivative of $h \mapsto \hat{f}_h$.

Lemma 7. *Let K be differentiable on $\mathbb{R}$ and square integrable. Let L be defined as $L(u) = K(u) + uK'(u)$. We must assume a certain decay assumption on L, namely $|L(v)| \leq C(1 + |v|)^{-\alpha}$ for some $C > 0$, $\alpha > 1/2$ and all $v \in \mathbb{R}$. The Fréchet-derivative of $h \mapsto \hat{f}_h$ at h is an operator $d\Psi(h) : \mathbb{R} \to L^2(\mathbb{R})$ given by $d\Psi(h)[\tau] = \tau \partial_h \hat{f}_h$, where*

$$\partial_h \hat{f}_h(x) = -\frac{1}{nh^2} \sum_{j=1}^{n} L\left(\frac{x - X_j}{h}\right), \quad x \in \mathbb{R}.$$

Additionally, the norm of the remainder term $r_h(\tau)$ can be bounded above by:

$$\|r_h(\tau)\|_2 \leq |\tau| \mu_J(|\tau|)$$

if $h, h + \tau \in J := [a, b] \subseteq (0, \infty)$. Also, $\mu_J \geq 0$ and $\mu_J(s) \to 0$ as $s \downarrow 0$.

Proof. See Appendix Section 6.2.2. □

Next, we state the main theorem of this thesis.

Proposition 8. *Let $D : \mathcal{P}_2 \times \mathcal{P}_2 \to \mathbb{R}_{\geq 0}$ be the L^2-divergence on $\mathcal{P}_2 := \mathcal{P} \cap L^2(\mathbb{R})$. Let $\mathcal{H}_N = \{0 < h_1 < \cdots < h_N\}$ be a r-linear grid for some $0 < r \leq 1$ with fixed endpoints $h_1 = h_{\min}$ and $h_N = h_{\max}$. Let λ_N be a sequence such that $\lambda_N \to +\infty$ and $\lambda_N / N^2 \to 0$. Define the scale similarity matrix $S^{(N)} = (S_{ij}^{(N)})_{i,j=1}^N$ with entries $S_{ij}^{(N)} = \exp\left(-\lambda_N D(\hat{f}_{h_i}, \hat{f}_{h_j})\right)$. Here $\hat{f}_h$ is the kernel density estimate with Gaussian kernel K and fixed data $X_1, \ldots, X_n \sim f$.*

Let i_N be an index sequence such that $h_{i_N} \to h_0 \in (h_{\min}, h_{\max})$. Let $R_{i_N}^{(N)} = \sum_{j=1}^N S_{i_N j}^{(N)}$ be the i_N-th row sum of $S^{(N)}$. Then

$$\sqrt{\lambda_N} \Delta_{r,N} R_{i_N}^{(N)} \to \sqrt{\frac{\pi}{\hat{Q}_r(h_0)}}, \quad N \to +\infty,$$

where $\hat{Q}_r(h) = 2^{-1} h^{2(1-r)} \|\partial_s \hat{f}_s|_{s=h}\|_2^2$ is an energy functional.

Proof. See Appendix Section 6.2.3. □

28.10: mention Laplace method as precedence!!

Let $v^{(N)}$ be the Perron vector of $S^{(N)}$, that is,

$$\sum_{j=1}^N S_{ij}^{(N)} v_j^{(N)} = \underbrace{\theta_{\max}^{(N)}}_{>0} v_i^{(N)}, \quad \text{for all } i = 1, \ldots, N.$$

The Perron vector $v^{(N)}$ has large entries $v_i^{(N)}$ if the weighted row sum $\sum_{j=1}^N S_{ij}^{(N)} v_j^{(N)}$ is large. Using the normalization $\|v^{(N)}\| = 1$ we can see the entries $v_i^{(N)}$ as weights, which allows us to say that the Perron vector gives a bandwidth h_i a higher weight if $\hat{f}_{h_i}$ is similar to $\hat{f}_{h_j}$ (in terms of L^2-distance) where the bandwidths h_j also have a large weight.

The Proposition 8 establishes that for a large N (finer and finer bandwidth grid) the unweighted row sum $\sum_{j=1}^N S_{i_N j}^{(N)}$ is approximately equal to $\lambda_N^{-1/2} \Delta_{r,N}^{-1} \pi^{1/2} \hat{Q}_r^{-1/2}(h_{i_N})$ (a properly scaled energy functional). This means that large row sums correspond to a low value of the energy functional $\hat{Q}_r$ and since the energy functional $\hat{Q}_r$ is the squared L^2 norm of the Fréchet derivative of the kernel density estimate, its minimizer can be interpreted as the point at which the L^2-curve $h \mapsto \hat{f}_h$ changes most slowly. This provides an informal explanation for why the Perron vector peaks at the minimizer of the energy functional $\hat{Q}_r$.

The question now is if the result of Proposition 8 can be extended to say that the actual Perron vector peaks at the minimizer of the energy functional (after proper scaling). The next result aims to give the answer.

Proposition 9. *Assume that the energy functional $\hat{Q}_r$ has a unique minimum on $[h_{\min}, h_{\max}]$ and let h^* be this unique minimizer. Let $v^{(N)}$ be the Perron vector of $S^{(N)}$ and let (k_N) be a sequence of indices defined by $k_N := \min \arg\max\{v_i^{(N)} : 1 \leq i \leq N\}$. Then the spectrally chosen bandwidth h_{k_N} converges to the unique minimizer h^* of $\hat{Q}_r$, that is, $h_{k_N} \to h^*$ as $N \to \infty$.*

Proof. See Appendix Section 6.2.4. □

The spectral machinery is asymptotically redundant: the spectral selector is asymptotically equivalent to a normalized curvature-energy criterion.

3.5.3 Asymptotic minimizer of expected energy functional is asymptotically equivalent to mean integrated squared error minimizer

29.10: finish this section

In this section a connection between the expectation of the energy functional $\hat{Q}_r$ and the MISE is presented. By our results from the previous Section 3.5.2 this connects the spectral scale selection directly with optimal bandwidth selection in the classical sense.

For the Gaussian kernel, $L(u) = (1-u^2)e^{-u^2/2}$ and the energy functional $\hat{Q}_r$ can be written down explicitly:

$$\begin{aligned}
\hat{Q}_r(h) &= \frac{1}{2}h^{2(1-r)}\|\partial_h \hat{f}_h\|_2^2 = \frac{1}{2n^2}h^{-2(1+r)}\int \sum_{i,j=1}^{n} L\left(\frac{x-X_i}{h}\right) L\left(\frac{x-X_j}{h}\right) dx \\
&= \frac{\sqrt{\pi}}{2n^2}h^{-2(1+r)+1}\sum_{i,j=1}^{n} e^{-d_{ij}^2/(4h^2)}\left(\frac{3}{4} - \left(1+\frac{d_{ij}^2}{4h^2}\right) + \left(1-\frac{d_{ij}^2}{4h^2}\right)^2\right) \\
&= \frac{\sqrt{\pi}}{32n^2}h^{-5-2r}\sum_{i,j=1}^{n} e^{-d_{ij}^2/(4h^2)}\left(d_{ij}^4 - 12h^2 d_{ij}^2 + 12h^4\right)
\end{aligned}$$

because

$$\int \underbrace{\left(1-\left(\frac{x-X_i}{h}\right)^2\right)\left(1-\left(\frac{x-X_j}{h}\right)^2\right)}_{=:P(x)} \exp\left(-\frac{1}{2}\left(\frac{x-X_i}{h}\right)^2\right)\exp\left(-\frac{1}{2}\left(\frac{x-X_j}{h}\right)^2\right)dx$$

$$= \exp\left(-\frac{(X_i-X_j)^2}{4h^2}\right)\int P(x)\exp\left(-\frac{\left(x-\frac{X_i+X_j}{2}\right)^2}{h^2}\right)dx$$

$$= e^{-d_{ij}^2/(4h^2)}\frac{h}{\sqrt{2}}\int P\left(\frac{h}{\sqrt{2}}u+\frac{X_i+X_j}{2}\right)\exp\left(-u^2/2\right)du$$

$$= e^{-d_{ij}^2/(4h^2)}h\sqrt{\pi}\left(\frac{3}{4}-\left(1+\frac{d_{ij}^2}{4h^2}\right)+\left(1-\frac{d_{ij}^2}{4h^2}\right)^2\right),$$

where we have used $\int u^n e^{-u^2/2}\,du = \sqrt{2\pi}n!/(2^{n/2}(n/2)!)$ for even n and have set $d_{ij} := X_i - X_j$.

We can characterize the behavior of $\hat{Q}_r$ for $h \to 0$ and $h \to \infty$. Namely,

$$\hat{Q}_r(h) \sim \frac{\|L\|_2^2}{2n}\frac{1}{h^{1+2r}}, \quad h \to 0,$$
$$\hat{Q}_r(h) \sim \frac{\|L\|_2^2}{2}\frac{1}{h^{1+2r}}, \quad h \to \infty.$$

This means that $\hat{Q}_r(h) \to \infty$ for $h \to 0$ and $\hat{Q}_r(h) \to 0$ for $h \to \infty$. Because $\hat{Q}_r(h) > 0$ for all $h > 0$ this implies that $\hat{Q}_r$ has no global minimum.

add short discussion about local minima

Using standard root finding methods, we can look for stationary points and check if they are local minima. Afterwards, we can compare with the energy function values at the borders $h_{\min}$ and $h_{\max}$ to see if the found local minimum is in fact a global minimum on the interval $[h_{\min}, h_{\max}]$.

The black dotted horizontal line in Figure 3.6 represents $h_* = \arg\min_{h\in[h_{\min},h_{\max}]} \hat{Q}_0(h)$. The green histograms inside Figures 3.7and 3.8 also represent the minimizer of the energy functional $\hat{Q}_0$.

Expected energy functional

The expectation of the energy functional is

$$\mathbb{E}[\hat{Q}_r(h)] = \frac{1}{2}h^{2(1-r)}\mathbb{E}[\|\partial_h \hat{f}_h\|_2^2]$$
$$\mathbb{E}[\|\partial_h \hat{f}_h\|_2^2] = \frac{1}{nh^4}\|L\|_2^2 + \left(1 - \frac{1}{n}\right)\|(\partial_h K_h * f)\|_2^2$$

It holds $\|h^{-1}\partial_h(K_h * f) - \mu_2(K)f''\|_2^2 \to 0$ as $h \downarrow 0$ for all suitable kernels K.

This implies $\|(\partial_h K_h * f)\|_2^2 = h^2\mu_2(K)^2\|f''\|_2^2 + \mathbf{o}(h^2)$, since

$$\frac{\|(\partial_h K_h * f)\|_2^2 - h^2\mu_2(K)^2\|f''\|_2^2}{h^2} = \|h^{-1}(\partial_h K_h * f)\|_2^2 - \mu_2(K)^2\|f''\|_2^2$$
$$\underbrace{\leq \|h^{-1}(\partial_h K_h * f) - \mu_2(K)f''\|_2^2}_{\to 0 \text{ as } h\downarrow 0} + \underbrace{\mu_2(K)^2\|f''\|_2^2 - \mu_2(K)^2\|f''\|_2^2}_{=0}.$$

We point out that if K is the Gaussian kernel, then this is the well known Gaussian heat equation: $\partial_h K_h * f = h(K_h * f'')$.

A leading term approximation yields

$$\mathbb{E}[\hat{Q}_r(h)] = \underbrace{\frac{1}{2n}h^{-2(1+r)}\|L\|_2^2 + \frac{n-1}{2n}h^{4-2r}\|f''\|_2^2}_{=:\mathrm{AMQ}_r(h)} + \mathbf{o}(h^{4-2r}).$$

The unique minimizer of $\mathrm{AMQ}_r(h)$ over $(0, \infty)$ is

$$h_{\mathrm{AMQ},r}(n) = \left(\frac{1+2r}{4-2r}\frac{\|L\|_{L^2}^2}{(n-1)\mu_2(K)^2\|f''\|_{L^2}^2}\right)^{1/5}$$

given that $n \geq 2$ and $-1/2 < r < 2$.

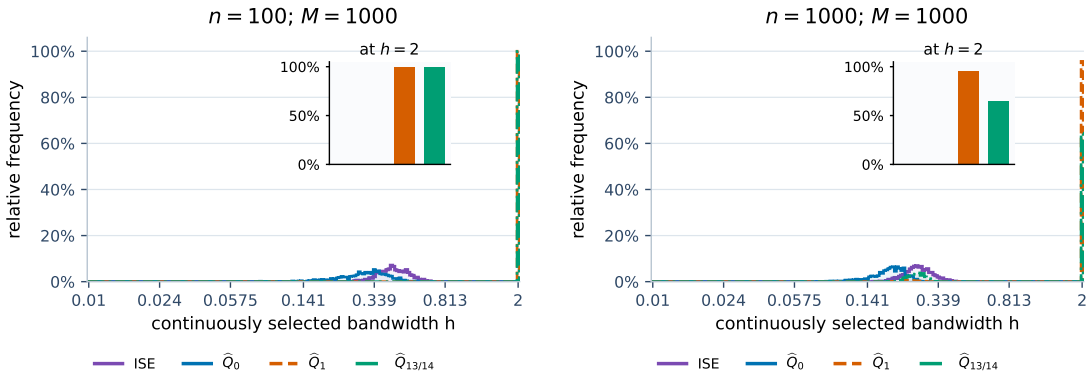
We recall that the asymptotic minimizer of the mean integrated squared error is given in Equation (3.1). We therefore get

$$\lim_{n\to\infty}\frac{h_{\mathrm{AMQ},r}(n)}{h_{\mathrm{AMISE}}(n)} = \frac{1+2r}{4-2r}\left(\frac{\|L\|_2^2}{\|K\|_2^2}\right)^{1/5} =: C_r(K, L). \tag{3.11}$$

If K is the Gaussian kernel, then $\|K\|_2^2 = 1/(2\sqrt{\pi})$ and $\|L\|_2^2 = 3/(8\sqrt{\pi})$. In this case, $C_r(K, L) = \frac{1+2r}{4-2r}(3/4)^{1/5}$ and it becomes 1 if set $r = 13/14$, that is, for

$r = 13/14$ (the bandwidths inside the grid are arranged slightly below being linear) the asymptotic minimizer of the mean integrated squared error and the asymptotic minimizer of the expected energy functional are asymptotically equivalent.

Figure 3.10 visualizes our theoretical considerations. It compares the distribution of $M = 1000$ selected bandwidths minimizing: 1) ISE (violet), 2) $\hat{Q}_0$ (blue), 3) $\hat{Q}_1$ (orange), 4) $\hat{Q}_{13/14}$ (green).



(a) $n = 100$

(b) $n = 1000$

(c) $n = 1500$

(d) $n = 2000$

Figure 3.10: Sampling distributions of the continuous minimizers of ISE, $\widehat{Q}_0$, $\widehat{Q}_1$, and $\widehat{Q}_{13/14}$, based on $M = 1000$ repetitions for each sample size. Insets show the relative frequency of minimizers at the upper endpoint $h = 2$.

3.6 Literature Positioning

29.10: Write it!

The paper by Goldenshluger and Lepski [24] can be seen as a precedent for a bandwidth selector that is based on the comparison across different smoothing scales.

They set $\mathcal{H} = [h_{\min}, h_{\max}]$ and define $\hat{f}_{h,\eta}(x) = n^{-1} \sum_{j=1}^{n} (K_h * K_\eta)(x - X_j)$ and

$$\hat{\mathcal{R}}_h := \sup_{\eta \in \mathcal{H}} \Big(\|\hat{f}_{h,\eta} - \hat{f}_\eta\|_2 - m_2(h,\eta) \Big)_+ + \sup_{\eta \in \mathcal{H}} m_2(\eta, h),$$

where $m_2(h,\eta) := g_2(K_\eta) + g_2(K_h * K_\eta)$ and $g_2(U) := (25/3\sqrt{n})\|U\|_2$. Their bandwidth selector is $\hat{h} = \arg\min\{\hat{R}_h : h \in \mathcal{H}\}$.

They then proceed to show the oracle inequality

$$\mathbb{E}[\|\hat{f}_{\hat{h}} - f\|_2] \leq C \inf_{h \in \mathcal{H}} \mathbb{E}[\|\hat{f}_h - f\|_2] + \delta_n,$$

where $C > 0$ is a constant independent of n and f and δ_n is a remainder independent of f. This means that the risk of their selector performs within a constant factor of the best candidate bandwidth plus a remainder.

Lacour and Massart [37] build upon the work of Goldenshluger and Lepski by providing considerations on how to make this procedure practically usable. Namely they use $\mathcal{H} = \{e^{-k} : 3 \leq k \leq 10\} \cup \{0.002 + k \cdot 0.02, 0 \leq k \leq 24\}$. That is, they sidestep the continuous optimization that would be needed to obtain $\hat{h}$ by discretization of the interval $\mathcal{H}$.

Another important paper dealing with analyzing the KDE across a family of bandwidths is by Chaudhuri and Marron [17]. Their “SiZer” (Significant Zero crossings) method simultaneously studies a wide range of bandwidths and avoids the classical need to choose a bandwidth. This is a visual data analysis tool that produces a colored map, where for each location-scale pair (x, h) the derivative $\hat{f}'_h(x)$ is computed. The corresponding point (x, h) on the color map is blue if the derivative is significantly positive, red if the derivative is significantly negative, and purple if the derivative is not significantly different from 0. It is a visual aid that lets the practitioner see at one glance how significant features like bumps depend on the chosen bandwidth.

energy functional precedence: physics, kde bandwidth

We also mention the paper by Lindeberg [39] which is a direct precedent for the form of our energy functional $\hat{Q}_r(h) = (1/2)h^{2(1-r)}\|\partial_h \hat{f}_h\|_2^2$. For a continuous signal f its convolution with the Gaussian kernel K is studied: $L(x,t) := (K_{\sqrt{t}} * f)(x)$. A γ-normalized derivative operator is defined by $\partial_\gamma L(x,t) := t^{\gamma/2}\partial_x L(x,t)$. The main principle of the paper is that a scale at which suitable combinations of these normalized derivatives assume a local maximum can reflect a characteristic length of structure in the data. The developed methods are applied in computer vision, for example, to detect blobs, junctions, edges, or ridges. By the Gaussian heat equation, $\partial_h \hat{f}_h(x) = h\partial_x^2 \hat{f}_h(x)$. Setting $t = h^2$ now gives

$$\partial_\gamma^2 L(x,h^2) = h^{2\gamma}\partial_x^2(K_h * \mu_n)(x) = h^{2\gamma}\partial_x^2 \hat{f}_h(x) = h^{2\gamma-1}\partial_h \hat{f}_h(x)$$

and for $\gamma = 1 - r/2$ we get $\partial_{1-r/2}^2 L(x,h^2) = h^{1-r}\partial_h \hat{f}_h(x)$. Lindeberg also introduces the energy measure $E_2(t) := \int |\partial_\gamma^2 L(x,t)|^2\,dx$. In our setting, this directly evaluates to

$$E_2(h^2) = \int |\partial_{1-r/2}^2 L(x,h^2)|^2\,dx = h^{2(1-r)}\int |\partial_h \hat{f}_h(x)|^2\,dx = 2\hat{Q}_r(h).$$

Computing an eigenvector is equivalent to minimizing an energy over vectors

As a parameter changes, that eigenvector concentrates near minima of an effective potential.

4 Conclusion

In the first part of this thesis we developed a new method for estimating the coefficients and determining the order of auto-regressive processes. It is based on the minimization of the width of the sliding window embedding of the AR process via projected sub-gradient descent. By establishing a connection to convex geometry, we were able to derive consistency results for this new method.

The second part of this thesis was concerned about applying the spectral scale selection methodology developed in the manuscript [6] to bandwidth selection in kernel density estimation. We developed a framework for how to construct pairwise similarity matrices using the notion of divergences between probability distributions. Our main result establishes an interesting connection between energy functional minimization and spectral scale selection.

5 Outlook

During the research many interesting questions arose that could not be explored further due to limited resources. We use this Outlook chapter to present those ideas that can be approached in future research.

Geometric Estimation of Autoregressive processes

31.10: Write outlook for AR part!

Other convex–geometric functionals

Seeing the support of the sliding window point cloud of an auto-regressive time series as a convex–geometric object, which is a zonoid in case of bounded innovations and using the minimum width as convex–geometric functional to infer information about the AR order and AR coefficients leads to the question if there could be other convex–geometric functionals that could recover something from the time series.

A natural candidate would be the mean width, given by integrating $w_k(u)$ over $\mathbb{S}^{k-1}$:

$$\overline{w}_k = \int_{\mathbb{S}^{k-1}} w_k(u)\, d\sigma(u).$$

In practice one would estimate the integral by Monte Carlo simulation. The estimator would take the form

$$\hat{\overline{w}}_k = \int_{\mathbb{S}^{k-1}} \hat{w}_{k,T}(u)\, d\sigma(u) \approx \frac{1}{M} \sum_{m=1}^{M} \max_{s,t} \langle Y_s^{(k)} - Y_t^{(k)}, U_k \rangle,$$

where $U_k \sim \mathcal{U}(\mathbb{S}^{k-1})$.

empirical version of intrinsic volumes? And also convergence to theoretical object

spherical harmonics: Q_0 (mean width), Q_2, Q_4, ...

numerical analogues $\hat{Q}_0$, $\hat{Q}_2$, $\hat{Q}_4$

General innovation distributions

A vital question is how to generalize the minimum width approach to a broader class of innovation distributions. The main difficulty to overcome is that $\mathrm{supp}(Y_t^{(k)})$ is generally not compact but usually unbounded. Thus, the minimum width is infinite.

Probability integral transform (PIT)

linear filter, Wold representation

The *minimum volume set* (compare [46]) is defined by

$$E_\alpha \in \arg\min\{\lambda_d(A) : \mathbb{P}(Y \in A) \geq \alpha, A \in \mathcal{B}(\mathbb{R}^d)\}.$$

The *highest density region* C_α, $\alpha \in (0,1)$ (see [33]) is defined by

$$\begin{aligned} C_\alpha &= C_{h_\alpha} \\ C_h &= \{x \in \mathbb{R}^d : f(x) \geq h\} \\ h_\alpha &= \sup\{h \in \mathbb{R} : \mathbb{P}(Y \in C_h) \geq 1 - \alpha\} \end{aligned}$$

connect these concepts with each other. For many distributions they are the same. Using minimum volume sets would remove dependence on having a smooth density

estimate HDR directly from the points $Y_t^{(k)}$; non parametric estimation, empirical HDR

Maybe find a class of distributions where the highest density region (or minimum volume set) has a nice geometric shape and compute the intrinsic volumes of that shape

log concave distributed innovations give convex highest density regions; VERY INTERESTING

A probability density is log concave if

$$f(x) = \frac{e^{-V(x)}}{\int e^{-V(u)}\, du}$$

for some convex function V.

Normally distributed innovations

Next, we turn our attention to Gaussian innovations, which is often seen as the default case, that is, $Z_t \sim N(0, \sigma^2)$ for some $\sigma > 0$. Using the MA(∞) representation, we see that $Y_t^{(k)} \sim \mathcal{N}(0, \Gamma_k)$ with covariance matrix

$$\Gamma_k = \sigma^2 \sum_{j=0}^{\infty} g_j^{(k)} (g_j^{(k)})^T$$

The covariance matrix plays an important role in describing the geometry of the point cloud $\mathcal{P}_k$. Let $E_c = \{y \in \mathbb{R}^k : y^T \Gamma_k^{-1} y \leq c\}$ be the ellipsoid of size $c > 0$. Then for all $\alpha \in (0, 1)$ we have

$$E_{c_\alpha} \in \arg\min \left\{ \mathrm{Vol}(A) : \mathbb{P}(Y \in A) \geq \alpha : A \subseteq \mathbb{R}^k \right\}$$

That is, the ellipsoid with size c_α is the set with minimum volume, such that the probability that Y is inside E_{c_α} is at least α.

find reference or prove

Real-world Data

Talk about applications, argue where AR models with bounded innovations might be used

- Wind or solar power generation
- traffic occupancy
- industrial control, system identification

Spectral Scale Selection for Kernel Density Estimation

Regarding the spectral scale methodology that we invented in Chapter 3, the natural next step is to first test the other symmetric divergences: Hellinger, Jensen-

Shannon, and Total Variation.

Bounded Divergences

These three divergence also have the special property that they are bounded. From [] we have the uniform bound $D_f(p,q) \leq C_f := f(0) + f^*(0)$ over all $p, q \in \mathcal{P}$. cite! Here, $f^*(0) = \lim_{t \downarrow 0} t f(t^{-1})$. It follows that a f-divergence is bounded if and only if $f(0) + f^*(0) < \infty$. In case f is symmetric, this is equivalent to $2f(0) < \infty$. For bounded divergences, we can use a more straightforward transformation approach by setting $\Sigma_{ij}^{(N)} = C_f - D_f(\hat{f}_{h_i}, \hat{f}_{h_j})$ and normalizing $\tilde{\Sigma}_{ij}^{(N)} = C_f^{-1}\Sigma_{ij}^{(N)} = 1 - D_f(\hat{f}_{h_i}, \hat{f}_{h_j})/C_f$. For example, the total variation divergence can be written as

$$\begin{aligned} d_{\mathsf{TV}}(p,q) &= \frac{1}{2}\int |p-q| = 1 - 1 + \frac{1}{2}\int |p-q| \\ &= 1 - \frac{1}{2}\int p - \frac{1}{2}\int q + \frac{1}{2}\int |p-q| \\ &= 1 - \int \frac{p+q-|p-q|}{2} \\ &= 1 - \int \min\{p,q\}. \end{aligned}$$

Thus, $d_{\mathsf{TV}}(p,q) \leq 1$ and $1 - d_{\mathsf{TV}}(p,q) = \int \min\{p,q\}$ is a similarity measure between p and q. Similarly for the squared Hellinger distance we can write

$$H^2(p,q) = \frac{1}{2}\int (\sqrt{p} - \sqrt{q})^2 = 1 - \int \sqrt{pq}.$$

So again $H^2(p,q) \leq 1$ and $1 - H^2(p,q) = \int \sqrt{pq}$ is a similarity measure between p and q. For total variation, we use the $\min$ kernel and for squared Hellinger Bhattacharyya kernel (see [35]). The JS divergence can also be bounded:

$$f(t) = \frac{1}{2} t \log \underbrace{\frac{2t}{1+t}}_{\leq 2} + \frac{1}{2}\log \underbrace{\frac{2}{1+t}}_{\leq 2} \leq \frac{1}{2}\log 2(t+1), \quad t \geq 0,$$

$$\mathrm{JS}(p,q) = \int q f(p/q) \leq \int q \frac{1}{2}\log 2 \left(\frac{p}{q} + 1\right) = \log 2.$$

Rényi divergence

Another popular class of parameterized divergences is the class of so called Rényi

divergences. The class of Rényi divergences is defined as

$$R_\alpha(p\|q) = \frac{1}{\alpha - 1} \log \left(\int p(x)^\alpha q(x)^{1-\alpha} \, dx \right),$$

where $\alpha \in \mathbb{R}_{>0} \setminus \{1\}$ is the order. In case of $\alpha > 1$, we read $f^\alpha g^{1-\alpha} = f^\alpha / g^{\alpha-1}$. The limiting case $\alpha \to 1$ recovers the Kullback-Leibler divergence. The Rényi divergence is not a f-divergence, but the Rényi and Amari divergences are both strictly decreasing transformations of the same integral $C_\alpha(p, q) = \int p^\alpha q^{1-\alpha}$. We have

$$\begin{aligned} R_\alpha(p\|q) &= g(C_\alpha(p, q)), \quad g(t) = \frac{1}{\alpha - 1} \log t, \quad \text{and} \\ D_\alpha(p\|q) &= h(C_\alpha(p, q)), \quad h(t) = \frac{1}{\alpha(\alpha - 1)}(t - 1). \end{aligned}$$

Thus, the Rényi and Amari divergences can be transformed into each other using a strictly increasing function.

Rényi is not conditionally negative definite?

Rényi is unbounded?

Numerical Approximation of the α-divergence and total variation

Following the need for rigorous error bounds when approximating the divergences, we give truncation bounds for the α-divergence and the total variation divergence.

We decide to use the classical quadrature formula

$$\int_a^b f(x) \, dx \approx Q_M(f) = (b - a) \sum_{m=0}^{M} w_m f(x_m),$$

where w_m are the quadrature weights and x_m the (equidistant) quadrature nodes. As quadrature weights we choose $w_m = 1/(2M)$ for $m = 0, M$ and $w_m = 1/M$ for $m = 1, \ldots, M - 1$. This choice corresponds to the composite trapezoidal rule.

Let $A = [a, b]$ and $A^c = \mathbb{R} \setminus A$. We first aim to bound the truncation error $E_{\text{trunc}} = |\int_{A^c} f(p(x), q(x)) \, dx|$, where f is any of the functions that appear in a divergence,

and $p = \hat{f}_h$ and $q = \hat{f}_g$ are kernel density estimates with a Gaussian kernel. It holds

$$\tau_h := \int_{A^c} \hat{f}_h = \frac{1}{n} \sum_{j=1}^{n} \underbrace{\int_{\mathbb{R}\setminus(a,b)} \frac{1}{h} K\left(\frac{x - X_j}{h}\right) dx}_{=\Phi((a-X_j)/h)+\Phi((X_j-b)/h)} .$$

Setting $a = X_{\min} - Lh_{\max}$ and $b = X_{\max} + Lh_{\max}$, where $X_{\min} = \min X_j$, $X_{\max} = \max X_j$ and $h_{\max} = \max h_i$, we obtain

$$\Phi((a - X_j)/h) + \Phi((X_j - b)/h) \leq 2\Phi(-L) =: \delta_L$$

and thus $\tau_h \leq 2\Phi(-L)$ for any $h \leq h_{\max}$. Using the upper bound for the above truncation error, we choose L such that $\delta_L \leq \epsilon$. This can be achieved by setting $L \geq \Phi^{-1}(1 - \epsilon/2)$.

Replacing $\int p^\alpha q^{1-\alpha}$ in Equation (3.2) with the truncated integral $\int_a^b p^\alpha q^{1-\alpha}$ leads to the truncation error

$$E_{\text{trunc}} = \frac{1}{\alpha(1-\alpha)} \int_{A^c} p^\alpha q^{1-\alpha} \overset{\text{Hölder}}{\leq} \frac{1}{\alpha(1-\alpha)} \left(\int_{A^c} p\right)^\alpha \left(\int_{A^c} q\right)^{1-\alpha} \leq \frac{\delta_L}{\alpha(1-\alpha)}$$

for the α-divergence and truncating the total variation yields

$$\frac{1}{2} \int_{A^c} |p - q| \leq \frac{1}{2} \int_{A^c} p + \frac{1}{2} \int_{A^c} q \leq \delta_L.$$

As a consequence, we see that approximating α-divergence and total variation using the composite trapezoidal rule allows clean error control.

Numerical Approximation of Jensen-Shannon divergence

For KL and JS divergence it is natural to use the Gauss-Hermite quadrature, which is a quadrature rule that approximates integrals of the form $\int_{-\infty}^{\infty} e^{-x^2} f(x)\, dx$ by $\sum_{m=1}^{M} w_m f(x_m)$ with the nodes x_m being the M-th Hermite Polynomial H_M and weights $w_m = (2^{M-1} M! \sqrt{\pi})/(M^2 H_{M-1}^2(x_m))$. For reference on Gauss-Hermite quadrature, see [1, Section 25.4.46]. We can write (and consequently approximate)

the KL-divergence between two kernel density estimates $\hat{f}_h$ and $\hat{f}_g$ as

$$D_{\mathrm{KL}}(\hat{f}_h \| \hat{f}_g) = \frac{1}{\sqrt{\pi}} \frac{1}{n} \sum_{i=1}^{n} \int_{-\infty}^{+\infty} e^{-x^2} \log\left(\hat{f}_h(X_j + h\sqrt{2}x) - \hat{f}_g(X_j + h\sqrt{2}x)\right) dx$$
$$\approx \frac{1}{\sqrt{\pi}} \frac{1}{n} \sum_{i=1}^{n} \sum_{m=1}^{M} w_m \log\left(\hat{f}_h(X_j + h\sqrt{2}x_m) - \hat{f}_g(X_j + h\sqrt{2}x_m)\right).$$

Using the formula in Equation (3.3) we can also approximate the JS-divergence between $\hat{f}_h$ and $\hat{f}_g$.

One advantage of the Gauss-Hermite quadrature is that we do not need to care about the truncation error because there is none.

Extension to asymmetric divergences

Stationary distribution construction for asymmetric scale similarity matrices

The fact that the similarity matrix S is not symmetric in general is not a problem because the Doob h transform also works in this case. The resulting stationary distribution π is given by

$$\pi_i = \frac{l_i r_i}{\sum_{j=1}^{N} l_j r_j},$$

where r and l are the right and left principal eigenvectors of S, that is, $Sr = \lambda_{\max} r$ and $S^T l = \lambda_{\max} l$.

This construction gives rise to the study of our methodology for generally asymmetric divergences, most prominently the Kullback-Leibler divergence: $D_{\mathrm{KL}}(p, q) = \int p \log(p/q)$ (as a limit of α-divergence for $\alpha \to 1$).

Perron vector stability analysis

Mention Davis Kahan theorem and spectral gap

Matrix pertubation theory

Davis-Kahan: Stability $\leftrightarrow$ informativeness

Power method (compute Perron vector through Power iteration)

Data coming from the Gaussian mixture model

In cases where the data comes from a Gaussian mixture model, there might not exist a unique "optimal" bandwidth, and our spectral approach could serve as a ranking procedure for candidate bandwidths. We will be particularly interested in cases where the globally optimal bandwidth (with respect to some error criterion) would not be picked by the Perron vector, and instead another bandwidth is picked because it has a more central position.

Bandwidth grid that scales with sample size

grid must scale with n? Otherwise spectrally selected bandwidth will not be consistent (outlook)

6 Appendix

6.1 Proofs Chapter 2

6.1.1 Proof of Proposition 1

Lemma 8. *There exist $C > 0$ and $\rho \in (0,1)$ such that $|\psi_j| \leq C\rho^j$. In particular, $\lim_{j\to\infty} \psi_j = 0$.*

Proof. Define $r_\phi := \min\{|z| : z \in \mathbb{C}, \phi(z) = 0\}$. Since $\{X_t\}$ is causal, $r_\phi > 1$. Let $R \in (1, r_\phi)$ be arbitrary. From complex analysis, it follows that $1/\phi$ is analytical on the disk $D_0(R) = \{z \in \mathbb{C} : |z| \leq R\}$. Let $1/\phi(z) = \sum_{j=0}^{\infty} \psi_j z^j$ be its representation. By Cauchy's integral inequality, see [47, Corollary 4.3], $|\psi_j| \leq R^{-j} \max_{|z|=R} |1/\phi(z)|$. Setting $C = \max_{|z|=R} |1/\phi(z)|$ and $\rho = R^{-1}$ yields the statement. □

Proof. First, notice that $\mathcal{Z}_n^{(k)} \subseteq \mathcal{Z}$ for all $n \in \mathbb{N}$ and therefore $d(y, \mathcal{Z}^{(k)}) = 0$ for all $y \in \mathcal{Z}_n^{(k)}$. By definition of the Hausdorff distance $d_H(\mathcal{Z}_n^{(k)}, \mathcal{Z}) = \sup\{d(\mathcal{Z}_n^{(k)}, z) : z \in \mathcal{Z}^{(k)}\}$. Let $y \in \mathcal{Z}_n^{(k)}$ and $z \in \mathcal{Z}^{(k)}$ be arbitrary. They have representations $\sum_{j=0}^{n} a_j g_j^{(k)}$ and $\sum_{j=0}^{\infty} b_j g_j^{(k)}$ respectively for some $a_j, b_j \in [-1, 1]$. Now we have

$$\begin{aligned} d(y,z) &= \left\| \sum_{j=0}^{n} a_j g_j^{(k)} - \sum_{j=0}^{\infty} b_j g_j^{(k)} \right\| = \left\| \sum_{j=n+1}^{\infty} b_j g_j^{(k)} + \sum_{j=0}^{n} (b_j - a_j) g_j^{(k)} \right\| \\ &\leq \sum_{j=n+1}^{\infty} |b_j| \|g_j^{(k)}\| + \sum_{j=0}^{n} |b_j - a_j| \|g_j^{(k)}\| \end{aligned}$$

and consequently

$$\begin{aligned} d(\mathcal{Z}_n^{(k)}, z) &= \inf_{a_0,\dots,a_n} d\left(\sum_{j=0}^{n} a_j g_j^{(k)}, z\right) \\ &\leq \inf_{a_0,\dots,a_n} \left(\sum_{j=n+1}^{\infty} |b_j| \|g_j^{(k)}\| + \sum_{j=0}^{n} |b_j - a_j| \|g_j^{(k)}\|\right) \\ &= \sum_{j=n+1}^{\infty} |b_j| \|g_j^{(k)}\| \end{aligned}$$

Finally,

$$d_H(\mathcal{Z}_n^{(k)}, \mathcal{Z}) = \sup_{b_0,b_1,\dots} d\left(\mathcal{Z}_{\setminus}^{(\|)}, \sum_{j=0}^{\infty} b_j g_j^{(k)}\right) \leq \sup_{b_0,b_1,\dots} \sum_{j=n+1}^{\infty} |b_j| \|g_j^{(k)}\| = \sum_{j=n+1}^{\infty} \|g_j^{(k)}\|$$

Using Lemma 8 we obtain

$$\|g_j^{(k)}\| = \sqrt{\sum_{r=0}^{k-1} |\psi_{j-r}|^2} \leq \sqrt{\sum_{r=0}^{k-1} C^2 \rho^{2j} \rho^{-2r}} = C\rho^{j-k+1} \sqrt{\frac{1-\rho^{2k}}{1-\rho^2}} \leq C\sqrt{k}\rho^{j-k+1}$$

In the end

$$d_H(\mathcal{Z}_n^{(k)}, \mathcal{Z}^{(k)}) \leq \sum_{j=n+1}^{\infty} \|g_j^{(k)}\| \leq C\sqrt{k} \frac{1}{\rho^{k-2}} \rho^n \sum_{j=0}^{\infty} \rho^j \leq C \frac{\sqrt{k}}{1-\rho} \frac{1}{\rho^{k-2}} \rho^n$$

□

6.1.2 Proof of Proposition 2

Lemma 9. *Let* $a = (a_n)_{n\in\mathbb{Z}} \in l^2(\mathbb{Z}) \setminus \{0\}$ *and* $b = (b_n)_{n\in\mathbb{Z}} \in l^1(\mathbb{Z})$ *be sequences. Then it holds* $\|a * b\|_2 = \|a\|_2 \|b\|_1$ *if and only if* $b = 0$ *or* $b = c\delta_r$ *for some* $c \neq 0$ *and* $r \in \mathbb{Z}$.

Proof. First, assume that the equality holds. We define the shifted sequence $\tau_m a$ by $(\tau_m a)[n] = a_{n-m}$. Then the convolution can be seen as an infinitely weighted sum of shifts, that is, $a * b = \sum_m b_m \tau_m a$. We set $x_m = b_m \tau_m a$. By assumption $\{x_m\}$ is

absolutely summable, which implies $\| \sum_{|m|\leq N} x_m - \sum_m x_m \| \to 0$ for $N \to \infty$ and by the reverse triangle inequality $|\| \sum_{|m|\leq N} x_m \|_2 - \| \sum_m x_m \|_2| \leq \| \sum_{|m|\leq N} x_m - \sum_m x_m \|_2$. Therefore,

$$\sum_{m,k} \langle x_m, x_k \rangle = \lim_{N\to\infty} \sum_{|m|\leq N} \sum_{|k|\leq N} \langle x_m, x_k \rangle = \lim_{N\to\infty} \left\| \sum_{|m|\leq N} x_m \right\|_2^2 = \left\| \sum_m x_m \right\|_2^2.$$

By assumption we also have

$$\sum_m \|x_m\|_2 = \|a\|_2 \|b\|_1 = \|a * b\|_2 = \left\| \sum_m x_m \right\|_2$$

Thus,

$$\sum_{m,k} \langle x_m, x_k \rangle = \left\| \sum_m x_m \right\|_2^2 = \sum_{m,k} \|x_m\|_2 \|x_k\|_2$$

By Cauchy Schwarz, $\langle x_m, x_k \rangle \leq \|x_m\|_2 \|x_k\|_2$ and because of the above equality it follows $\langle x_m, x_k \rangle = \|x_m\|_2 \|x_k\|_2$. This means that x_m and x_k are linearly dependent for all m, k with $b_m, b_k \neq 0$ and therefore $x_m = c_{m,k} x_k$ for some $c_{m,k} \neq 0$. It further follows that $|a_{n-m}| = |a_{n-k}|$ for all n. Now $m \neq k$ would lead to $a \notin l^2(\mathbb{Z})$. This shows that b must either be a null sequence or can only have one nonzero entry.

The other direction simply follows by plugging in and checking. □

Proof. Recall that $g_j^{(k)} = (\psi_j, \dots, \psi_{j-k+1})^T$ where $\{\psi_j\}_{j\in\mathbb{Z}}$ is a sequence given by

$$\psi_j = \begin{cases} \sum_{i=1}^p \phi_i \psi_{j-i} & , j \geq 1 \\ 1 & , j = 0 \\ 0 & , j < 0. \end{cases}$$

Let $u = (u_0, \dots, u_{k-1}) \in \mathbb{S}^{k-1}$ and consider the extension of u to a sequence $\{\tilde{u}_l\}_{l\in\mathbb{Z}}$ where $\tilde{u}_l = \mathbf{1}\{0 \leq l \leq k-1\} u_l$. Define the sequence $a = \{a_j\}_{j\in\mathbb{Z}}$ by

$$a_j = \langle g_j^{(k)}, u \rangle = \sum_{l=0}^{k-1} u_l \psi_{j-l} = \sum_{l\in\mathbb{Z}} \tilde{u}_l \psi_{j-l} = (\tilde{u} * \psi)[j].$$

By Definition of $\{\psi_j\}_{j\in\mathbb{Z}}$ we have $(q_\phi * \psi) = \delta_0$ and thus $q_\phi * a = q_\phi * (\tilde{u} * \psi) = (q_\phi * \psi) * \tilde{u} = \delta_0 * \tilde{u} = \tilde{u}$ by properties of the convolution. By Young's convolution inequality (see [26, Theorem 1.2.12])

$$1 = \|u\|_2 = \|q_\phi * a\|_2 \leq \|q_\phi\|_2 \|a\|_1. \tag{6.1}$$

Next, we use the simple formula for the support function of the zonoid $\mathcal{Z}_0^{(k)}$ to obtain

$$w_k(u) = (b-a)h_{\mathcal{Z}_0^{(k)}}(u) = (b-a)\sum_{j=0}^{\infty} |\langle g_j^{(k)}, u\rangle| = (b-a)\|a\|_1.$$

Therefore, we obtain

$$w_k(u) = (b-a)\|a\|_1 \geq \frac{b-a}{\|q_\phi\|_2}.$$

Plugging in the directions from the set in Equation (2.6) in w_k we easily see that the above lower bound is actually attained. It remains to show that there are no other minimizing directions. Inequality (6.1) becomes an equality iff a has exactly one nonzero entry (see Lemma 9 in the Appendix 6). This means $a = c\delta_r$ and it follows $\tilde{u} = c\tau_r q_\phi$, which forces u to have the form $\pm q_r^{(k)}/\|q_\phi\|_2$ for some $r = 0, \ldots, k-(p+1)$.

Now let $k < p+1$. We have shown above that $w_k(u)$ attains the lower bound $(b-a)/\|q_\phi\|_2$ if and only if u has the specific form above. But if $k < p+1$ there are no permitted r that would lead to the specific form. Since w_k is continuous it attains its minimum on $\mathbb{S}^{k-1}$ and therefore $m_k > (b-a)/\|q_\phi\|_2$.

We are left to show $m_{k-1} \geq m_k$ for $k = 2, \ldots, p$. Let $v = (v_0, \ldots, v_{k-2}) \in \mathbb{S}^{k-2}$ and $u = (v_0, \ldots, v_{k-2}, 0) \in \mathbb{S}^{k-1}$ and thus

$$w_k(u) = (b-a)\sum_j \left|\sum_{l=0}^{k-1} u_l \psi_{j-l}\right| = (b-a)\sum_j \left|\sum_{l=0}^{k-2} v_l \psi_{j-l}\right| = w_{k-1}(v).$$

Let v be a minimizing direction. Then $m_{k-1} = w_{k-1}(v) = w_k(u) \geq \min_{u\in\mathbb{S}^{k-1}} w_k(u) = m_k$. □

6.1.3 Proof of Lemma 2

Lemma 10. *Let A, B, C be subsets of a metric space with $A \subseteq B \subseteq C$. Then $d_H(B, C) \leq d_H(A, C)$.*

Proof. Since $A \subseteq B$, we have $\inf_{b \in B} d(b, c) \leq \inf_{a \in A} d(a, c)$. Therefore,

$$d_H(B, C) = \sup_{c \in C} \inf_{b \in B} d(b, c) \leq \sup_{c \in C} \inf_{a \in A} d(a, c) = d_H(A, C).$$

□

Proof. Let $\epsilon > 0$ be arbitrary. Let $\Omega_0 = \{\omega \in \Omega : \overline{\{Y_t(\omega) : t \geq 1\}} = \mathcal{Z}\}$. By assumption, $\mathbb{P}(\Omega_0) = 1$. Fix $\omega \in \Omega_0$. In what follows, we abbreviate $Y_t = Y_t(\omega)$ and $\mathcal{P}_T = \{Y_1(\omega), \ldots, Y_T(\omega)\}$. Because $\{Y_t : t \geq 1\}$ is dense in $\mathcal{Z}$, it holds $\mathcal{Z} \subseteq \bigcup_{t \geq 1} B(Y_t, \epsilon)$ and since $\mathcal{Z}$ is compact, there exists a finite sub-cover $\mathcal{Z} \subseteq \bigcup_{j=1}^{m(\epsilon)} B(Y_{t_j(\epsilon)}, \epsilon)$. Set $T_\epsilon = \max\{t_1(\epsilon), \ldots, t_{m(\epsilon)}(\epsilon)\}$. Now, let $z \in \mathcal{Z}$ be arbitrary. It follows that $d(z, \mathcal{P}_T) = \inf\{\|z - Y_t\| : t = 1, \ldots, T\} \leq \epsilon$ for all $T \geq T_\epsilon$. Thus, $\sup_{z \in \mathcal{Z}} d(z, \mathcal{P}_T) \leq \epsilon$, since z was arbitrary. Since $\mathcal{P}_T \subseteq \mathcal{Z}$, we also have $\sup_{t=1,\ldots,T} d(Y_t, \mathcal{Z}) = 0$. By definition of the Hausdorff distance, we obtain $d_H(\mathcal{P}_T, \mathcal{Z}) = \sup_{z \in \mathcal{Z}} d(z, \mathcal{P}_T) \leq \epsilon$ and therefore $d_H(\mathcal{P}_T, \mathcal{Z}) \to 0$ for $T \to \infty$, since $\epsilon > 0$ was arbitrary. By Lemma 10 we also get $d_H(\mathrm{conv}(\mathcal{P}_T), \mathcal{Z}) \to 0$ for $T \to \infty$. Since we worked on Ω_0, this proves almost sure convergence. □

6.1.4 Proof of Lemma 3

Proof. It is well known that the family $\mathcal{B}$ of open balls with rational centers $q \in \mathbb{Q}^k$ and radii $r \in \mathbb{Q}_{>0}$ forms a countable base for $\mathbb{R}^k$. That is, $\mathcal{B}$ is countable and for any open set $U \subseteq \mathbb{R}^k$ and every $x \in U$ there exists $B \in \mathcal{B}$ such that $x \in B \subseteq U$. Define $\mathcal{B}_\mathcal{Z} = \{B \in \mathcal{B} : B \cap \mathcal{Z} \neq \emptyset\}$. Let $B \in \mathcal{B}_\mathcal{Z}$ and choose some $z \in B$. Because B is open, there exists $\delta > 0$ with $B(z, \delta) \subseteq B$. Since $z \in \mathcal{Z} = \mathrm{supp}(Y_t)$, by the definition of the support, we have $\mathbb{P}(Y_t \in B) \geq \mathbb{P}(Y_t \in B(z, \delta)) > 0$. By Birkhoff's ergodic theorem

$$\frac{1}{T} \sum_{t=1}^{T} 1\{Y_t \in B\} \overset{T \to \infty}{\to} \mathbb{P}(Y_t \in B) > 0$$

This implies that $Y_t \in B$ for infinitely many $t \geq 1$ almost surely. In particular, there exists $t \geq 1$ such that $Y_t \in B$. Let $z \in \mathcal{Z}$ and $\epsilon > 0$ be arbitrary. There exists $B \in \mathcal{B}$ such that $z \in B \subseteq B(z, \epsilon)$. From above we also know that there exists $t \geq 1$ such that $Y_t \in B$. Thus $\|Y_t - z\| \leq \epsilon$ □

6.1.5 Proof of Proposition 3

Proof. Let $\{U_t\} \overset{\text{iid}}{\sim} \text{Beta}(\alpha, \beta)$, $Z_t = a + (b-a)U_t$ and write $X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$. Set $c = (b+a)/2$ and $d = (b-a)/2$. Also set $\kappa := \min\{\alpha, \beta\}$. Then, $X_t = \sum_{\psi_j \geq 0} \psi_j Z_{t-j} + \sum_{\psi_j < 0} \psi_j Z_{t-j} \leq b \sum_{\psi_j \geq 0} \psi_j + a \sum_{\psi_j < 0} \psi_j = c \sum_{j=0}^{\infty} \psi_j + d \sum_{j=0}^{\infty} |\psi_j|$ and similarly $X_t \geq c \sum_{j=0}^{\infty} \psi_j - d \sum_{j=0}^{\infty} |\psi_j|$. Thus, $\text{supp}(X_t) = [G - H, G + H]$, where $G := c \sum_{j=0}^{\infty} \psi_j$ and $H := d \sum_{j=0}^{\infty} |\psi_j|$. Consequently,

$$G + H - X_t = \sum_{j=0}^{\infty} |\psi_j| \left(\underbrace{d - \text{sgn}(\psi_j)(Z_{t-j} - c)}_{=: D_{t-j}} \right),$$

where $D_{t-j} = (b-a)(\frac{1}{2} - \text{sgn}(\psi_j)(U_{t-j} - \frac{1}{2})) \geq 0$. There exist infinitely many nonzero ψ_j. (Assume, for contradiction, that only finitely many ψ_j are nonzero. Then, the set $\{j \in \mathbb{N}_0 : \psi_j \neq 0\}$ is nonempty ($\psi_0 = 1$) and finite. Let J be its maximum. This would mean that $1/\phi(z) = \sum_{j=0}^{J} \psi_j z^j$ is a polynomial of order J. By assumption, the polynomial ϕ has degree p. But now we have $0 = \deg(\phi \cdot (1/\phi)) = \deg(\phi) + \deg(1/\phi) = p + J$. This would force $p = 0$ and $J = 0$, which contradicts our assumption that $p \geq 1$.) Thus, for each $q > 0$ no matter how large we find distinct indices $j_1, \ldots, j_{M(q)}$ where $\kappa M(q) > q$ such that $\psi_{j_m} \neq 0$ for all $m = 1, \ldots, M(q)$. Let $0 < \epsilon < 2^{-1}(b-a) \min_m |\psi_{j_m}|$. Then $G + H - X_t \leq \epsilon$ implies $D_{t-j_m} \leq \epsilon |\psi_{j_m}|^{-1}$ for all $m = 1, \ldots, M(q)$. Now, it follows

$$\begin{aligned} \mathbb{P}(G + H - X_t \leq \epsilon) &\leq \mathbb{P}\left(\bigcap_{m=1}^{M(q)} \left\{ D_{t-j_m} \leq \epsilon |\psi_{j_m}|^{-1} \right\} \right) \overset{(*)}{=} \prod_{m=1}^{M(q)} \mathbb{P}(D_{t-j_m} \leq \epsilon |\psi_{j_m}|^{-1}) \\ &\overset{(**)}{\leq} C_q \epsilon^{\kappa M(q)}, \end{aligned}$$

where in step $(*)$ we have used the independence of the innovations and in step $(**)$ the following reasoning.

For $U \sim \text{Beta}(\alpha, \beta)$ and $0 < s \leq 1/2$ we have $\mathbb{P}(U \leq s) \leq C_{\alpha,\beta} s^{\alpha}$ for some constant $C_{\alpha,\beta} > 0$. It is also well known that $1 - U \sim \text{Beta}(\beta, \alpha)$. By definition of D_{t-j_m} we have $D_{t-j_m}/(b-a) = 1 - U_{t-j_m}$ for $\psi_{j_m} > 0$ and $D_{t-j_m} = U_{t-j_m}$ for $\psi_{j_m} < 0$ and therefore $\mathbb{P}(D_{t-j_m}/(b-a) \leq s) \leq \max\{C_{\alpha,\beta}, C_{\beta,\alpha}\} s^{\min\{\alpha,\beta\}}$. Set $D_{\alpha,\beta} = \max\{C_{\alpha,\beta}, C_{\beta,\alpha}\}$ and $\kappa = \min\{\alpha, \beta\}$. Applying these considerations to our setting yields

$$\begin{aligned}\mathbb{P}(D_{t-j_m} \leq \epsilon |\psi_{j_m}|^{-1}) &= \mathbb{P}(D_{t-j_m}/(b-a) \leq \epsilon (b-a)^{-1} |\psi_{j_m}|^{-1}) \\ &\leq D_{\alpha,\beta} \left(\frac{\epsilon}{(b-a)|\psi_{j_m}|} \right)^{\kappa}.\end{aligned}$$

Because $G + H - X_t \leq \epsilon \iff Y_t^{(k)} \in C_{\mathcal{Z}^{(k)}}(e_1, \epsilon)$, we conclude

$$\frac{\mathbb{P}(Y_t^{(k)} \in C_{\mathcal{Z}^{(k)}}(e_1, \epsilon))}{\epsilon^q} = \frac{\mathbb{P}(G + H - X_t \leq \epsilon)}{\epsilon^q} \leq C_q \epsilon^{\kappa M(q) - q} \overset{\epsilon \to 0}{\to} 0.$$

□

6.1.6 Proof of Proposition 4

Proof. Let $u, v \in S^{k-1}$. Write $\hat{w}_k(u) = \max_{s,t} \langle Y_s^{(k)} - Y_t^{(k)}, u \rangle$ and we obtain

$$\begin{aligned}\hat{w}_k(u) - \hat{w}_k(v) &= \max_{s,t} \langle Y_s^{(k)} - Y_t^{(k)}, u \rangle - \max_{s,t} \langle Y_s^{(k)} - Y_t^{(k)}, v \rangle \\ &\leq \max_{s,t} \left(\langle Y_s^{(k)} - Y_t^{(k)}, u \rangle - \langle Y_s^{(k)} - Y_t^{(k)}, v \rangle \right) \\ &= \max_{s,t} \langle Y_s^{(k)} - Y_t^{(k)}, u - v \rangle \\ &\leq \left(\max_{s,t} \| Y_s^{(k)} - Y_t^{(k)} \| \right) \| u - v \|\end{aligned}$$

for all $u, v \in \mathbb{R}^k$.

Convexity of $\hat{w}_k$ follows from the convexity of linear functions and the pointwise maximum of convex functions. Piecewise linearity also follows directly from the standard fact that the pointwise maximum of linear functions is piecewise linear. Differentiability almost everywhere follows directly from Rademacher's theorem (Lipschitz-continuous functions are differentiable almost everywhere). The set of non-differentiable

points is just the set of all points u where two distinct linear functions $\langle Y_s^{(k)} - Y_t^{(k)}, u\rangle$ and $\langle Y_q^{(k)} - Y_p^{(k)}, u\rangle$ coincide. The gradient of $\hat{w}_k$ in all directions $u \notin \mathcal{N}$ is obtained by straightforward differentiation.

By Weierstrass' extreme value theorem, $\hat{w}_k$ has a global minimum on S^{k-1}. The set of minimizers

$$\mathcal{X}_k = \arg\min\{\hat{w}_k(u) : u \in \mathbb{S}^{k-1}\} \tag{6.2}$$

is not empty. We also note that $\mathcal{X}_k$ is compact, again due to the continuity of $\hat{w}_k$ and the compactness of $\mathbb{S}^{k-1}$.

Denote by $\hat{m}_{k,T}$ the minimum of $\hat{w}_k$ on $\mathbb{S}^{k-1}$. We have $\hat{m}_{k,T} > 0$ if and only if the affine hull of $\mathcal{P}_k$ is the whole space $\mathbb{R}^k$. Let $\mathrm{aff}(\mathcal{P}_{k,T}) = \mathbb{R}^k$. Assume for a contradiction that $\hat{m}_{k,T} = 0$. Then there would exist a direction $u \in \mathbb{S}^{k-1}$ with $\langle Y_t^{(k)}, u\rangle = c_{k,u}$ constant for all observations t. Hence, all points $Y_t^{(k)}$ lie inside the affine hyperplane $H_u = \{x \in \mathbb{R}^k : \langle u, x\rangle = c_{k,u}\}$ and therefore $\mathrm{aff}(\mathcal{P}_{k,T}) \subseteq H_u$. But H_u is not the whole space, leading to a contradiction. If $\mathrm{aff}(\mathcal{P}_{k,T}) \neq \mathbb{R}^k$, then $V := \{Y_t^{(k)} - Y_K^{(k)} : t = K, \ldots, T\}$ is a proper subspace since $Y_K^{(k)} + V = \mathrm{aff}(\mathcal{P}_{k,T})$. This means that we can find a direction $u' \in V^\perp$, $\|u'\| = 1$ and this direction fulfills $\langle Y_t^{(k)} - Y_K^{(k)}, u\rangle = 0$ for all $t = K, \ldots, T$, implying that $\langle Y_t^{(k)}, u\rangle$ is constant for all t and thus $\hat{m}_{k,T} = 0$.

Take $Y_K^{(k)}, \ldots, Y_{K+k}^{(k)}$ and let the matrix M be defined by having columns $Y_{K+1}^{(k)} - Y_K^{(k)}, \ldots, Y_{K+k}^{(k)} - Y_K^{(k)}$. The determinant $D = \det M$ is a nonzero polynomial in the observations $W = (X_{K-k+1}, \ldots, X_K, X_{K+1}, \ldots, X_{K+k})$, implying that $\{x : D(x) = 0\}$ has Lebesgue measure zero; see [15]. Note that we need at least $T \geq K + k$ to form this vector W. The random vector W has a joint density if the innovations have a density. Therefore, it follows $\mathbb{P}(D(W) \neq 0) = 1$, that is, the determinant of M is almost surely nonzero. This immediately implies that the differences $Y_{K+1}^{(k)} - Y_K^{(k)}, \ldots, Y_{K+k}^{(k)} - Y_K^{(k)}$ span the whole space $\mathbb{R}^k$ and thus $\mathrm{aff}(\mathcal{P}_{k,T}) = \mathbb{R}^k$. Combining the arguments, we find that $\hat{m}_{k,T} > 0$ is almost surely the case given that $T \geq K + k$. □

6.1.7 Proof of Lemma 4

Proof. The function $\hat{w}_k$ is a pointwise maximum over finitely many linear functions. In the following, we write $A = \{Y_s^{(k)} - Y_t^{(k)} : s, t\}$ and $f(u) = \max\{\langle a, u\rangle : a \in A\}$. Also, let $m = \min\{f(u) : u \in \mathbb{S}^{k-1}\}$ and $\mathcal{X} = \{u \in \mathbb{S}^{k-1} : f(u) = m\}$. Let $u^* \in \mathcal{X}$ and $T_{u^*}\mathbb{S}^{k-1}$ be the tangent space at u^*. Furthermore, let $A_* = \{a \in A : \langle a, u^*\rangle = m\}$ be the set of active vectors. Define $g(d) = \max\{\langle a, d\rangle : a \in A_*\}$ for all unit tangent directions $d \in T_{u^*}\mathbb{S}^{k-1} \cap \mathbb{S}^{k-1} =: S_{u^*}\mathbb{S}^{k-1}$.

Let $t > 0$. We can see that

$$f(u^* + td) = \max_{a\in A}\langle a, u^* + td\rangle = \underbrace{\max_{a\in A}\langle a, u^*\rangle}_{=m} + t\max_{a\in A}\langle a, d\rangle.$$

We want to assert that $f(u^* + td) = m + tg(d)$ for sufficiently small $t > 0$. In general, $\max\{\langle a, d\rangle : a \in A\} \geq \max\{\langle a, d\rangle : a \in A_*\} = g(d)$ since $A_* \subseteq A$. In case of equality there is nothing to show. Therefore we assume $\max\{\langle a, d\rangle : a \in A\} > g(d)$. If this is the case, there must exist an inactive vector $b \in A \setminus A_*$ with $\langle b, d\rangle > g(d)$. We now write

$$\langle b, u^* + td\rangle = \langle b, u^*\rangle + t\langle b, d\rangle = m - (\underbrace{m - \langle b, u^*\rangle}_{=:\delta_b>0}) + t\langle b, d\rangle.$$

We see that the difference

$$m + tg(d) - \langle b, u^* + td\rangle = \delta_b - t\left(\langle b, d\rangle - g(d)\right)$$

stays positive for $t < \delta_b/(\langle b, d\rangle - g(d)) =: r_b > 0$. Set $r(u^*)$ to be the minimum over all $b \in A \setminus A_*$ with $\langle b, d\rangle > g(d)$. It follows that

$$f(u^* + td) = \max_{a\in A}\langle a, u^* + td\rangle = \max_{a\in A_*}\langle a, u^* + td\rangle = m + tg(d)$$

for all $t < r(u^*)$ and $d \in S_{u^*}\mathbb{S}^{k-1}$ and therefore also

$$f\left(\frac{u^* + td}{\sqrt{1+t^2}}\right) = \frac{m + tg(d)}{\sqrt{1+t^2}} \tag{6.3}$$

for all $d \in S_{u^*}\mathbb{S}^{k-1}$ and all $0 < t < r(u^*)$.

From this it follows that $g(d) > 0$ and therefore also $c^* = \min\{g(d) : d \in S_{u^*}\mathbb{S}^{k-1}\} > 0$ since g is a continuous function and the unit tangent sphere $S_{u^*}\mathbb{S}^{k-1}$ is compact. Now write $u_t(d) = (u^* + td)/\sqrt{1+t^2}$. For $t \in (0, r(u^*))$ we have

$$\begin{aligned} f(u_t(d)) - m &\overset{(6.3)}{=} \frac{m + tg(d)}{\sqrt{1+t^2}} - m \geq \frac{m + tc^*}{\sqrt{1+t^2}} - m \\ &= tc^* + (m + tc^*)\left(\frac{1}{\sqrt{1+t^2}} - 1\right) \\ &= tc^* - t^2 \underbrace{\frac{m + tc^*}{\sqrt{1+t^2}(1+\sqrt{1+t^2})}}_{\to m/2 \text{ for } t \downarrow 0}. \end{aligned}$$

Denote the remainder term by r_t. We have $|r_t|t^{-2} \to m/2$ for $t \downarrow 0$ and in particular there exist constants $C > 0$ and $r'_* > 0$ such that $|r_t| \leq Ct^2$ for all $t \in (0, r'_*)$.

We continue working on $t < \min\{r(u^*), r'_*, \sqrt{2}, c^*/(2C)\}$. Because $\|u_t(d) - u^*\|^2 = 2 - 2/\sqrt{1+t^2} \leq \min\{t^2, 2\} = t^2$ it follows

$$f(u_t(d)) - m \geq tc^* + r(t) \geq tc^* - Ct^2 = t(c^* - Ct) \geq \frac{c^*}{2}t \geq \frac{c^*}{2}\|u_t(d) - u^*\|, \tag{6.4}$$

which lets us see that $\mathcal{X}$ must be finite.

To see this, assume that $u_t(d) \in \mathcal{X}$ is another minimizer. It would follow from Equation (6.3) that $f(u_t(d)) - m \geq (c^*/2)\|u_t(d) - u^*\| > 0$, which is a contradiction to $u_t(d) \in \mathcal{X}$. Thus for each $u^* \in \mathcal{X}$ there exists a radius $s(u^*) = \min\{\dots\}$ such that $B(u^*, s(u^*)) \cap \mathcal{X} = \{u^*\}$. Obviously $\{B(u^*, s(u^*)) : u^* \in \mathcal{X}\}$ covers $\mathcal{X}$. Since $\mathcal{X}$ is compact, there exists a finite subcover. It follows that $\mathcal{X}$ must be finite and we denote by $u_1^*, \dots, u_q^* \in \mathcal{X}$ the finitely many minimizing directions. We can thus choose $\alpha = \min_{1 \leq i \leq q} c_i^*/2 > 0$ and $\rho = \min_{1 \leq i \leq q} s(u_i^*)$. For all $u \in \mathbb{S}^{k-1}$, $\operatorname{dist}(u, \mathcal{X}) < \rho$ now follows

$$f(u) - m \geq \|u - u_i^*\|\frac{c_i^*}{2} \geq \alpha \operatorname{dist}(u, \mathcal{X})$$

□

6.2 Proofs Chapter 3

6.2.1 Proof of Lemma 5

Proof. Since the points x_j are distinct, all entries of the matrix D are positive and by Perron-Frobenius, its Perron vector v has positive entries and is unique up to scaling. We fix v so that $v^T\mathbf{1} = 1$. We interpret v as a discrete probability distribution on $\{x_1, \dots, x_N\}$. Let X be a discrete random variable with $\mathbb{P}(X = x_j) = v_j$. Write the expected value $\mathbb{E}[X] = \mu = \sum_{j=1}^N v_j x_j$ and the variance $\mathrm{Var}[X] = \sigma^2 = \sum_{j=1}^N v_j \|x_j - \mu\|^2$. Now we have

$$(Dv)_i = \sum_{j=1}^N D_{ij} v_j = \sum_{j=1}^N \|x_i - x_j\|^2 v_j = \mathbb{E}[\|X - x_i\|^2] \overset{(*)}{=} \|\mu - x_i\|^2 + \sigma^2, \quad (6.5)$$

where at $(*)$ we have used the bias-variance decomposition. Since v is the Perron vector of D, the left hand side of Equation (6.5) is equal to ρv_i with $\rho > 0$ being the unique maximum eigenvalue of D.

Next, we write

$$\begin{aligned}
\mu &= \sum_{j=1}^N v_j x_j = v_1 x_1 + v_2 x_2 + \cdots + v_N x_N \\
&= x_1 - (v_2 + \cdots + v_N) x_1 \\
&+ v_2 x_2 + v_3 x_2 + \cdots + v_N x_2 - (v_3 x_2 + \cdots + v_N x_2) \\
&\vdots \\
&+ v_{N-2} x_{N-2} + v_{N-1} x_{N-2} + v_N x_{N-2} - (v_{N-1} x_{N-2} + v_N x_{N-2}) \\
&+ v_{N-1} x_{N-1} + v_N x_{N-1} - v_N x_{N-1} \\
&+ v_N x_N \\
&= x_1 + (v_2 + \cdots + v_N)(x_2 - x_1) + \cdots + (v_{N-1} + v_N)(x_{N-1} - x_{N-2}) + v_N (x_N - x_{N-1}) \\
&= x_1 + \sum_{k=1}^{N-1} \sum_{j=k+1}^{N} v_j (x_{k+1} - x_k) = x_1 + \sum_{k=1}^{N-1} a_k \delta_k,
\end{aligned}$$

where $a_k = \sum_{j=k+1}^N v_j$ and $\delta_k = x_{k+1} - x_k$. Note that because of the positivity of the Perron vector, it holds $1 > a_1 > a_2 > \cdots > a_{N-1} = v_N > 0$.

As a next step, write

$$x_i - \mu = x_i - x_1 - (\mu - x_1) = \sum_{k=1}^{i-1} \delta_k - \sum_{k=1}^{N-1} a_k \delta_k = \sum_{k=1}^{i-1} \delta_k - \sum_{k=1}^{i-1} a_k \delta_k - \sum_{k=i}^{N-1} a_k \delta_k$$
$$= \underbrace{\sum_{k=1}^{i-1} (1 - a_k)\delta_k}_{=P_i} - \underbrace{\sum_{k=i}^{N-1} a_k \delta_k}_{=Q_i}$$

Consequently,

$$\begin{aligned} \|x_i - \mu\|^2 = \|P_i - Q_i\|^2 &= \|P_i\|^2 + \|Q_i\|^2 - 2\langle P_i, Q_i \rangle \\ &\leq \|P_i\|^2 + \|Q_i\|^2 + 2\langle P_i, Q_i \rangle = \|P_i + Q_i\|^2 \end{aligned}$$

for all $i = 2, \ldots, N-1$, where we have used

$$\langle P_i, Q_i \rangle = \sum_{1 \leq k \leq i-1} \sum_{i \leq l \leq N-1} \underbrace{(1 - a_k)}_{>0} \underbrace{a_l}_{>0} \underbrace{\langle \delta_k, \delta_l \rangle}_{\geq 0} > 0.$$

In particular, this is the step where we use the acute chain condition (3.9).

Now we see that

$$\|P_i + Q_i\| \leq \begin{cases} \|\sum_{k=1}^{N-1} a_k \delta_k\| = \|\mu - x_1\| & , a_{i-1} \geq \frac{1}{2} \\ \|\sum_{k=1}^{N-1} (1 - a_k)\delta_k\| = \|x_N - \mu\| & , a_{i-1} < \frac{1}{2}. \end{cases} \tag{6.6}$$

Finally, by combining Equations (6.5) and (6.6) we arrive at

$$\begin{aligned} \rho v_i - \sigma^2 = \|\mu - x_i\|^2 &\leq \|P_i + Q_i\|^2 \leq \max\{\|\mu - x_1\|, \|x_N - \mu\|\} \\ &= \max\{\rho v_1 - \sigma^2, \rho v_N - \sigma^2\} \end{aligned}$$

and therefore $v_i \leq \max\{v_1, v_N\}$ for all interior coordinates $i = 2, \ldots, N-1$. □

6.2.2 Proof of Lemma 7

Proof. Let $h > 0$. For every $x \in \mathbb{R}$ we have

$$\begin{aligned}\lim_{\tau\to 0}\frac{1}{\tau}(\hat{f}_{h+\tau}(x)-\hat{f}_h(x)) &= \frac{1}{n}\sum_{j=1}^{n}\lim_{\tau\to 0}\frac{K_{h+\tau}(x-X_j)-K_h(x-X_j)}{\tau}\\ &= \frac{1}{n}\sum_{j=1}^{n}\partial_h K_h(x-X_j)\\ &= -\frac{1}{nh^2}\sum_{j=1}^{n}\left(K\left(\frac{x-X_j}{h}\right)+\frac{x-X_j}{h}K'\left(\frac{x-X_j}{h}\right)\right)\\ &=: \partial_h\hat{f}_h(x).\end{aligned}$$

Let $0 < |\tau| < h/2$. Using the mean value inequality, we bound

$$\begin{aligned}\left|\frac{K_{h+\tau}(u)-K_h(u)}{\tau}\right| &\le \sup_{s\in(\min\{h,h+\tau\},\max\{h,h+\tau\})}|\partial_s K_s(u)|\\ &\le \sup_{s\in(h/2,3h/2)}\left|\frac{1}{s^2}L\left(\frac{u}{s}\right)\right|\\ &\le \frac{4}{h^2}\underbrace{\sup_{a\in(2/3,2)}|L(au/h)|}_{=:G(u/h)}\end{aligned}$$

and

$$\begin{aligned}\left|\frac{\hat{f}_{h+\tau}(x)-\hat{f}_h(x)}{\tau}\right| &\le \frac{1}{n}\sum_{j=1}^{n}\left|\frac{K_{h+\tau}(x-X_j)-K_h(x-X_j)}{\tau}\right|\\ &\le \frac{4}{nh^2}\sum_{j=1}^{n}G\left(\frac{x-X_j}{h}\right) =: M(x),\end{aligned}$$

where $M \in L^2(\mathbb{R})$ because by the Cauchy-Schwarz inequality $\|M\|_2^2 \le 16h^{-3}\|G\|_2^2$ and by the decay assumption on L we get $\|G\|_2^2 \le 3C^2/(2\alpha-1) < +\infty$. By the dominated convergence theorem

$$\left\|\frac{\hat{f}_{h+\tau}-\hat{f}_h}{\tau}-\partial_h\hat{f}_h\right\|_2 \to 0, \quad \tau\to 0. \tag{6.7}$$

Set $d\Psi(h)[\tau] := \tau\partial_h\hat{f}_h$. Now it holds

$$\hat{f}_{h+\tau} - \hat{f}_h = \tau\partial_h\hat{f}_h + \underbrace{\left(\hat{f}_{h+\tau} - \hat{f}_h - \tau\partial_h\hat{f}_h\right)}_{=:r_h(\tau)},$$

where the remainder $r_h(\tau)$ fulfills $\|r_h(\tau)\|_2/|\tau| \to 0$ as $\tau \to 0$ as established in Equation (6.7).

The continuity of the operator $d\Psi : (0,\infty) \to \mathcal{L}(\mathbb{R}, L^2(\mathbb{R}))$ can be established by defining the dilation operator T_h and the shift operator g_a and computing

$$\begin{aligned}
\|\partial_h\hat{f}_{h_1} - \partial\hat{f}_{h_2}\|_2 &= \|\frac{1}{nh_2}\sum_j \tau_{X_j}T_{h_2}L - \frac{1}{nh_1}\sum_j \tau_{X_j}T_{h_1}L\|_2 \\
&\leq \|\frac{1}{h_2}T_{h_2}L - \frac{1}{h_1}T_{h_1}L\|_2 \leq \|\frac{1}{h_2}T_{h_2}L - \frac{1}{h_2}T_{h_1}L\|_2 + \|\frac{1}{h_2}T_{h_1}L - \frac{1}{h_1}T_{h_1}L\|_2 \\
&= \frac{1}{h_2}\|T_{h_2}L - T_{h_1}L\|_2 + |\frac{1}{h_2} - \frac{1}{h_1}|\|T_{h_1}L\|_2
\end{aligned}$$

Both terms on the right hand side vanish for $h_1 \to h_2$. (A well known fact (see [20]) is that $(U(s))_{s\in\mathbb{R}}$, $U(s) : L^2 \to L^2$, $U(s)f(x) := e^{s/2}f(e^s x)$ is a strongly continuous unitary group. In particular, $\|U(s)f - U(s_0)f\|_2 \to 0$ as $s \to s_0$ for all $f \in L^2$. We now note that we can write $T_h f = h^{-1/2}U(-\log h)f$ and thus $\|T_h f - T_{h_0}f\|_2 \to 0$ as $h \to h_0$.)

Since $\|d\Psi(h) - d\Psi(s)\| = \sup_{|\tau|\leq 1}\|\tau\partial_h\hat{f}_h - \tau\partial_h\hat{f}_s\|_2 = \|\partial_h\hat{f}_h - \partial_h\hat{f}_s\|_2$, the continuity of $d\Psi$ is established.

Fix a closed nonempty interval $J := [a,b] \subseteq (0,\infty)$. Applying the bound from above:

$$\begin{aligned}
\|r_h(\tau)\|_2 &\leq |\tau| \sup_{t\in[0,1]} \|d\Psi(h+\tau t) - d\Psi(h)\|_{\mathsf{op}} \\
&\leq |\tau| \underbrace{\sup_{h,s\in J, |h-s|\leq|\tau|} \|d\Psi(s) - d\Psi(h)\|_{\mathsf{op}}}_{=:\mu_J(|\tau|)}
\end{aligned}$$

for all $h \in [a,b]$ and $\tau \in \mathbb{R}$ with $h + \tau \in [a,b]$. $\mu_J(s) = \sup_{x,y\in J, |x-y|\leq s}\|d\Psi(x) - d\Psi(y)\|_{\mathsf{op}}$ is known in the literature as the modulus of continuity. Because $d\Psi$ is continuous, $d\Psi : J \to \mathcal{L}(\mathbb{R}, L^2(\mathbb{R}))$ is uniformly continuous, and in this case it

is well known that $\mu_J(s) \to 0$ for $s \downarrow 0$. Also note that $\|d\Psi(s) - d\Psi(h)\|_{\mathsf{op}} = \|\partial_h \hat{f}_h|_{h=s} - \partial_h \hat{f}_h|_{h=h}\|_2$. Since we have established the continuity of $d\Psi$ before, the proof is finished. □

6.2.3 Proof of Proposition 8

Proof. Set $\epsilon_N := \sqrt{\lambda_N}\Delta_{r,N}$ and $t_{N,j} := \epsilon_N(j - i_N)$. Note that $\epsilon_N = t_{N,j+1} - t_{N,j}$ for all $j = 1, \dots, N-1$. Fix $A > 0$ and set $I_N(A) := \{j = 1, \dots, N : |t_{N,j}| \le A\}$.

We have $\#I_N(A) \le 2A\epsilon_N^{-1} + 1$ because $\#I_N(A) \le \#\{j \in \mathbb{Z} : -A\epsilon_N^{-1} + i_N \le j \le A\epsilon_N^{-1} + i_N\} = \lfloor A\epsilon_N^{-1} \rfloor - \lceil -A\epsilon_N^{-1} \rceil + 1 = 2\lfloor A\epsilon_N^{-1} \rfloor + 1 \le 2A\epsilon_N^{-1} + 1$.

We also set $\delta_{N,j} := (j - i_N)\Delta_{r,N} = \psi_r(h_j) - \psi_r(h_{i_N})$.

Split the row sum:

$$R_{i_N}^{(N)} = \sum_{j \in I_N(A)} S_{i_N j}^{(N)} + \underbrace{\sum_{j \notin I_N(A)} S_{i_N j}^{(N)}}_{=:T_N(A)}.$$

We have $D_{i_N j} \ge c(\delta_{N,j})^2$ for some constant $c > 0$. To see this, first write

$$\left| e^{-\frac{h_j^2\omega^2}{2}} - e^{-\frac{h_{i_N}^2\omega^2}{2}} \right| = \int_{\min\{h_j, h_{i_N}\}}^{\max\{h_j, h_{i_N}\}} t\omega^2 e^{-\frac{t^2\omega^2}{2}}\, dt \ge |h_{i_N} - h_j| h_{\min} \omega^2 e^{-h_{\max}^2\omega^2/2}$$

and, furthermore

$$\begin{aligned} D_{i_N j} &= \frac{1}{4\pi} \int |P_n(\omega)|^2 \left| e^{-\frac{h_j^2\omega^2}{2}} - e^{-\frac{h_{i_N}^2\omega^2}{2}} \right|^2 d\omega \\ &\ge |h_{i_N} - h_j|^2 \frac{h_{\min}^2}{4\pi} \int \underbrace{|P_n(\omega)|^2 \omega^4 e^{-h_{\max}^2\omega^2}}_{=:g(\omega)}\, d\omega. \end{aligned}$$

Now, $|\delta_{N,j}| = |h_j^r - h_{i_N}^r|/r = \int_{\min\{h_j, h_{i_N}\}}^{\max\{h_j, h_{i_N}\}} s^{r-1}\, ds \le h_{\min}^{r-1}|h_j - h_{i_N}|$. Thus, it follows $D_{i_N j} \ge c(\delta_{N,j})^2$ with positive constant $c = h_{\min}^{4-2r}(4\pi)^{-1} \int g(\omega)\, d\omega$. It follows $S_{i_N j}^{(N)} = e^{-\lambda_N D_{i_N j}} \le e^{-c\lambda_N(\delta_{N,j})^2} = e^{-ct_{N,j}^2}$. Note that the constant is, in fact, positive due to the continuity of the integrand $g(\omega)$ and because $g(\omega) \neq 0$ on a disk $\{\omega : |\omega| \le \delta\} \setminus \{0\}$ for some $\delta > 0$.

Continuing

$$\begin{aligned}\epsilon_N T_N(A) &\le \epsilon_N \sum_{1\le j\le N:\epsilon_N|j-i_N|>A} e^{-ct_{N,j}^2} = \epsilon_N \sum_{1-i_N\le k\le N-i_N:|k|>\epsilon_N^{-1}A} e^{-c\epsilon_N^2 k^2} \\ &\le \epsilon_N \sum_{k\in\mathbb{Z}:|k|>\epsilon_N^{-1}A} e^{-c\epsilon_N^2 k^2} = 2\sum_{k=\lfloor\epsilon_N^{-1}A\rfloor+1}^{\infty} \epsilon_N e^{-c\epsilon_N^2 k^2} \\ &\le 2\sum_{k=\lfloor\epsilon_N^{-1}A\rfloor+1}^{\infty} \int_{(k-1)\epsilon_N}^{k\epsilon_N} e^{-ct^2}\,dt \le 2\int_{A-\epsilon_N}^{\infty} e^{-ct^2}\,dt\end{aligned}$$

We see that $\epsilon_N T_N(A)$ is asymptotically bounded and therefore

$$\limsup_{N\to\infty} \epsilon_N T_N(A) \le 2\int_A^\infty e^{-ct^2}\,dt \le \frac{2}{A}\int_A^\infty te^{-ct^2}\,dt = \frac{e^{-cA^2}}{cA} \to 0, \quad A\to\infty.$$

Now, let $j \in I_N(A)$, that is, $j - i_N \le |j - i_N| \le \epsilon_N^{-1}A = \mathbf{O}(N/\sqrt{\lambda_N}) = \mathbf{o}(N)$. Thus, by Lemma 6 we have $\tau_{N,j} := h_j - h_{i_N} = h_{i_N}^{1-r}\delta_{N,j} + \mathbf{O}(\delta_{N,j}^2)$ and by Lemma 7 follows

$$\begin{aligned}\hat f_{h_j} - \hat f_{h_{i_N}} &= \hat f_{h_{i_N}+\tau_{N,j}} - \hat f_{h_{i_N}} = \tau_{N,j}\partial_h \hat f_h|_{h=h_{i_N}} + u_{N,j} \\ &= h_{i_N}^{1-r}\delta_{N,j}\partial_h \hat f_h|_{h=h_{i_N}} + r_{N,j},\end{aligned}$$

where $u_{N,j} \in L^2$ is the remainder with $\|u_{N,j}\|_2 \le |\tau_{N,j}|\mu_{[h_{\min},h_{\max}]}(|\tau_{N,j}|)$ according to Lemma 7 and $r_{N,j} := u_{N,j} + \alpha_{N,j}\partial_h \hat f_{h_{i_N}}$, where $|\alpha_{N,j}| \le C\delta_{N,j}^2$ for some constant $C > 0$ and all N sufficiently large.

We have $|\tau_{N,j}| \le h_{\max}^{1-r}|\delta_{N,j}|$ and $|\delta_{N,j}| \le A\lambda_N^{-1/2}$ for all $j \in I_N(A)$. Since $h \mapsto \partial_h \hat f_h$ is continuous in $L^2(\mathbb{R})$, it follows that $h \mapsto \|\partial_h \hat f_h\|_2$ is continuous in $\mathbb{R}$. Thus, $M_0 := \max_{h\in[h_{\min},h_{\max}]} \|\partial_h \hat f_h\|_2$ is finite.

We have $\|r_{N,j}\|_2 \le |\delta_{N,j}|a_N(A)$ for some sequence $a_N(A) \to 0$ for $N \to \infty$, be-

cause

$$\begin{aligned}\|r_{N,j}\|_2 &\leq \|u_{N,j}\|_2 + C\delta_{N,j}^2 \|\partial_h \hat{f}_h|_{h=h_{i_N}}\|_2 \\ &\leq |\tau_{N,j}| \mu_{[h_{\min},h_{\max}]}(|\tau_{N,j}|) + C\delta_{N,j}^2 M_0 \\ &\leq |\delta_{N,j}| \left(h_{\max}^{1-r} \mu_{[h_{\min},h_{\max}]}(h_{\max}^{1-r}|\delta_{N,j}|) + CM_0|\delta_{N,j}|\right) \\ &\leq |\delta_{N,j}| \left(\underbrace{h_{\max}^{1-r} \mu_{[h_{\min},h_{\max}]}(h_{\max}^{1-r} A\lambda_N^{-1/2}) + CM_0 A\lambda_N^{-1/2}}_{=:a_N(A)} \right)\end{aligned}$$

and $a_N(A) \to 0$ since $\lambda_N^{-1/2} \to 0$ by assumption and $\mu_{[h_{\min},h_{\max}]}(s) \to 0$ for $s \downarrow 0$ by Lemma 7.

Consequently,

$$D_{i_N j} = \frac{1}{2}\|\hat{f}_{h_j} - \hat{f}_{h_{i_N}}\|_2^2 = \frac{1}{2} h_{i_N}^{2(1-r)} (\delta_{N,j})^2 \|\partial_h \hat{f}_h|_{h=h_{i_N}}\|_2^2 + \rho_{N,j} = (\delta_{N,j})^2 \hat{Q}_r(h_{i_N}) + \rho_{N,j},$$

where $\rho_{N,j} = h_{i_N}^{1-r} \delta_{N,j} \langle \partial_h \hat{f}_h|_{h=h_{i_N}}, r_{N,j}\rangle + (1/2)\|r_{N,j}\|_2^2$ is yet another remainder term for which we have

$$\begin{aligned}|\rho_{N,j}| &\leq |\delta_{N,j}| h_{i_N}^{1-r} |\langle \partial_h \hat{f}_h|_{h=h_{i_N}}, r_{N,j}\rangle| + (1/2)\|r_{N,j}\|_2^2 \\ &\leq |\delta_{N,j}| h_{\max}^{1-r} \underbrace{\|\partial_h \hat{f}_h|_{h=h_{i_N}}\|_2}_{\leq M_0} \underbrace{\|r_{N,j}\|_2}_{\leq |\delta_{N,j}| a_N(A)} + (1/2) \underbrace{\|r_{N,j}\|_2^2}_{\leq (\delta_{N,j})^2 a_N(A)^2} \\ &\leq (\delta_{N,j})^2 \underbrace{\left(h_{\max}^{1-r} M_0 a_N(A) + (1/2) a_N(A)^2\right)}_{=:b_N(A)}.\end{aligned}$$

Since $a_N(A) \to 0$, we have shown that $\rho_{N,j} = \mathbf{o}(\delta_{N,j}^2)$. Thus,

$$S_{i_N j} = e^{-\lambda_N D_{i_N j}} = e^{-\lambda_N \delta_{N,j}^2 \hat{Q}_r(h_{i_N})} e^{-\lambda_N \rho_{N,j}}.$$

Define $\eta_{N,j} = e^{-\lambda_N \rho_{N,j}} - 1$. It follows $S_{i_N j} = e^{-\lambda_N (\delta_{N,j})^2 \hat{Q}_r(h_{i_N})}(1 + \eta_{N,j})$. Now write

$$\epsilon_N \sum_{j \in I_N(A)} S_{i_N j}^{(N)} = \epsilon_N \sum_{j \in I_N(A)} e^{-\lambda_N \delta_{N,j}^2 \hat{Q}_r(h_{i_N})} + \epsilon_N \sum_{j \in I_N(A)} e^{-\lambda_N \delta_{N,j}^2 \hat{Q}_r(h_{i_N})} \eta_{N,j}.$$

The local error can be bounded:

$$\left|\epsilon_N \sum_{j\in I_N(A)} \underbrace{e^{-\lambda_N \delta_{N,j}^2 \hat{Q}_r(h_{i_N})}}_{\leq 1} \eta_{N,j}\right| \leq \underbrace{\epsilon_N \#I_N(A)}_{\leq 2A+\epsilon_N} \sup_{j\in I_N(A)} |\eta_{N,j}| \to 0, \quad N\to\infty.$$

We must show that $\sup_{j\in I_N(A)} |\eta_{N,j}| \to 0$ for $N\to\infty$. We have $\lambda_N|\rho_{N,j}| \leq \lambda_N(\delta_{N,j})^2 b_N(A) \leq A^2 b_N(A)$ for all $j\in I_N(A)$. Consequently, $|\eta_{N,j}| = |e^{-\lambda_N\rho_{N,j}} - 1| \leq \max\{e^{A^2 b_N(A)} - 1, 1 - e^{-A^2 b_N(A)}\} \to 0$ as $N\to\infty$.

Since $\hat{Q}_r$ is continuous in h_0, we have $\hat{Q}_r(h_{i_N}) \to \hat{Q}_r(h_0)$ for $N\to\infty$. Therefore

$$\left|\epsilon_N \sum_{j\in I_N(A)} e^{-t_{N,j}^2 \hat{Q}_r(h_{i_N})} - \epsilon_N \sum_{j\in I_N(A)} e^{-t_{N,j}^2 \hat{Q}_r(h_0)}\right| \leq (2A+\epsilon_N)A^2 |\hat{Q}_r(h_{i_N}) - \hat{Q}_r(h_0)| \to 0.$$

We continue working with the remaining non-vanishing term:

$$\begin{aligned}
&\left|\epsilon_N \sum_{j\in I_N(A)} e^{-t_{N,j}^2 \hat{Q}_r(h_{i_N})} - \int_{-A}^{A} e^{-t^2 \hat{Q}_r(h_0)}\, dt\right| \\
&\leq \underbrace{\left|\epsilon_N \sum_{j\in I_N(A)} e^{-t_{N,j}^2 \hat{Q}_r(h_{i_N})} - \epsilon_N \sum_{j\in I_N(A)} e^{-t_{N,j}^2 \hat{Q}_r(h_0)}\right|}_{\to 0} + \left|\epsilon_N \sum_{j\in I_N(A)} e^{-t_{N,j}^2 \hat{Q}_r(h_0)} - \int_{-A}^{A} e^{-t^2 \hat{Q}_r(h_0)}\, dt\right|.
\end{aligned}$$

In order to show that the second summand above also converges to 0 for $N\to\infty$ we first rewrite the sum over $I_N(A)$. Before that, set $q = \hat{Q}_r(h_0)$, $g(t) := e^{-qt^2}$, $m_N := \lfloor A\epsilon_N^{-1}\rfloor$ and $x_k := \epsilon_N(-m_N + k - 1)$. For sufficiently large N we can now

write

$$\begin{aligned}\epsilon_N \sum_{j\in I_N(A)} g(t_{N,j}) &= \epsilon_N \sum_{j=1,\dots,N:-A\epsilon_N^{-1}+i_N\le j\le A\epsilon_N^{-1}+i_N} g(t_{N,j}) \\ &= \epsilon_N \sum_{k=-\lfloor A\epsilon_N^{-1}\rfloor}^{\lfloor A\epsilon_N^{-1}\rfloor} g(k\epsilon_N) \\ &= \sum_{k=-m_N}^{m_N} (k\epsilon_N-(k-1)\epsilon_N)g(k\epsilon_N) \\ &= \sum_{k=1}^{2m_N+1} (x_k-x_{k-1})g(x_k)\end{aligned}$$

We have used $t_{N,1}=\sqrt{\lambda_N}\delta_{N,1}=\sqrt{\lambda_N}r^{-1}(h_{\min}^r-h_{i_N}^r)\to-\infty$ and $t_{N,N}\to+\infty$ for $N\to\infty$ because we assumed that $h_{i_N}\to h_0$ for some interior scale h_0 and because $\lambda_N\to\infty$.

Now $\{x_0,\dots,x_{2m_N+1}\}$ is a partition of the interval $[-\epsilon_N m_N-\epsilon_N,\epsilon_N m_N]$.

Thus,

$$\begin{aligned}\left|\sum_{k=1}^{2m_N+1}(x_k-x_{k-1})g(x_k)-\int_{-\epsilon_N m_N-\epsilon_N}^{\epsilon_N m_N} g(x)\,dx\right| &\le \sum_{k=1}^{2m_N+1}\int_{x_{k-1}}^{x_k}|g(x_k)-g(x)|\,dx \\ &\le 2A\hat{Q}_r(h_0)(2A\epsilon_N+\epsilon_N^2)\to 0,\quad N\to\infty,\end{aligned}$$

because $\epsilon_N\to 0$ since we have assumed $\lambda_N/N^2\to 0$ and

$$\left|\int_{-\epsilon_N m_N-\epsilon_N}^{\epsilon_N m_N} g(t)\,dt-\int_{-A}^{A} g(t)\,dt\right|\to 0,\quad N\to\infty,$$

because $-\epsilon_N(m_N+1)\to -A$ and $\epsilon_N m_N\to A$.

We have now shown that the aforementioned second summand also converges to zero.

Finally,

$$\begin{aligned} L_A &= \lim_{N\to\infty} \epsilon_N \sum_{j\in I_N(A)} S^{(N)}_{i_N j} \le \lim_{N\to\infty} \epsilon_N \sum_{j\in I_N(A)} S^{(N)}_{i_N j} + \liminf_{N\to\infty} \epsilon_N T_N(A) \\ &\le \liminf_{N\to\infty} \epsilon_N R^{(N)}_{i_N} \le \limsup_{N\to\infty} \epsilon_N R^{(N)}_{i_N} \le \lim_{N\to\infty} \epsilon_N \sum_{j\in I_N(A)} S^{(N)}_{i_N j} + \limsup_{N\to\infty} \epsilon_N T_N(A) \\ &\le L_A + \frac{e^{-cA^2}}{cA} \end{aligned}$$

Letting $A \to \infty$, we see that the limit of $\epsilon_N R^{(N)}_{i_N}$ as $N \to \infty$ exists and is given by

$$\lim_{N\to\infty} \epsilon_N R^{(N)}_{i_N} = \lim_{A\to\infty} L_A = \int_{-\infty}^{\infty} e^{-t^2 \hat{Q}_r(h_0)}\, dt = \sqrt{\frac{\pi}{\hat{Q}_r(h_0)}}.$$

We must make sure that $\hat{Q}_r(h_0) > 0$. This follows from previously established results inside this proof. For all sufficiently large N we have $0 < c \le \delta^{-2}_{N,i_N+1} D_{i_N,i_N+1} = \hat{Q}_r(h_{i_N}) + \delta^{-2}_{N,i_N+1}\rho_{N,i_N+1}$ and taking the limit gives $\hat{Q}_r(h_0) \ge c > 0$. □

6.2.4 Proof of Proposition 9

Proof. Define the matrix $A^{(N)} := \sqrt{\lambda_N}\Delta_{r,N} S^{(N)}$. Denote the i-th row sum of $A^{(N)}$ by $a_{i,N}$. Consequently, by definition of $A^{(N)}$, we have $a_{i,N} = \sqrt{\lambda_N}\Delta_{r,N} R^{(N)}_i$, where $R^{(N)}_i$ is the i-th row sum of $S^{(N)}$. Denote by ρ_N the principal eigenvalue of $A^{(N)}$ and by $v^{(N)} = (v^{(N)}_1, \ldots, v^{(N)}_N)$ the corresponding principal eigenvector. That is, we have $A^{(N)} v^{(N)} = \rho_N v^{(N)}$. Let $i^{(N)}_{\text{crit}} \in \arg\max\{v^{(N)}_i : 1 \le i \le N\}$. We have

$$\rho_N v^{(N)}_{i^{(N)}_{\text{crit}}} = \sqrt{\lambda_N}\Delta_{r,N} \sum_{j=1}^{N} S^{(N)}_{i^{(N)}_{\text{crit}} j} v_j \le \sqrt{\lambda_N}\Delta_{r,N} v_{i^{(N)}_{\text{crit}}} R^{(N)}_{i^{(N)}_{\text{crit}}}$$

and because every entry of the Perron vector is positive, $\rho_N \le a_{i^{(N)}_{\text{crit}},N}$.

There exists a sequence $\epsilon_N \to 0$ such that $a_{i,N} \le m(t_i) + \epsilon_N$ for all $1 \le i \le N$. Assume for a contradiction that for all null sequences (ϵ_N) and for all $N_0 \in \mathbb{N}$ there exist $N \ge N_0$ and $1 \le i \le N$ such that $a_{i,N} > m(t_i) + \epsilon_N$. This implies the existence of $d > 0$ and $N_1 < N_2 < \ldots$ and indices $i_1, i_2, \ldots$ such that $a_{i_l,N_l} > m(t_{i_l}) + d$. By compactness and continuity arguments $\liminf_{l\to\infty} a_{i_l,N_l} \ge m(t_0) + d$ for some

$t_0 \in [\alpha, \beta]$.

But on the other hand $\limsup_{l\to\infty} a_{i_l,N_l} \leq m(t_0)$, because

$$a_{i_l,N_l} = \sigma_{N_l} \underbrace{\sum_{j:|t_{N_l,j}|\leq A} S_{i_l,j}^{(N_l)}}_{=:P_{N_l}(A)} + \sigma_{N_l} \underbrace{\sum_{j:|t_{N_l,j}|>A} S_{i_l,j}^{(N_l)}}_{=:T_{N_l}(A)}.$$

In the proof of Proposition 8 we have already shown $\limsup_{l\to\infty} \sigma_{N_l} T_{N_l}(A) \leq e^{-cA^2}/(cA)$.

In the proof of Proposition 8 we have also seen that

$$\sigma_{N_l} P_{N_l}(A) = \sigma_{N_l} \sum_{j:|t_{N_l,j}|\leq A} e^{-t_{N_l,j}^2 \hat{Q}_r(h_0)} + \mathbf{o}(1).$$

Now we have

$$\sum_{j:|t_{N_l,j}|\leq A} e^{-q_0 t_{N_l,j}^2} \leq \sum_{k\in\mathbb{Z},|\sigma_{N_l}k|\leq A} e^{-q_0(\sigma_{N_l}k)^2} =: G_{N_l}(A).$$

In the proof of Proposition 8 we have seen that $\sigma_{N_l} G_{N_l}(A) \to \int_{-A}^{A} e^{-q_0 z^2}\, dz$ and thus we get $\limsup_{l\to\infty} \sigma_{N_l} P_{N_l}(A) \leq \int_{-A}^{A} e^{-q_0 z^2}\, dz$. In total we have $\limsup_{l\to\infty} a_{i_l,N_l} \leq \int_{-A}^{A} e^{-q_0 z^2}\, dz + e^{-cA^2}/(cA)$, and taking $A \to \infty$ yields $\limsup_{l\to\infty} a_{i_l,N_l} \leq m(t_0)$.

Combining this would yield the contradiction $m(t_0) + d \leq m(t_0)$.

The main difficulty lies in showing that $\rho_N \to M := \sqrt{\pi/\hat{Q}_r(h^*)}$. It can be divided into showing $\limsup \rho_N \leq M$ and $\liminf \rho_N \geq M$.

$\limsup \rho_N \leq M$. It holds $\rho_N \leq a_{i_N,N} \leq m(t_{i_N}) + \epsilon_N \leq M + \epsilon_N$ and therefore $\limsup \rho_N \leq M$.

$\liminf \rho_N \geq M$ We use the Rayleigh variational representation

$$\begin{aligned}\rho_N &= \max_{x\neq 0}(x^T A^{(N)} x)/(x^T x) = (v^{(N)})^T A^{(N)} v^{(N)}/(v^{(N)})^T (v^{(N)}) \\ &\geq (y^{(N)})^{(T)} A^{(N)} y^{(N)}/((y^{(N)})^T (y^{(N)}))\end{aligned}$$

for all $y^{(N)} \in \mathbb{R}^N$, $y^{(N)} \neq 0$. Let $0 < \eta < M$. By the continuity of m we can find a non-empty closed interval $J \subseteq (\alpha, \beta)$ such that $m(t) \geq M - \eta$ for all $t \in J$.

Denote the interval by $J = [u, v]$, $u < v$. Define $y^{(N)}$ by $y_i^{(N)} = \mathbf{1}\{t_i \in J\}$. Let $I_N = \{i : t_i \in J\}$.

For all sufficiently large N, I_N is nonempty (this follows from $\Delta_{r,N}$ being eventually smaller than $v - u$, which is why for $i_N = 1 + \lceil (u - \alpha)/\Delta_{r,N} \rceil$ we have $t_{i_N} \in J$). Because of that, $y^{(N)} \neq 0$ for sufficiently large N. It follows

$$\begin{aligned}\rho_N &\geq (y^{(N)})^{(T)} A^{(N)} y^{(N)} / ((y^{(N)})^T (y^{(N)})) = \frac{1}{\#I_N} \sum_{i,j \in I_N} A_{ij}^{(N)} \\ &= \frac{1}{\#I_N} \sum_{i \in I_N} a_{i,N} - \underbrace{\frac{1}{\#I_N} \sum_{i \in I_N} \sum_{j \notin I_N} A_{ij}^{(N)}}_{=:E_N}.\end{aligned}$$

Set $b_N := \min\{a_{i,N} : i \in I_N\}$. Then $(1/\#I_N) \sum_{i \in I_N} a_{i,N} \geq b_{i,N}$. For a contradiction, assume that there exists a subsequence (b_{N_l}) such that b_{N_l} converges towards a limit $b < M - \eta$. Choose $i_l \in I_{N_l}$ so that $b_{N_l} = \min_{i \in N_l} a_{i,N_l} = a_{i_l,N_l}$. Since $t_{i_l} \in J$ and J is compact, there exists a further subsequence $t_{i_{l_k}}$ such that $t_{i_{l_k}} \to t_0 \in J$ for $k \to \infty$. Now we are in a position to apply Proposition 8 and obtain $b_{i_{l_k}} \to m(t_0)$, but $m(t_0) \geq M - \eta$ by the definition of the set J, which is a contradiction. We have thus shown that $\liminf b_N \geq M - \eta$.

Next, we show that $E_N \to 0$ for $N \to \infty$. From the proof of Proposition 8 we know that $A_{ij}^{(N)} \leq \sigma_N e^{-c\sigma_N^2 (i-j)^2}$, where we have set $\sigma_N := \sqrt{\lambda_N} \Delta_{r,N}$. Now we see

$$\begin{aligned}(\#I_N) E_N &= \sum_{i \in I_N, j \notin I_N} A_{ij}^{(N)} \leq \sigma_N \sum_{k=1}^{\infty} \sum_{i \in I_N, j \notin I_N, |i-j|=k} e^{-c\sigma_N^2 k^2} \\ &= \sigma_N \sum_{k=1}^{\infty} \underbrace{\#\{(i,j) : i \in I_N, j \notin I_N, |i-j| = k\}}_{\leq 2k} e^{-c\sigma_N^2 k^2} \\ &\leq 2\sigma_N \sum_{k=1}^{\infty} k e^{-c\sigma_N^2 k^2} \leq 2\sigma_N \int_0^{\infty} (x+1) e^{-c\sigma_N^2 x^2} \, dx \\ &= \frac{1}{c\sigma_N} + \frac{\sqrt{\pi}}{\sqrt{c}}.\end{aligned}$$

During the above calculations, we have bounded the number of index pairs (i, j) with $i \in I_N$, $j \in I_N$ and $|i - j| = k$ above by $2k$. This can be done because I_N is a set of consecutive indices which we can denote by $t_{j_1}, \ldots, t_{j_K}$ where only one of

the first k indices $i \in \{j_1, \ldots, j_k\}$ can have an index j such that t_j is outside of I_N and $|i-j| = k$. Similarly, only the last k indices $i \in \{j_{K-k+1}, \ldots, j_K\}$ can have an index j such that t_j is outside of I_N and $|i-j| = k$. In the last inequality, we have also used $\int_{k-1}^{k}(x+1)e^{-c\sigma_N^2 x^2}\,dx \geq ke^{-c\sigma_N^2 k^2}$ for all $k \in \mathbb{N}$.

The number of nodes inside I_N can be bounded below by $\#I_N \geq (v-u)/\Delta_{r,N} - 1 \to \infty$. Hence, $1/(\#I_N) \to 0$ and $\sigma_N(\#I_N) \geq (u-v)\frac{\sigma_N}{\Delta_{r,N}} - \sigma_N = (u-v)\sqrt{\lambda_N} - \sqrt{\lambda_N}\Delta_{r,N} \to \infty$ and hence $1/\sigma_N(\#I_N) \to 0$ too. Thus $E_N \to 0$.

Bringing the previous arguments together, we arrive at $\liminf \rho_N \geq \liminf b_n \geq M - \eta$ and since η was arbitrary, $\liminf \rho_N \geq M$. Thus, we have $\rho_N \to M$ for $N \to \infty$.

We now have $\rho_N \leq a_{i_N,N} \leq m(t_{i_N}) + \epsilon_N$. It follows $M \leftarrow \rho_N - \epsilon_N \leq m(t_{i_N}) \leq M$ and thus $m(t_{i_N}) \to M$. Since m is continuous, defined on a compact interval and by assumption has a unique maximizer, it follows that $t_{i_N} \to t^*$. By continuity of ψ_r^{-1} follows $h_{i_N} \to h^*$, which is the claim of this Proposition. □

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Declaration

Ich erkläre, dass ich die Arbeit selbständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel verwendet habe.

AI used for code, idea generation, and literature search

Ulm, den .....

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